DELPHIX CONTINUOUS COMPLIANCE · DISCOVERY POLICY

United Kingdom — UK GDPR and Data Protection Act 2018

Discovery and masking of UK personal data: the National Insurance number, the NHS number and other identifiers, names and contact, bank details, quasi-identifiers such as the postcode, the special categories of Article 9, criminal offence data and financial data.

73	130	105
DATA CLASSES	CLASSIFIERS	ALGORITHMS

Discovery policy `UK – UK GDPR and DPA 2018 – v1`

Version 1 of October 12, 2026 · Assignment threshold 60%

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About this discovery policy

This discovery policy finds the personal data in a UK database and masks it, so that a non-production copy — for testing, development, training or analytics — no longer identifies anyone.

Masking only names and National Insurance numbers does not get there: a birth date, a sex and a full postcode are enough to point at one person. The policy works on the layers where the risk is:

- **Direct identifiers** — National Insurance number, NHS, CHI and H&C numbers, Unique Taxpayer Reference, passport, driving licence, immigration documents, civil registration, names, contact, bank account, sort code, IBAN and cards, property references, devices, vehicles, free text.
- **Quasi-identifiers** — birth date, age, sex, postcode, town, nationality, marital and civil partnership status, occupation, employer, qualifications.
- **Special category data** — the categories of Article 9 of the UK GDPR: racial or ethnic origin, political opinions, religious or philosophical beliefs, trade union membership, genetic and biometric data, sex life and sexual orientation — and gender identity, which the Gender Recognition Act 2004 protects.
- **Health** — data concerning health, also special category: hospital numbers, diagnoses (ICD-10 and SNOMED CT), medications, clinical notes, test results, blood group, disability, reproductive and mental health.
- **Financial data** — credit files, County Court Judgments and insolvency, debt, income, assets and pensions, benefits, deprivation indices, housing tenure.
- **Criminal offence data** — convictions, cautions and the alleged commission of offences, PNC and case references, DBS checks.

In Data Control Tower the policy is one discovery policy with its classifiers, the data classes they vote for, the algorithms those data classes name — and the algorithms those build on — and the lookup files they read. The appendices list every one of them with its full configuration. Some data classes use algorithms that come with Delphix — CM Numeric, Credit Card, BloodType SL — and those are named but not included.

The assignment threshold is 60%. An unambiguous column name reaches it alone; an ambiguous one — `NAME`, `RESULT`, `CITY` — needs the values to agree. A column with a meaningless name (`FIELD1`, `PFNOM`, `REMARKS`) is still found when its values have a recognizable shape or vocabulary: a National Insurance number, an NHS number that passes its check digit, a UK phone number, a postcode, a plate, an IBAN, a town, a full name.

Column names are recognized in English and in the abbreviations of UK systems (`NINO`, `NI_NO`, `NHS_NO`, `CHI`, `UTR`, `UPRN`, `MPAN`, `PNC_ID`, `DBS`, `DOB`).

Where a number has a check digit, the masked number has a valid one too: the NHS number (and the CHI and H&C numbers, which use the same check) and the Unique Taxpayer Reference are masked digit by digit and their check digit recomputed, and the IBAN gets new check digits. An application that validates them still accepts the masked copy.

> Before masking, inventory the binary columns (BLOBs, images, PDFs). Onboarding and right-to-work processes often store a scan of the passport, the driving licence or the biometric residence permit — and a facial image processed to identify someone is biometric data under Article 4(14). Report those columns as residual risk.

The UK GDPR and the Data Protection Act 2018 at a glance

The United Kingdom has no single data protection statute. The **UK GDPR** — the EU General Data Protection Regulation as retained in UK law after Brexit and amended since — sets the principles, rights and duties; the **Data Protection Act 2018** supplements it, sets the exemptions and the offences, and covers law enforcement and the intelligence services. Both were amended by the **Data (Use and Access) Act 2025**, whose data protection changes came into force in stages in 2026. On 30 September 2026 the **Information Commission** replaced the Information Commissioner's Office as the regulator.

What matters for a copy of production data:

- **Personal data** is any information relating to an identified or identifiable living individual — directly or indirectly, by reference to an identifier such as a name, an identification number, location data or an online identifier, or to factors specific to the person's physical, physiological, genetic, mental, economic, cultural or social identity (UK GDPR Article 4(1); DPA 2018 s. 3).
- **Pseudonymisation** is processing personal data so that it can no longer be attributed to a specific person without additional information, kept separately and under technical and organisational measures (Article 4(5)). Pseudonymised data is still personal data; only data that is effectively anonymous, taking account of the means reasonably likely to be used to identify someone, falls outside the law (Recital 26).
- **Processing** includes copying, adapting, altering, combining and erasing data (Article 4(2)). Copying production data to a test environment is processing; so is masking it.
- **Principles** (Article 5): purpose limitation — no further processing incompatible with the original purposes —, **data minimisation** — adequate, relevant and limited to what is necessary —, storage limitation, and integrity and confidentiality. A full copy of production data in a test environment fails minimisation unless it is needed there.
- **Data protection by design and by default** (Article 25) asks controllers to implement measures such as pseudonymisation that put the principles into effect, and to process by default only the data each purpose needs.
- **Security of processing** (Article 32(1)(a)) names **the pseudonymisation and encryption of personal data** among the appropriate technical measures, with a process for regularly testing their effectiveness (Article 32(1)(d)).
- **Special categories** (Article 9): racial or ethnic origin, political opinions, religious or philosophical beliefs, trade union membership, genetic data, biometric data processed to uniquely identify a person, data concerning health, and data concerning a person's sex life or sexual orientation. Processing them needs a condition of Article 9(2) and, for most of them, one of Schedule 1 to the 2018 Act, with an appropriate policy document.
- **Criminal offence data** (Article 10) may be processed only under official authority or a condition of Schedule 1. Section 11(2) of the Act extends it to **the alleged commission of offences** and to proceedings and their disposal — a charge that ended in acquittal is criminal offence data.
- **Re-identifying de-identified data is a criminal offence.** A person who knowingly or recklessly re-identifies information that is de-identified personal data, without the consent of the controller responsible for de-identifying it, commits an offence, and so does a person who knowingly or recklessly processes data re-identified that way (DPA 2018 s. 171). Testing the effectiveness of de-identification is a defence when done in the public interest and notified to the Commission (s. 172).
- **Enforcement.** The higher maximum fine is £17.5 million or 4% of annual worldwide turnover, whichever is higher (Article 83; DPA 2018 s. 157). Personal data breaches are reported to the Commission within 72 hours when they pose a risk to people (Article 33).
- **Children.** For online services offered directly to children, consent is valid from the age of 13 (Article 8 and DPA 2018 s. 9).

A test environment with production data is further processing for purposes the data was not collected for, by people who have no need to see it, and it holds special category and criminal offence data outside the

controls built for them. Masking is the pseudonymisation Articles 25 and 32 name: what goes to the test environment no longer identifies anyone without the key, and section 171 makes undoing it without the controller's consent an offence.

The policy goes beyond Article 9 where the risk is the same: it treats gender identity with the same care as sexual orientation — section 22 of the Gender Recognition Act 2004 makes disclosing, from official information, that a person applied for or holds a gender recognition certificate an offence — and masks financial data, which banks keep under a duty of confidence and credit reference agencies share only under the Consumer Credit Act 1974. Convictions follow the Rehabilitation of Offenders Act 1974: a spent conviction is treated as if it never happened, and a copy that keeps it undoes that.

> Whether data is anonymous depends on the means reasonably likely to be used to identify someone, by the holder or by anyone it may reach — the Commission's anonymisation guidance tests it against a motivated intruder. Masking is deterministic for the engine key, so whoever has both the production data and the key can link a masked row back to its source: for them the copy is pseudonymised data, still personal data. Keep the key, and access to production, away from the people who use the masked copy. Whether a copy counts as anonymous is a legal assessment of the whole environment, not of the discovery policy alone.

3

How sensitive data is discovered

A discovery job looks at every column of its rule set and asks each classifier of the discovery policy how confident it is that the column belongs to its data class. The policy uses the four classifier frameworks of Delphix:

- **Column name** (Path) — regular expressions on the column name. Each rule has a match strength: the confidence it gives when it matches.
- **Data type** (Type) — accepts the column type and length, or rules the data class out. It never assigns a data class on its own.
- **Value format** (Regex) — tests a sample of the values against patterns, some with a check digit. The confidence follows the share of values that match.
- **Known values** (List) — looks the sampled values up in lists; for names, word by word.

Evidence from the name and from the values is combined per data class, and the column is assigned to the data class with the highest confidence when it reaches the assignment threshold of the discovery policy — **60%** here. Values that do not match a format or a list count against their data class by the classifier's reject strength.

These are the names Data Control Tower gives the objects: data classes, classifiers and a discovery policy, run by a discovery job. A Continuous Compliance Engine used on its own calls the same objects domains, classifiers and a profile set, run by a profiling job; everything in this document applies to both.

Databases pad values: a CHAR column fills them with spaces up to its length, and a numeric code kept as text often comes with zeros on the left. Discovery tests values as they come, so every format pattern first removes the spaces around the value, patterns of digits also accept zeros on the left, lists of single words are looked up word by word, and lists of names made of several words (places, categories) are held as a single value-format pattern. A column whose name says nothing — `CAMP01`, `FLD_07`, the short names of packaged systems — is found by its values alone wherever their format is distinctive: documents with a check digit, e-mail addresses, telephone numbers, cards, places, names.

Column-name rules are shown as their alternatives. They ignore case and must not be glued to other letters: `rut` finds `RUT_CLIENTE` and `CLIENTE_RUT` but not `RUTA`. In them, `_?` is an optional underscore, `(a|b)` either form, `s?` an optional letter and `[a-z_]*` any continuation.

> Discovery proposes; it does not decide. Review the data classes assigned in the rule set before running a masking job, and look for columns the policy left unassigned.

4

How data is masked

Each data class names one algorithm, which a masking job applies to every column the rule set assigns to the data class. Algorithms can be composed: a **Regex Decompose** splits a value into parts and gives each part its own action; a **String Algorithm Chain** applies algorithms one after another. The tree under each data class shows everything its algorithm uses; Appendix A has every configuration in full.

The algorithms are deterministic for their key: the same value masks the same way in every table, database and job, so joins between masked systems keep working — as long as every engine and compliance node uses the same keys.

Where an algorithm that comes with Delphix already does the job — CM Numeric for amounts, Credit Card, the blood group lookup — the policy names it instead of creating one of its own. Those algorithms appear in the trees by their `dlpx-core:` name, the one an engine uses; Data Control Tower lists them without the prefix (`CM Numeric` , `CreditCard` , `BloodType SL`). They are on every engine and in every Data Control Tower, and are not listed in Appendix A.

Padding stays where it is: the spaces around a value are kept, and zeros on the left of a number are not masked — `0012345678` becomes `00` followed by what `12345678` becomes. A value kept in a zero-filled CHAR column still joins the same value kept as a number.

> The masked examples in this document were produced with one particular key. With other keys the masked values differ; their shape does not.

5

Essential and extended packs

The policy can be adopted whole or in part. The **extended pack** is everything this document describes. The **essential pack** is the minimum for the law — identity documents, names, contact details, address and birth date. To adopt only the essential pack, create just its data classes, the classifiers that vote for them and the algorithms and files those use, and put only those classifiers in the discovery policy.

PACK	DATA CLASSES	CLASSIFIERS	ALGORITHMS	FILES
Essential	13	27	27	7
Extended	73	130	105	45

The essential pack holds these data classes, marked **Essential** in the tables and chapters:

- National Insurance number — `UK_L1_NINO`
- NHS number, CHI number, H&C number — `UK_L1_NHS_NUMBER`
- Passport — `UK_L1_PASSPORT`
- Driving licence — `UK_L1_DRIVING_LICENCE`
- Given names — `UK_L1_GIVEN_NAME`
- Surnames — `UK_L1_SURNAME`

- Full name — `UK_L1_FULL_NAME`
- E-mail — `UK_L1_EMAIL`
- Phone — `UK_L1_PHONE`
- Address — `UK_L1_ADDRESS`
- Address detail — `UK_L1_ADDRESS_DETAIL`
- Date of birth — `UK_L2_BIRTH_DATE`
- Postcode — `UK_L2_POSTCODE`

> With fewer data classes competing, a column that a data class of the extended pack would claim by its name can be proposed for an essential data class instead — a numeric code, for instance, can look like a telephone number. Review the discovery result.

6

Data classes at a glance

Every data class of the policy, how it is found and the algorithm that masks it.

DATA CLASS	FOUND BY	ALGORITHM
DIRECT IDENTIFIERS		
National Insurance number ESSENTIAL UK_L1_NINO	COLUMN NAME · VALUE FORMAT	UK_NINO
NHS number, CHI number, H&C number ESSENTIAL UK_L1_NHS_NUMBER	COLUMN NAME · VALUE FORMAT	UK_NHS_COLUMN
Unique Taxpayer Reference (UTR) UK_L1_UTR	COLUMN NAME · VALUE FORMAT	UK_UTR
Passport ESSENTIAL UK_L1_PASSPORT	COLUMN NAME · VALUE FORMAT	UK_CM_ALNUM
Driving licence ESSENTIAL UK_L1_DRIVING_LICENCE	COLUMN NAME · VALUE FORMAT	UK_CM_ALNUM
Immigration documents UK_L1_IMMIGRATION_DOCUMENT	COLUMN NAME	UK_CM_ALNUM
Civil registration UK_L1_VITAL_RECORD	COLUMN NAME	UK_CM_ALNUM
Given names ESSENTIAL UK_L1_GIVEN_NAME	COLUMN NAME · KNOWN VALUES	UK_GIVEN_NAMES
Surnames ESSENTIAL UK_L1_SURNAME	COLUMN NAME · KNOWN VALUES	UK_SURNAMES
Full name ESSENTIAL UK_L1_FULL_NAME	COLUMN NAME · KNOWN VALUES	UK_FULL_NAME
E-mail ESSENTIAL UK_L1_EMAIL	COLUMN NAME · DATA TYPE · VALUE FORMAT	UK_EMAIL
Phone ESSENTIAL UK_L1_PHONE	COLUMN NAME · VALUE FORMAT	UK_PHONE
Address ESSENTIAL UK_L1_ADDRESS	COLUMN NAME · DATA TYPE · VALUE FORMAT	UK_ADDRESS
Address detail ESSENTIAL UK_L1_ADDRESS_DETAIL	COLUMN NAME	UK_CM_ALNUM
Bank account UK_L1_BANK_ACCOUNT	COLUMN NAME	UK_CM_ALNUM
Sort code UK_L1_SORT_CODE	COLUMN NAME · VALUE FORMAT	UK_SORT_CODE
IBAN UK_L1_IBAN	COLUMN NAME · VALUE FORMAT	UK_IBAN
Payment card UK_L1_PAYMENT_CARD	COLUMN NAME · VALUE FORMAT	UK_CARD
IP address UK_L1_IP_ADDRESS	COLUMN NAME · VALUE FORMAT	UK_IP

DATA CLASS	FOUND BY	ALGORITHM
Device identifier `UK_L1_DEVICE_ID`	COLUMN NAME · VALUE FORMAT	`UK_CM_HEX`
Vehicle registration `UK_L1_LICENCE_PLATE`	COLUMN NAME · VALUE FORMAT	`UK_PLATE`
VIN and V5C reference `UK_L1_VIN`	COLUMN NAME · VALUE FORMAT	`UK_CM_ALNUM`
Property and meter references `UK_L1_PROPERTY`	COLUMN NAME	`UK_CM_ALNUM`
Customer, contract or account number `UK_L1_CONTRACT_NUMBER`	COLUMN NAME	`UK_CM_ALNUM`
Username and social media `UK_L1_USERNAME`	COLUMN NAME	`UK_CM_ALNUM`
Credential `UK_L1_CREDENTIAL`	COLUMN NAME · VALUE FORMAT	`UK_REDACT`
Geolocation `UK_L1_GEOLOCATION`	COLUMN NAME · VALUE FORMAT	`UK_COORDINATE`
Free text `UK_L1_FREE_TEXT`	COLUMN NAME · DATA TYPE · VALUE FORMAT	`UK_FREE_TEXT_ALL`
QUASI-IDENTIFIERS		
Date of birth ESSENTIAL `UK_L2_BIRTH_DATE`	COLUMN NAME · DATA TYPE	`UK_BIRTH_DATE`
Year of birth `UK_L2_BIRTH_YEAR`	COLUMN NAME · DATA TYPE	`UK_YEAR`
Age `UK_L2_AGE`	COLUMN NAME · DATA TYPE	`UK_AGE`
Dates of life events `UK_L2_EVENT_DATE`	COLUMN NAME · DATA TYPE	`UK_EVENT_DATE`
Sex `UK_L2_SEX`	COLUMN NAME · KNOWN VALUES	`UK_SEX`
Postcode ESSENTIAL `UK_L2_POSTCODE`	COLUMN NAME · VALUE FORMAT	`UK_POSTCODE`
Town `UK_L2_TOWN`	COLUMN NAME · VALUE FORMAT	`UK_TOWN`
Nationality `UK_L2_NATIONALITY`	COLUMN NAME · VALUE FORMAT	`UK_NATIONALITY`
Marital and civil partnership status `UK_L2_MARITAL_STATUS`	COLUMN NAME · VALUE FORMAT	`UK_MARITAL_STATUS`
Occupation `UK_L2_OCCUPATION`	COLUMN NAME	`UK_OCCUPATION`
Employer `UK_L2_EMPLOYER`	COLUMN NAME	`UK_EMPLOYER`
Highest qualification `UK_L2_EDUCATION`	COLUMN NAME · VALUE FORMAT	`UK_EDUCATION`
School `UK_L2_SCHOOL`	COLUMN NAME	`UK_SCHOOL`

DATA CLASS	FOUND BY	ALGORITHM
Number of children or dependants `UK_L2_DEPENDANTS`	COLUMN NAME DATA TYPE	`UK_CAPPED_COUNT`
SPECIAL CATEGORY DATA		
Genetic data `UK_L3_GENETIC`	COLUMN NAME VALUE FORMAT	`UK_REDACT`
Biometric data `UK_L3_BIOMETRIC`	COLUMN NAME DATA TYPE	`UK_REDACT`
Racial or ethnic origin `UK_L3_ETHNICITY`	COLUMN NAME VALUE FORMAT	`UK_ETHNICITY`
Religious beliefs `UK_L3_RELIGION`	COLUMN NAME VALUE FORMAT	`UK_RELIGION`
Philosophical beliefs `UK_L3_BELIEF`	COLUMN NAME	`UK_SUPPRESSED_CATEGORY`
Political opinions `UK_L3_POLITICAL_OPINION`	COLUMN NAME VALUE FORMAT	`UK_POLITICAL_OPINION`
Political party `UK_L3_POLITICAL_AFFILIATION`	COLUMN NAME VALUE FORMAT	`UK_POLITICAL_PARTY`
Trade union membership `UK_L3_TRADE_UNION`	COLUMN NAME VALUE FORMAT	`UK_SUPPRESSED_CATEGORY`
Sexual orientation `UK_L3_SEXUAL_ORIENTATION`	COLUMN NAME VALUE FORMAT	`UK_SEXUAL_ORIENTATION`
Sex life `UK_L3_SEXUAL_LIFE`	COLUMN NAME	`UK_SUPPRESSED_CATEGORY`
Gender identity `UK_L3_GENDER_IDENTITY`	COLUMN NAME VALUE FORMAT	`UK_GENDER_IDENTITY`
DATA CONCERNING HEALTH		
Health record identifiers `UK_HEALTH_ID`	COLUMN NAME	`UK_CM_ALNUM`
Diagnosis `UK_HEALTH_DIAGNOSIS`	COLUMN NAME VALUE FORMAT	`UK_DIAGNOSIS`
Medication `UK_HEALTH_MEDICATION`	COLUMN NAME VALUE FORMAT	`UK_MEDICATION`
Clinical notes `UK_HEALTH_CLINICAL_TEXT`	COLUMN NAME DATA TYPE	`UK_FREE_TEXT_ALL`
Test results `UK_HEALTH_TEST_RESULT`	COLUMN NAME	`UK_CM_ALNUM`
Blood group `UK_HEALTH_BLOOD_GROUP`	COLUMN NAME VALUE FORMAT	`dlpx-core:BloodTypeSL`
Disability `UK_HEALTH_DISABILITY`	COLUMN NAME VALUE FORMAT	`UK_DISABILITY`
Reproductive and sexual health `UK_HEALTH_REPRODUCTIVE`	COLUMN NAME	`UK_SUPPRESSED_CATEGORY`
Mental health `UK_HEALTH_MENTAL`	COLUMN NAME	`UK_SUPPRESSED_CATEGORY`

DATA CLASS	FOUND BY	ALGORITHM
FINANCIAL DATA		
Credit file and insolvency `UK_FIN_CREDIT_REFERENCE`	COLUMN NAME	`UK_CREDIT`
Debt `UK_FIN_DEBT`	COLUMN NAME DATA TYPE	`dlpx-core:CM Numeric`
Income `UK_FIN_INCOME`	COLUMN NAME DATA TYPE	`dlpx-core:CM Numeric`
Assets `UK_FIN_ASSETS`	COLUMN NAME DATA TYPE	`dlpx-core:CM Numeric`
Pension `UK_FIN_PENSION`	COLUMN NAME DATA TYPE	`dlpx-core:CM Numeric`
Benefits `UK_FIN_BENEFIT`	COLUMN NAME VALUE FORMAT	`UK_BENEFIT`
Deprivation and socio-economic class `UK_FIN_DEPRIVATION`	COLUMN NAME VALUE FORMAT	`UK_DEPRIVATION`
Housing tenure `UK_FIN_HOUSING`	COLUMN NAME VALUE FORMAT	`UK_HOUSING`
CRIMINAL OFFENCE DATA		
Criminal record `UK_PENAL_CRIMINAL_RECORD`	COLUMN NAME	`UK_CRIMINAL_RECORD`
Police and court references `UK_PENAL_CASE_NUMBER`	COLUMN NAME VALUE FORMAT	`UK_CM_ALNUM`
DBS checks `UK_PENAL_DBS`	COLUMN NAME	`UK_DBS`

PART 7

Direct identifiers

Data that identifies a person on its own. The United Kingdom has no national identity card; a person is known to the state by several numbers instead — the National Insurance number for work, tax and benefits, the NHS, CHI or H&C number for health care, the Unique Taxpayer Reference for Self Assessment — and to everyone else by name, address, phone, bank account and card. Each number is masked the same way everywhere it appears, so it still joins its tables after masking, and the check digits that applications validate are recomputed.

7.1 National Insurance number

UK_L1_NINO

ESSENTIAL · COLUMN NAME · VALUE FORMAT · REGEX DECOMPOSE

WHAT IT IS

The National Insurance number of HMRC and the Department for Work and Pensions: two prefix letters, six digits and a suffix letter from A to D, usually written `QQ 12 34 56 C`. It follows a person through work, tax, benefits and the State Pension, and employers, banks, pension schemes and the NHS hold it. Found by its name (`NINO` , `NI_NO` , `NATIONAL_INSURANCE_NUMBER`) and by its shape, with the prefixes HMRC does not issue left out.

HOW IT IS FOUND

COLUMN NAME UK_L1_NINO — Path

90% `nino` `ni_?(no|num|number|nbr)` `nin` `national_?insurance(_?(no|num|number|nbr))?` `nat_?ins(_?(no|num|number))?` `ni_?num` `insurance_?number_?ni`

VALUE FORMAT UK_L1_NINO — Regex

PATTERN	STRENGTH	NOTES
`[A-CEGHJ-PR-TW-Z][A-CEGHJ-NPR-TW-Z]\s?\d{2}\s?\d{2}\s?\d{2}\s?[A-D]`	95%	
`[A-CEGHJ-PR-TW-Z][A-CEGHJ-NPR-TW-Z]\s?\d{2}\s?\d{2}\s?\d{2}`	40%	

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

The six digits are mapped digit by digit; the prefix and suffix letters, which say nothing about the person, and the spacing stay. The result is still a well-formed National Insurance number, and the same number masks the same way in every table.

`UK_NINO` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `UK_CM_ALNUM` .

> `UK_CM_DIGITS` CHARACTER MAPPING
>
> Maps each character to another of its group (1 groups), keeping length and format.
>
> `UK_CM_ALNUM` CHARACTER MAPPING
>
> Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
AB 12 34 56 C	→	**AB 90 42 92 C**
JK123456D	→	**JK904292D**

7.2 NHS number, CHI number, H&C number

UK_L1_NHS_NUMBER

ESSENTIAL | COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

WHAT IT IS

The ten-digit number of the patient in England, Wales and the Isle of Man (NHS number, written `943 476 5919`), in Scotland (Community Health Index number, whose first six digits are the date of birth) and in Northern Ireland (Health and Care number). All three end in the same modulus 11 check digit, and the classifier checks it: a column of valid NHS numbers is found even under a meaningless name.

HOW IT IS FOUND

COLUMN NAME UK_L1_NHS_NUMBER – Path

90% `nhs_?(no|num|number|nbr|id)?` `nhsno` `nhs_?number` `chi(_?(no|num|number))?` `community_?health_?index(_?(no|number))?` `h_?and_?c(_?(no|number))?` `hcn` `health_?and_?care_?(no|num|number)` `patient_?nhs_?(no|number)`

VALUE FORMAT UK_L1_NHS_NUMBER – Regex

PATTERN	STRENGTH	NOTES
`0*(?:\d{3}[\s\-]?\d{3}[\s\-]?\d{4})`	80%	MOD11_NHS check digit; after removing `^0+(?=(?:[\s\-]?\d){10}\s*$)|[\s\-]`

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

The nine digits are masked and the check digit recomputed, so the result is a valid number. The first digit stays — it tells the issuing system — and a CHI number keeps a real date (day 01–28, month 01–12). The rule never issues a number whose check would be ten; when the masked digits give one, the ninth digit moves up by one and the check is computed again. One masked number in about eleven is moved that way, so a few different numbers may mask alike.

`UK_NHS_COLUMN` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `UK_CM_ALNUM`.

> `UK_NHS_NUMBER` STRING ALGORITHM CHAIN
>
> Applies `UK_NHS_CD` → `UK_NHS_WIDEN` → `UK_NHS_AUX` → `UK_NHS_RESOLVE` → `UK_NHS_REPAIR` → `UK_NHS_RECOMPUTE`, in this order.
>
> > `UK_NHS_CD` CHECK DIGIT
> >
> > Masks the digits with `UK_NHS_BODY` and recomputes a modulus 11 check digit with weights `2 3 4 5 6 7 8 9 10` from the right.
> >
> > > `UK_NHS_BODY` REGEX DECOMPOSE
> > >
> > > Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_CM_DIGITS`.
> > >
> > > > `UK_CHI_DAY` SECURE LOOKUP
> > > >
> > > > Replaces the value with one of the 28 values in `uk-chi-day.txt`, chosen by a keyed hash.
> > > >
> > > > `UK_CHI_MONTH` SECURE LOOKUP
> > > >
> > > > Replaces the value with one of the 12 values in `uk-chi-month.txt`, chosen by a keyed hash.
> > > >
> > > > `UK_CM_DIGITS` CHARACTER MAPPING
> > > >
> > > > Maps each character to another of its group (1 groups), keeping length and format.
> > >
> > > `UK_NHS_WIDEN` REGEX DECOMPOSE
> > >
> > > Tries 10 patterns in order and gives each part of the value its own action; when none matches: keep.
> > >
> > > `UK_NHS_AUX` CHECK DIGIT
> > >
> > > Masks the digits with `UK_IDENTITY` and recomputes a modulus 11 check digit with weights `0 4 6 8 10 1 3 5 7 9` from the right.
> > >
> > > > `UK_IDENTITY` REGEX DECOMPOSE
> > > >
> > > > Matches one pattern and gives each part of the value its own action; when none matches: keep.

`UK_NHS_RESOLVE` REGEX DECOMPOSE

Tries 11 patterns in order and gives each part of the value its own action; when none matches: keep.

`UK_NHS_REPAIR` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

> `UK_NHS_STEP` REGEX DECOMPOSE
>
> Tries 10 patterns in order and gives each part of the value its own action; when none matches: keep.

`UK_NHS_RECOMPUTE` CHECK DIGIT

Masks the digits with `UK_IDENTITY` and recomputes a modulus 11 check digit with weights `2 3 4 5 6 7 8 9 10` from the right.

`UK_CM_ALNUM` CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
943 476 5919	→	**917 340 1153**
1503851238	→	**1502946521**

7.3 Unique Taxpayer Reference (UTR)

UK_L1_UTR

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

The ten-digit reference of HMRC for Self Assessment, Corporation Tax and partnerships, often written as two groups of five and sometimes followed by a `K`. Its check digit is the first digit: the other nine weighted 6 7 8 9 10 5 4 3 2, modulus 11.

HOW IT IS FOUND

COLUMN NAME UK_L1_UTR — Path

90% `utr` `sa_?utr` `ct_?utr` `unique_?tax(payer)?_?ref(erence)?(_?(no|num|number))?` `self_?assessment_?(ref|reference|utr|no|number)` `tax_?ref(erence)?(_?(no|num|number))?` `taxpayer_?ref(erence)?` `corporation_?tax_?(ref|reference|utr)`

VALUE FORMAT UK_L1_UTR — Regex

PATTERN	STRENGTH	NOTES
`0*(?:\d{5}\s?\d{5}\s?K?)`	40%	

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

The nine digits are masked and the first one recomputed by HMRC's rule, so the masked reference passes validation; the spacing and the `K` stay.

`UK_UTR` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `UK_CM_ALNUM`.

`UK_UTR_DIGITS` STRING ALGORITHM CHAIN

Applies `UK_UTR_CD` → `UK_UTR_WIDEN` → `UK_UTR_AUX` → `UK_UTR_RESOLVE`, in this order.

> `UK_UTR_CD` CHECK DIGIT
>
> Masks the digits with `UK_CM_DIGITS` and recomputes a modulus 11 check digit with weights `2 3 4 5 10 9 8 7 6` from the right.
>
> > `UK_CM_DIGITS` CHARACTER MAPPING
> >
> > Maps each character to another of its group (1 groups), keeping length and format.

`UK_UTR_WIDEN` REGEX DECOMPOSE

Tries 10 patterns in order and gives each part of the value its own action; when none matches: keep.

`UK_UTR_AUX` CHECK DIGIT

Masks the digits with `UK_IDENTITY` and recomputes a modulus 11 check digit with weights `4 6 8 10 9 7 5 3 1 0` from the right.

`UK_IDENTITY` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

`UK_UTR_RESOLVE` REGEX DECOMPOSE

Tries 11 patterns in order and gives each part of the value its own action; when none matches: keep.

`UK_CM_ALNUM` CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
12345 67890	→	**43218 87983**
1234567890K	→	**4321887983K**

7.4 Passport

UK_L1_PASSPORT

ESSENTIAL · COLUMN NAME · VALUE FORMAT · CHARACTER MAPPING

WHAT IT IS

Passport numbers, British (nine digits, issued by HM Passport Office) or foreign.

HOW IT IS FOUND

COLUMN NAME UK_L1_PASSPORT — Path

85% `passport[a-z_]*` `pass_?port_?(no|num|number)` `pp_?(no|num|number)` `travel_?document(_?(no|num|number))?`

VALUE FORMAT UK_L1_PASSPORT — Regex

PATTERN	STRENGTH	NOTES
`0*(?:\d{9}|[A-Z]{1,2}\d{6,8})`	30%	

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Character by character: letters stay letters and digits stay digits.

`UK_CM_ALNUM` CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
123456789	→	**110858568**

7.5 Driving licence

UK_L1_DRIVING_LICENCE

ESSENTIAL · COLUMN NAME · VALUE FORMAT · CHARACTER MAPPING

WHAT IT IS

The DVLA driving licence number, sixteen characters that carry the holder in plain sight: five letters of the surname, six digits for the date of birth (the month plus five for a woman), two initials, a digit and two check letters — `MORGA753116SM9IJ`. Northern Ireland licences are eight digits. Found by its name and by its shape.

HOW IT IS FOUND

COLUMN NAME UK_L1_DRIVING_LICENCE — Path

85% `drivers?_?licen[cs]e(_?(no|num|number))?` `driving_?licen[cs]e(_?(no|num|number))?` `dl_?(no|num|number)` `dvla_?(no|number|licen[cs]e)` `licen[cs]e_?(no|num|number)` *(except when followed by list E1)*

VALUE FORMAT UK_L1_DRIVING_LICENCE — Regex

PATTERN	STRENGTH	NOTES
`[A-Z9]{5}\d[0156]\d([0][1-9]|[12]\d|3[01])\d[A-Z9]{2}\d[A-Z]{2}`	95%	

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Character by character, keeping the format: the surname letters, the date and the initials all change.

`UK_CM_ALNUM` CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
MORGA753116SM9IJ	→	**SEQCA384830KM4IK**

7.6 Immigration documents

UK_L1_IMMIGRATION_DOCUMENT

COLUMN NAME · CHARACTER MAPPING

WHAT IT IS

Biometric residence permits, eVisa and right-to-work share codes, Home Office and UKVI references, visas, EU Settlement Scheme decisions, certificates of sponsorship, asylum references and naturalisation certificates. Immigration status is not special category, but it exposes a person to serious harm when it leaks.

HOW IT IS FOUND

COLUMN NAME UK_L1_IMMIGRATION_DOCUMENT — Path

85% `brp(_?(no|num|number))?` `biometric_?residen(ce|cy)_?permit(_?(no|number))?` `residen(ce|cy)_?permit(_?(no|number))?` `share_?code` `right_?to_?(work|rent)_?(code|share_?code|check)` `evisa(_?(no|number|ref))?` `home_?office_?(ref|reference|no|number)` `ho_?ref` `visa_?(no|num|number|ref)` `euss(_?(ref|reference|no|number))?` `settled_?status_?(ref|reference)` `immigration_?(status|permission|document|ref|reference|no|number)` `ukvi_?(ref|reference|no|number)` `certificate_?of_?sponsorship` `cos_?(no|number|ref)` `asylum_?(ref|reference|no|number)` `arc_?(no|number|card)` `naturali[sz]ation_?(certificate|cert|no|number)` `registration_?certificate_?(no|number)`

HOW IT IS MASKED

Character by character, keeping the format.

`UK_CM_ALNUM` CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
ZR1234567	→	**DC8262159**

7.7 Civil registration

UK_L1_VITAL_RECORD

COLUMN NAME | CHARACTER MAPPING

WHAT IT IS

Birth, death, marriage and civil partnership certificate and entry numbers, and the General Register Office index references of England and Wales, Scotland and Northern Ireland.

HOW IT IS FOUND

> COLUMN NAME `UK_L1_VITAL_RECORD – Path`
>
> 85% `birth_?cert(ificate)?(_?(no|num|number|entry|ref))?` `death_?cert(ificate)?(_?(no|num|number|ref))?` `marriage_?cert(ificate)?(_?(no|num|number|ref))?` `civil_?partnership_?cert(ificate)?(_?(no|number))?` `gro_?(index|ref|reference|no|number)` `registry_?entry(_?(no|number))?` `vital_?(record|statistics)_?(no|num|number)` `registration_?district_?(no|number|entry)`
>
> 55% `entry_?(no|number)`

HOW IT IS MASKED

Character by character, keeping the format.

`UK_CM_ALNUM` CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
A123456	→	**E854099**

7.8 Given names

UK_L1_GIVEN_NAME

ESSENTIAL | COLUMN NAME | KNOWN VALUES | REGEX DECOMPOSE

WHAT IT IS

First and middle names (`FIRST_NAME` , `FORENAME` , `MIDDLE_NAME` , `KNOWN_AS`), including two given names in one column such as `Mary Elizabeth` .

HOW IT IS FOUND

COLUMN NAME UK_L1_GIVEN_NAME — Path

85% `first_?names?` `given_?names?` `fore_?names?` `christian_?name` `fname` `name_?first` `middle_?names?` `mname` `name_?middle` `preferred_?(first_?)?name` `known_?as`

50% `name` *(except when followed by list E2)*

KNOWN VALUES UK_L1_GIVEN_NAME — List

LIST	VALUES	STRENGTH	FIRST VALUES
`uk-detect-given-names.txt`	356	70%	Aarav, Abena, Abigail, Adaeze, Adam, Agnieszka

Values are split into words, and each word is looked up.

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Each word is replaced by a name from a list of 350 given names used across the United Kingdom — English, Scottish, Welsh, Irish, South Asian, African and Eastern European — keeping the number of words and the case.

`UK_GIVEN_NAMES` **REGEX DECOMPOSE**

Tries 3 patterns in order and gives each part of the value its own action; when none matches: `UK_CM_ALNUM` .

`UK_GIVEN_NAME` **NAME**

Replaces the name with one of the 349 names in `uk-given-names.txt` .

`UK_CM_ALNUM` **CHARACTER MAPPING**

Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
`Oliver`	→	**`Zain`**
`Mary Elizabeth`	→	**`Niall Ian`**

7.9 Surnames

UK_L1_SURNAME

ESSENTIAL **COLUMN NAME** **KNOWN VALUES** **REGEX DECOMPOSE**

WHAT IT IS

Surnames, maiden and previous names, including double-barrelled surnames joined by a hyphen (`Bowes-Lyon`).

HOW IT IS FOUND

COLUMN NAME UK_L1_SURNAME — Path

85% `last_?names?` `surnames?` `family_?names?` `lname` `name_?last` `maiden_?name` `mothers?_?maiden_?name` `maiden` `birth_?surname` `previous_?surname` `married_?name` `former_?(sur)?name`

KNOWN VALUES UK_L1_SURNAME — List

LIST	VALUES	STRENGTH	FIRST VALUES
`uk-detect-surnames.txt`	295	70%	Abdi, Adams, Adeyemi, Ahmed, Akhtar, Ali

Values are split into words, and each word is looked up.

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Each part becomes a surname from a list of 300 common surnames of the United Kingdom; the hyphen stays.

`UK_SURNAMES` REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `UK_CM_ALNUM` .

> `UK_SURNAME` NAME
>
> Replaces the name with one of the 294 names in `uk-surnames.txt` .
>
> `UK_CM_ALNUM` CHARACTER MAPPING
>
> Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
Smith	→	**Collins**
Bowes-Lyon	→	**Russell-Palmer**

7.10 Full name

UK_L1_FULL_NAME

ESSENTIAL COLUMN NAME KNOWN VALUES REGEX DECOMPOSE

WHAT IT IS

Given names and surname in one column, in the English order, with or without a title — `Mrs Mary E. Smith` — or written as `SMITH, MARY` .

HOW IT IS FOUND

> COLUMN NAME UK_L1_FULL_NAME – Path
>
> **90%** `full_?name` `fullname` `complete_?name` `person_?name` `name_?of_?(customer|cust|client|patient|employee|emp|staff|worker|member|student|pupil|insured|beneficiary|holder|account_?holder|policy_?holder|mother|father|parent|guardian|spouse|partner|next_?of_?kin|doctor|gp|physician|nurse|carer|contact|emergency_?contact|guarantor|debtor|borrower|applicant|signatory|witness|representative|owner|tenant|landlord|victim|complainant|accused|defendant|donor|heir|person|driver|agent|citizen|elector|voter|taxpayer|child|dependant|dependent|relative|officer|teacher|contributor|pensioner|claimant|resident|occupier)(customer|cust|client|patient|employee|emp|staff|worker|member|student|pupil|insured|beneficiary|holder|account_?holder|policy_?holder|mother|father|parent|guardian|spouse|partner|next_?of_?kin|doctor|gp|physician|nurse|carer|contact|emergency_?contact|guarantor|debtor|borrower|applicant|signatory|witness|representative|owner|tenant|landlord|victim|complainant|accused|defendant|donor|heir|person|driver|agent|citizen|elector|voter|taxpayer|child|dependant|dependent|relative|officer|teacher|contributor|pensioner|claimant|resident|occupier)_?name`
>
> **60%** `beneficiary` `guardian` `next_?of_?kin` `spouse` `emergency_?contact` `guarantor` `account_?holder` `policy_?holder` `legal_?representative` `attorney` `lasting_?power_?of_?attorney` `patient` `victim` `complainant` `accused` `defendant` `insured_?person` `nominee` `executor` `appointee`
>
> **50%** `name` *(except when followed by list E2)*

KNOWN VALUES UK_L1_FULL_NAME — List

LIST	VALUES	STRENGTH	FIRST VALUES
uk-detect-given-names.txt	356	65%	Aarav, Abena, Abigail, Adaeze, Adam, Agnieszka
uk-detect-surnames.txt	295	65%	Abdi, Adams, Adeyemi, Ahmed, Akhtar, Ali

Values are split into words, and each word is looked up.

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Given names become given names and the surname becomes a surname, by position: the last word is the surname. A title (Mr , Mrs , Ms , Miss , Mx , Dr , Prof , Sir , Dame , Rev) and a middle initial stay; the comma form keeps its order.

UK_FULL_NAME REGEX DECOMPOSE

Tries 9 patterns in order and gives each part of the value its own action; when none matches: UK_CM_ALNUM .

UK_SURNAMES REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: UK_CM_ALNUM .

UK_SURNAME NAME

Replaces the name with one of the 294 names in uk-surnames.txt .

UK_CM_ALNUM CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

UK_GIVEN_NAMES REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: UK_CM_ALNUM .

UK_GIVEN_NAME NAME

Replaces the name with one of the 349 names in uk-given-names.txt .

EXAMPLES

VALUE		MASKED
Mrs Mary E. Smith	→	**Mrs Niall A. Collins**
Rhys Evans	→	**Mia Hughes**
SMITH, OLIVER	→	**COLLINS, ZAIN**

7.11 E-mail

UK_L1_EMAIL

ESSENTIAL COLUMN NAME DATA TYPE VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

Personal and work e-mail addresses.

HOW IT IS FOUND

COLUMN NAME UK_L1_EMAIL — Path

85% e_?mail[a-z0-9_]* mail[a-z0-9_]* email_?address

DATA TYPE UK_L1_EMAIL — Type

Accepts text (length 5 or more); any other type rules the data class out.

VALUE FORMAT UK_L1_EMAIL — Regex

PATTERN	STRENGTH	NOTES
[\p{L}\p{N}._%+'\-]+@[\p{L}\p{N}.\-]+\.[A-Za-z]{2,24}	100%	

A value that matches none counts 20% against the data class.

HOW IT IS MASKED

The name and domain are mapped character by character and the top-level domain becomes `.test`, reserved for tests: a masked address can never deliver mail.

`UK_EMAIL` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `UK_CM_ALNUM`.

> `UK_CM_ALNUM` CHARACTER MAPPING
>
> Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
`mary.smith@example.co.uk`	→	**`doyl.nyijg@esozcyu.ge.test`**

7.12 Phone

UK_L1_PHONE

ESSENTIAL · COLUMN NAME · VALUE FORMAT · REGEX DECOMPOSE

WHAT IT IS

UK numbers with `+44` or the trunk `0`: mobiles (`07700 900123`), geographic numbers (`020 7946 0018`, `0161 496 0000`), non-geographic ones, and the mixed form `+44 (0)20…`.

HOW IT IS FOUND

> COLUMN NAME — UK_L1_PHONE — Path
>
> 85% `tel` `telephone[a-z0-9_]*` `tel_?(no|num|number|home|work|mobile)` `phone[a-z0-9_]*` `ph_?(no|num|number)` `mobile[a-z0-9_]*` `mob(_?(no|num|number))?` `cell` `cell_?(phone|no|number)` `cellphone` `contact_?(no|num|number)` `landline` `fax` `whatsapp`

> VALUE FORMAT — UK_L1_PHONE — Regex
>
PATTERN	STRENGTH	NOTES
`(?=[+(0])(\+?44\s?(\(0\)\s?)?|\(?0)[1-35789]\d{1,3}\)?[\s\-.]?\d{3,4}[\s\-.]?\d{3,4}`	100%	
`0*(?:\+?\d[\d\s()\-.]{6,15})`	30%	
>
> A value that matches none counts 30% against the data class.

HOW IT IS MASKED

The country code, the trunk `0`, the next two digits (the mobile range or the start of the area code) and the separators stay; the rest is masked digit by digit. A number not shaped like a UK one is masked character by character.

`UK_PHONE` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_CM_ALNUM`.

> `UK_CM_ALNUM` CHARACTER MAPPING
>
> Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
`+44 7700 900123`	→	**`+44 7700 838369`**
`020 7946 0018`	→	**`020 6016 5896`**
`(0161) 496 0000`	→	**`(0168) 842 8155`**

7.13 Address

UK_L1_ADDRESS

ESSENTIAL | COLUMN NAME | DATA TYPE | VALUE FORMAT | SECURE LOOKUP

WHAT IT IS

The address lines: house number or house name and street, flat and building, with or without the town and postcode. Found by its name and by values with a number and a street word, a flat, or a postcode after a comma.

HOW IT IS FOUND

COLUMN NAME UK_L1_ADDRESS – Path

80% `address[a-z0-9_]*` *(except when preceded by* `mac`, `ip`, `e_?mail`, `web`, `url`; *except when followed by list E3)*

`addr(ess)?_?(line|1|2|3)?[a-z0-9_]*` *(except when preceded by* `mac`, `ip`, `e_?mail`, `web`, `url`*)* `street(_?(address|name))?` `residential_?address` `home_?address` `mailing_?address` `postal_?address` `correspondence_?address` `residence` `place_?of_?residence` `domicile` `po_?box`

DATA TYPE UK_L1_ADDRESS – Type

Accepts text (length 6 or more); any other type rules the data class out.

VALUE FORMAT UK_L1_ADDRESS – Regex

PATTERN	STRENGTH	NOTES
`(?i)\d+[A-Z]?,?\s+.*\b(street|st\.?|road|rd\.?|avenue|ave\.?|lane|ln\.?|drive|dr\.?|close|crescent|cres\.?|way|gardens|gdns\.?|place|pl\.?|court|ct\.?|terrace|grove|square|sq\.?|hill|mews|row|walk|park|green|view|rise|parade|broadway|approach)\b.*`	80%	
`(?i)(flat|apartment|apt\.?|unit)\s*\d+[A-Z]?\b.*`	80%	
`(?i).+,.*\b[A-Z]{1,2}\d[A-Z\d]?\s*\d[A-Z]{2}\b.*`	80%	

A value that matches none counts 20% against the data class.

HOW IT IS MASKED

It is replaced by one of 3,000 fictitious UK addresses built from common street names (High Street, Station Road, Heol y Felin, Lisburn Road), flats in named buildings and house names. The postcode column has a data class of its own.

`UK_ADDRESS` SECURE LOOKUP

Replaces the value with one of the 3,000 values in `uk-addresses.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
`10 Downing Street, London`	→	**`16 Albert Road`**

7.14 Address detail

UK_L1_ADDRESS_DETAIL

ESSENTIAL | COLUMN NAME | CHARACTER MAPPING

WHAT IT IS

House number or name, flat, building, unit and floor in separate columns.

HOW IT IS FOUND

COLUMN NAME UK_L1_ADDRESS_DETAIL — Path

70% `house_?(no|num|number|name)` `building_?(no|number|name)` `flat_?(no|number)?` `apt(_?(no|number))?` `apartment(_?(no|number))?` `unit_?(no|num|number)` `floor_?(no|number)?` `suite(_?no)?` `sub_?building(_?name)?`

HOW IT IS MASKED

Character by character, keeping the format.

`UK_CM_ALNUM` CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
`Flat 4B`	→	**`Ypox 6W`**

7.15 Bank account

UK_L1_BANK_ACCOUNT

COLUMN NAME CHARACTER MAPPING

WHAT IT IS

Bank and building society account numbers (eight digits) and building society roll numbers.

HOW IT IS FOUND

COLUMN NAME UK_L1_BANK_ACCOUNT — Path

85% `account_?(no|num|number|nbr)` `acct_?(no|num|number|nbr)?` `acc_?(no|num|number)` `bank_?(account|acct)(_?(no|num|number))?` `current_?(account|acct)(_?(no|number))?` `savings_?(account|acct)(_?(no|number))?` `building_?society_?(account|roll)(_?(no|number))?` `roll_?(no|number)`

HOW IT IS MASKED

Character by character, keeping length and separators; zeros on the left stay.

`UK_CM_ALNUM` CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
`31926819`	→	**`50681346`**

7.16 Sort code

UK_L1_SORT_CODE

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

The six-digit sort code, written `20-00-00`, which names a bank and a branch. With an account number it identifies an account.

HOW IT IS FOUND

COLUMN NAME UK_L1_SORT_CODE — Path

90% `sort_?code` `sortcode` `bank_?sort(_?code)?` `branch_?sort_?code` `nsc`

VALUE FORMAT UK_L1_SORT_CODE — Regex

PATTERN	STRENGTH	NOTES
`0*(?:\d{2}-\d{2}-\d{2})`	60%	

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

The first two digits, which tell the bank, stay; the four that tell the branch are masked.

`UK_SORT_CODE` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `UK_CM_ALNUM` .

`UK_CM_DIGITS` CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

`UK_CM_ALNUM` CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
20-00-00	→	**20-70-32**

7.17 IBAN

UK_L1_IBAN

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

UK IBANs (`GB82 WEST 1234 5698 7654 32`): the country, two check digits, the four-letter bank code, the sort code and the account number. The classifier checks the ISO 13616 check digits.

HOW IT IS FOUND

COLUMN NAME UK_L1_IBAN — Path

90% `iban(_?(no|number|code))?` `international_?bank_?account(_?number)?`

VALUE FORMAT UK_L1_IBAN — Regex

PATTERN	STRENGTH	NOTES
`GB\d{2}\s?[A-Z]{4}(\s?\d{4}){3}\s?\d{2}`	100%	MOD97 check digit; after removing `\s`

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

The IBAN algorithm of Delphix masks the sort code and account number — fourteen digits — and recomputes the two check digits, so the masked IBAN validates. The bank code and the spacing stay.

`UK_IBAN` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_CM_ALNUM` .

`UK_IBAN_COMPACT` IBAN

Masks the last 14 characters of the IBAN and recomputes its two ISO 13616 check digits; the country code and the characters before stay.

`UK_IBAN_PRINTED` IBAN

Masks the last 17 characters of the IBAN and recomputes its two ISO 13616 check digits; the country code and the characters before stay.

`UK_CM_ALNUM` **CHARACTER MAPPING**

Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
`GB82 WEST 1234 5698 7654 32`	→	**`GB54 WEST 4268 1399 9538 24`**
`GB82WEST12345698765432`	→	**`GB31WEST22532507119890`**

7.18 Payment card

`UK_L1_PAYMENT_CARD`

COLUMN NAME **VALUE FORMAT** **REGEX DECOMPOSE**

WHAT IT IS

Credit, debit and prepaid card numbers.

HOW IT IS FOUND

COLUMN NAME `UK_L1_PAYMENT_CARD – Path`

85% `card_?(no|num|number)` `credit_?card(_?(no|num|number))?` `debit_?card(_?(no|num|number))?` `cc_?(no|num|number)` `pan` `card_?pan` `masked_?pan` `primary_?account_?number`

VALUE FORMAT `UK_L1_PAYMENT_CARD – Regex`

PATTERN	STRENGTH	NOTES
`0*(?:3[47]\d{13}|[1–9]\d{12,13}|[1–9]\d{15,18})`	90%	LUHN check digit; after removing `[\s\-]`

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Credit Card, which comes with Delphix: another number that passes the Luhn check, keeping the first six digits (the issuer, not the holder).

`UK_CARD` **REGEX DECOMPOSE**

Matches one pattern and gives each part of the value its own action; when none matches: keep.

`dlpx-core:Credit Card` **PAYMENT CARD**

Comes with Delphix and is used as it ships. Replaces the card number with another that passes the Luhn check, keeping the first six digits — the issuer, not the holder.

EXAMPLES

VALUE		MASKED
`4929 1234 5678 9012`	→	**`4929 1298 3367 1542`**

7.19 IP address

`UK_L1_IP_ADDRESS`

COLUMN NAME **VALUE FORMAT** **REGEX DECOMPOSE**

WHAT IT IS

IPv4 and IPv6 addresses in access logs, sessions and transactions — online identifiers under Article 4(1).

HOW IT IS FOUND

COLUMN NAME UK_L1_IP_ADDRESS — Path

85% `ip` `ip_?(address|addr|origin|source|destination|client|user|remote|public|private|login)` `ipv[46]` `remote_?addr(ess)?` `client_?ip` `host_?ip` `source_?ip`

VALUE FORMAT UK_L1_IP_ADDRESS — Regex

PATTERN	STRENGTH	NOTES
`0*(?:((25[0-5]|2[0-4]\d|1?\d?\d)\.){3}(25[0-5]|2[0-4]\d|1?\d?\d))`	100%	
`0*(?:[0-9A-Fa-f]{1,4}(:[0-9A-Fa-f]{0,4}){2,7})`	90%	

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Each IPv4 octet is mapped into 1–254, so the address stays valid; an IPv6 address is mapped digit by digit in hexadecimal.

`UK_IP` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `UK_CM_HEX`.

> `UK_OCTET` NUMERIC MAPPING
>
> Maps the number to another between 1 and 254.
>
> `UK_CM_HEX` CHARACTER MAPPING
>
> Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
81.2.69.160	→	82.5.41.128

7.20 Device identifier

UK_L1_DEVICE_ID

COLUMN NAME VALUE FORMAT CHARACTER MAPPING

WHAT IT IS

IMEI, IMSI, ICCID, MAC addresses, advertising identifiers and session cookies.

HOW IT IS FOUND

COLUMN NAME UK_L1_DEVICE_ID — Path

80% `imei` `imsi` `iccid` `mac` `mac_?address` `device_?(id|uuid|serial)` `advertising_?id` `idfa` `gaid` `cookie(_?id)?` `session_?id` `serial_?(no|number)_?(device|phone|handset)` `handset_?serial`

VALUE FORMAT UK_L1_DEVICE_ID — Regex

PATTERN	STRENGTH	NOTES
`0*(?:([0-9A-Fa-f]{2}[:\-]){5}[0-9A-Fa-f]{2})`	100%	
`0*(?:\d{15})`	80%	LUHN check digit

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Character by character in hexadecimal: a MAC stays a MAC, an IMEI stays fifteen digits.

`UK_CM_HEX` CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
3c:22:fb:9a:10:e4	→	5e:83:bb:3c:31:f9

7.21 Vehicle registration

UK_L1_LICENCE_PLATE

COLUMN NAME · VALUE FORMAT · REGEX DECOMPOSE

WHAT IT IS

Number plates (vehicle registration marks): the current format since 2001 — two letters for the region, two digits for the age and three random letters, `AB12 CDE` — the older prefix and suffix formats (`A123 BCD` , `ABC 123D`) and Northern Ireland plates (`BAZ 1234`).

HOW IT IS FOUND

COLUMN NAME UK_L1_LICENCE_PLATE – Path

85% `licen[cs]e_?plates?(_?(no|num|number))?` `plates?(_?(no|num|number))?` `vrm` `vehicle_?(reg|registration|registration_?mark|plate)(_?(no|num|number))?` `registration_?(plate|mark)` `number_?plate` `reg_?(no|number|plate|mark)` `car_?reg(istration)?`

VALUE FORMAT UK_L1_LICENCE_PLATE – Regex

PATTERN	STRENGTH	NOTES
`[A-Z]{2}\d{2}\s?[A-Z]{3}`	90%	
`[A-Z]\d{1,3}\s?[A-Z]{3}|[A-Z]{3}\s?\d{1,3}[A-Z]`	60%	
`[A-Z]{2}[IZ]\s?\d{1,4}`	60%	

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

In the current format the region and the age stay — they say where and when a car was first registered, as a year of birth would — and the three random letters are masked; older formats are masked character by character. Masked letters never include I or Q, which plates do not use.

`UK_PLATE` REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: `UK_CM_PLATE` .

> `UK_CM_PLATE` CHARACTER MAPPING
>
> Maps each character to another of its group (3 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
AB12 CDE	→	AB12 BRL
A123 BCD	→	A957 GGN

7.22 VIN and V5C reference

UK_L1_VIN

COLUMN NAME · VALUE FORMAT · CHARACTER MAPPING

WHAT IT IS

Vehicle identification, chassis and engine numbers, and the reference of the V5C registration certificate (the logbook).

HOW IT IS FOUND

COLUMN NAME UK_L1_VIN – Path

80% `vin` `vin_?(no|number)` `vehicle_?identification_?(no|number)` `chassis(_?(no|number))?` `engine_?(no|number|serial)` `v5c(_?(ref|reference|no|number))?` `logbook_?(ref|reference|no|number)`

VALUE FORMAT UK_L1_VIN – Regex

PATTERN	STRENGTH	NOTES
`[A-HJ-NPR-Z0-9]{17}`		70%

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Character by character, keeping the 17-character format.

UK_CM_ALNUM CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
SAJAA01P9XH123456	→	**WEVIE42B4TY592357**

7.23 Property and meter references

UK_L1_PROPERTY

COLUMN NAME CHARACTER MAPPING

WHAT IT IS

References that lead to one household: the Unique Property Reference Number (UPRN), the HM Land Registry title number, the council tax and business rates account, and the electricity and gas meter numbers (MPAN, MPRN).

HOW IT IS FOUND

COLUMN NAME UK_L1_PROPERTY – Path

85% `uprn` `unique_?property_?ref(erence)?(_?(no|num|number))?` `title_?(no|number)` `land_?registry_?(title|ref|reference|no|number)` `registers_?of_?scotland_?(ref|title)` `council_?tax_?(account|ref|reference|no|number)(_?(no|number))?` `ctax_?(ref|account)` `business_?rates_?(account|ref)` `mpan` `mprn` `meter_?point_?(ref|reference|admin(istration)?_?number)` `meter_?(serial|no|number)` `property_?(ref|reference|id)`

HOW IT IS MASKED

Character by character, keeping the format, so references between tables still match.

UK_CM_ALNUM CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
100023336956	→	**532624755798**
NGL123456	→	**GKS222692**

7.24 Customer, contract or account number

UK_L1_CONTRACT_NUMBER

COLUMN NAME | CHARACTER MAPPING

WHAT IT IS

Customer, member, employee, payroll, policy, loan, mortgage, claim and tenancy numbers, and the pupil and learner numbers of the education system (UPN, ULN).

HOW IT IS FOUND

COLUMN NAME — UK_L1_CONTRACT_NUMBER — Path

70% `contract_?(no|num|number)` `policy_?(no|num|number)` `loan_?(no|num|number|id)` `mortgage_?(no|number|account)` `member(ship)?_?(no|num|number|id)` `customer_?(no|num|number|id|ref)` `client_?(no|num|number|id|ref)` `cust_?(no|num|id)` `employee_?(no|num|number|id)` `emp_?(no|num|id)` `staff_?(no|num|number|id)` `payroll_?(no|number)` `works_?(no|number)` `claim_?(no|num|number|ref)` `student_?(no|num|number|id)` `upn` `unique_?pupil_?number` `uln` `unique_?learner_?number` `pupil_?(no|number|id)` `library_?card_?(no|number)` `tenancy_?(ref|reference|no|number)` `housing_?(ref|reference)`

HOW IT IS MASKED

Character by character, keeping the format, so references between tables still match.

`UK_CM_ALNUM` CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
POL-123456	→	**MIF-194239**

7.25 Username and social media

UK_L1_USERNAME

COLUMN NAME | CHARACTER MAPPING

WHAT IT IS

Logins, nicknames and handles on social networks.

HOW IT IS FOUND

COLUMN NAME — UK_L1_USERNAME — Path

70% `user_?name` `username` `login(_?(name|id))?` `user_?id_?login` `nick_?name` `nickname` `alias` `screen_?name` `handle` `facebook` `instagram` `twitter` `tiktok` `linkedin` `social_?media(_?(handle|account|profile))?` `telegram`

HOW IT IS MASKED

Character by character.

`UK_CM_ALNUM` CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
@mary_s	→	**@hipg_c**

7.26 Credential

UK_L1_CREDENTIAL

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

WHAT IT IS

Passwords, PINs, memorable words, password hashes, security answers, tokens and API keys. A hash is also found by its scheme — bcrypt, crypt, Argon2, PBKDF2, LDAP.

HOW IT IS FOUND

COLUMN NAME UK_L1_CREDENTIAL — Path

90%

`password` `passwd` `pwd` `pass_?word` `pass` `pin(_?(code|no|number))?` `passcode` `hash_?password` `password_?hash` `secret` `secret_?(answer|question)` `security_?(answer|question)` `memorable_?(word|information|date)` `access_?token` `refresh_?token` `api_?key` `otp`

VALUE FORMAT UK_L1_CREDENTIAL — Regex

PATTERN	STRENGTH	NOTES
`\$2[abxy]?\$\d{2}\$[./A-Za-z0-9]{53}`	95%	
`\$(1|5|6)\$(rounds=\d+\$)?[./A-Za-z0-9]{1,16}\$[./A-Za-z0-9]{22,86}`	95%	
`\$argon2(id|i|d)\$v=\d+\$m=\d+,t=\d+,p=\d+\$\S+`	95%	
`pbkdf2_sha(1|256|512)\$\d+\$\S+\$\S+`	95%	
`\{(SSHA|SHA|SSHA256|SSHA512|SMD5|MD5|CRYPT)\}\S+`	95%	

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Every character becomes `X`. A credential has no reason to look real in a test environment.

`UK_REDACT` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

EXAMPLES

VALUE		MASKED
`Secret#2024`	→	**XXXXXXXXXX**

7.27 Geolocation

UK_L1_GEOLOCATION

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

WHAT IT IS

Latitude and longitude, together or in separate columns, and Ordnance Survey eastings, northings and grid references. Great Britain and Northern Ireland lie between 49° and 61° north and 8° west and 2° east.

HOW IT IS FOUND

COLUMN NAME UK_L1_GEOLOCATION — Path

85%

`lat` `latitude` `lon` `lng` `longitude` `coordinates?` `coord_?[xy]` `geo_?(lat|lon|lng|loc|location|position)` `geolocation` `gps(_?(location|position|coordinates))?` `easting` `northing` `grid_?ref(erence)?`

50%

`long`

VALUE FORMAT UK_L1_GEOLOCATION – Regex

PATTERN	STRENGTH	NOTES
`0*(?:(49|5\d|60)\.\d{3,}\s*,\s*-?[0-8]\.\d{3,})`	90%	
`0*(?:(49|5\d|60)\.\d{4,})`	40%	

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

The integer part and first decimal stay — about 11 km — and the rest is masked: the place is generalized, not moved.

`UK_COORDINATE` **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

> `UK_CM_ALNUM` **CHARACTER MAPPING**
>
> Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
`51.501009,-0.141588`	→	**`51.506862,-0.153030`**

7.28 Free text

UK_L1_FREE_TEXT

COLUMN NAME **DATA TYPE** **VALUE FORMAT** **STRING ALGORITHM CHAIN**

WHAT IT IS

Remarks, comments, complaints, case notes and statements where personal data is written without structure.

HOW IT IS FOUND

COLUMN NAME UK_L1_FREE_TEXT – Path

60% `remarks?` `comments?` `notes?` `observations?` `narrative` `free_?text` `memo` `messages?` *(except when followed by list E4)* `complaint(_?(text|details|description))?` `description_?(of_?)?` `(case|complaint|incident|request|issue|problem|events)` `incident_?` `(details|description|narrative)` `case_?(notes?|summary|details)` `statement` `details` `summary`

DATA TYPE UK_L1_FREE_TEXT – Type

Accepts text (length 30 or more); any other type rules the data class out.

VALUE FORMAT UK_L1_FREE_TEXT – Regex

PATTERN	STRENGTH	NOTES
`[A-CEGHJ-PR-TW-Z][A-CEGHJ-NPR-TW-Z]\s?\d{2}\s?\d{2}\s?\d{2}\s?[A-D]`	70%	anywhere in the value
`[\w.+\-]+@[\w\-]+\.[A-Za-z]{2,}`	70%	anywhere in the value
`0*(?:(\+44\s?|0)7\d{3}\s?\d{6})`	60%	anywhere in the value

A value that matches none does not count against the data class.

HOW IT IS MASKED

UK numbers are written with spaces, and Free Text Redaction looks at one word at a time, so a Regex Decompose first turns the spaced forms — National Insurance and NHS numbers, UTRs, phone numbers, IBANs, postcodes and plates, up to four in a value — into `[DATA]` . Free Text Redaction then does the rest

word by word: the same identifiers written without spaces, e-mails, card numbers and IPs become `[DATA]`, and the given names and surnames of the lists become `[NAME]`.

`UK_FREE_TEXT_ALL` **STRING ALGORITHM CHAIN**

Applies `UK_TEXT_SPACED` → `UK_FREE_TEXT`, in this order.

> `UK_TEXT_SPACED` **REGEX DECOMPOSE**
>
> Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> `UK_FREE_TEXT` **FREE TEXT REDACTION**
>
> Replaces the text matching 10 patterns with `[DATA]`, and the 1,266 terms of `uk-text-names.txt` with `[NAME]`.

EXAMPLES

VALUE		MASKED
`Customer Smith, NI number AB 12 34 56 C, tel 07700 900123, SW1A 2AA.`	→	**`Customer [NAME], NI number [DATA], tel [DATA], [DATA].`**

PART 8

Quasi-identifiers

Data that identifies no one on its own, but does in combination. A full postcode covers about fifteen addresses; with a birth date and a sex it usually points at one person. Of the 4,643 towns and villages of more than 1,000 people in the United Kingdom, 3,993 have fewer than 20,000. These attributes are generalized or shifted rather than replaced: the postcode keeps its outward code, small places become the main town of their local authority, and the data keeps its analytical value.

8.1 Date of birth

UK_L2_BIRTH_DATE

ESSENTIAL · COLUMN NAME · DATA TYPE · DATE SHIFT

WHAT IT IS

Date of birth as a date, a timestamp or text.

HOW IT IS FOUND

COLUMN NAME UK_L2_BIRTH_DATE – Path

90% `dob` `d_?o_?b` `date_?of_?birth` `birth_?date` `birthdate` `birthday` `born_?(on|date)` `bdate`

DATA TYPE UK_L2_BIRTH_DATE – Type

Accepts date, TIMESTAMP or text (length 6 or more); any other type rules the data class out.

HOW IT IS MASKED

Shifted by up to 60 days, the same way for the same date: ages stay nearly right, and few people cross a birthday that matters (16, 18).

`UK_BIRTH_DATE` DATE SHIFT

Moves the date by -60 to 60 days.

EXAMPLES

VALUE		MASKED
1985-07-20	→	**1985-08-05T00:00**

8.2 Year of birth

UK_L2_BIRTH_YEAR

COLUMN NAME · DATA TYPE · REGEX DECOMPOSE

WHAT IT IS

The year of birth in a column of its own.

HOW IT IS FOUND

COLUMN NAME UK_L2_BIRTH_YEAR – Path

90% `birth_?year` `year_?of_?birth` `yob` `yr_?of_?birth`

DATA TYPE UK_L2_BIRTH_YEAR – Type

Accepts number (length up to 4) or text (length up to 4); any other type rules the data class out.

HOW IT IS MASKED

The decade stays and the last digit is masked: 1985 becomes 198x.

`UK_YEAR` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

`UK_CM_ALNUM` **CHARACTER MAPPING**

Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
1985	→	**1980**

8.3 Age

UK_L2_AGE

COLUMN NAME **DATA TYPE** **REGEX DECOMPOSE**

WHAT IT IS

Age in years, at admission, at death, at referral or in bands.

HOW IT IS FOUND

COLUMN NAME UK_L2_AGE – Path

85% `age(_?(years|yrs|in_?years|at_?(admission|death|entry|diagnosis|registration|referral)|group|range|band|bracket|last_?birthday))?` `years_?old`

DATA TYPE UK_L2_AGE – Type

Accepts number (length up to 6) or text (length up to 6); any other type rules the data class out.

HOW IT IS MASKED

The decade stays (37 becomes 3x); a single digit is masked; from 100 on, every age becomes 90 — very old people are few, and an exact age identifies them.

`UK_AGE` **REGEX DECOMPOSE**

Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.

`UK_CM_ALNUM` **CHARACTER MAPPING**

Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
37	→	**35**
104	→	**90**

8.4 Dates of life events

UK_L2_EVENT_DATE

COLUMN NAME **DATA TYPE** **DATE SHIFT**

WHAT IT IS

Dates of death, marriage, civil partnership, divorce, hiring, leaving, admission, discharge, diagnosis, referral, appointment, arrest, caution and conviction — each one, with a birth date, narrows the person down.

HOW IT IS FOUND

COLUMN NAME UK_L2_EVENT_DATE — Path

80% `date_?of_?(death|marriage|civil_?partnership|divorce|dissolution|hire|employment|engagement|termination|dismissal|resignation|retirement|admission|discharge|diagnosis|referral|visit|appointment|consultation|delivery|arrest|charge|caution|conviction|sentence|sentencing|naturali[sz]ation|injury|surgery)` `(death|marriage|divorce|hire|hiring|employment|employment_?start|start_?of_?employment|termination|leaving|resignation|retirement|admission|discharge|diagnosis|referral|appointment|visit|delivery|arrest|caution|conviction|sentencing|naturali[sz]ation|pension_?start|injury|vaccination|surgery|arrival)_?date` `dod`

DATA TYPE UK_L2_EVENT_DATE — Type

Accepts date, TIMESTAMP or text (length 6 or more); any other type rules the data class out.

HOW IT IS MASKED

Shifted by up to 30 days.

`UK_EVENT_DATE` DATE SHIFT

Moves the date by -30 to 30 days.

EXAMPLES

VALUE		MASKED
2021-03-15	→	**2021-03-05T00:00**

8.5 Sex

UK_L2_SEX

COLUMN NAME KNOWN VALUES REGEX DECOMPOSE

WHAT IT IS

Sex or gender as a code or a word: `F` / `M`, `Female` / `Male`, `Woman` / `Man`.

HOW IT IS FOUND

COLUMN NAME UK_L2_SEX — Path

85% `sex(_?(code|at_?birth|registered|registered_?at_?birth))?` `gender(_?code)?` *(except when followed by* `identity`, `expression`, `recognition`*)*

KNOWN VALUES UK_L2_SEX — List

LIST	VALUES	STRENGTH	FIRST VALUES
`uk-detect-sex.txt`	5	80%	female, male, woman, man, fem
`uk-detect-sex-letter.txt`	2	60%	f, m

Values are split into words, and each word is looked up.

A value that matches none counts 50% against the data class.

HOW IT IS MASKED

Replaced within its own vocabulary by keyed hash: `F` / `M` stays `F` / `M`, a word stays a word, and the proportions stay about the same.

`UK_SEX` REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: keep.

`UK_SEX_FEMALE_MALE` SECURE LOOKUP

Replaces the value with one of the 2 values in `uk-sex-female-male.txt`, chosen by a keyed hash.

`UK_SEX_WOMAN_MAN` SECURE LOOKUP

Replaces the value with one of the 2 values in `uk-sex-woman-man.txt`, chosen by a keyed hash.

`UK_SEX_F_M` SECURE LOOKUP

Replaces the value with one of the 2 values in `uk-sex-f-m.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
F	→	**M**
Male	→	**Female**

8.6 Postcode

UK_L2_POSTCODE

ESSENTIAL COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

The full postcode (`SW1A 2AA`): an outward code — area and district — and an inward code of a digit and two letters, about fifteen addresses in all. Found by its name and by its shape, which allows only the letters the inward code uses.

HOW IT IS FOUND

COLUMN NAME UK_L2_POSTCODE — Path

90% `post_?code` `postcode` `postal_?code` `post_?cd` `zip(_?code)?` `pcode`

VALUE FORMAT UK_L2_POSTCODE — Regex

PATTERN	STRENGTH	NOTES
`[A-Z]{1,2}\d[A-Z\d]?\s*\d[ABD-HJLNP-UW-Z]{2}`	95%	

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

The outward code — a postcode district of thousands of addresses, the level at which the ONS and the NHS publish data — and the spacing stay; the inward code is masked within the postcode alphabet, which leaves out C, I, K, M, O and V. The result is shaped like a real postcode of the same district.

`UK_POSTCODE` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

`UK_CM_INWARD` CHARACTER MAPPING

Maps each character to another of its group (3 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
SW1A 2AA	→	**SW1A 7SB**
M1 1AE	→	**M1 9ZA**

8.7 Town

UK_L2_TOWN

COLUMN NAME | VALUE FORMAT | DATA CLEANSING

WHAT IT IS

Town, village, place of birth, locality and post town, in columns of their own. The values are recognized against the 4,643 places of the list.

HOW IT IS FOUND

COLUMN NAME — UK_L2_TOWN – Path

85% `town` `village` `hamlet` `community` *(except when followed by* `service`, `group`, `organi[sz]ation`, `work`, `council`, `interest`*)* `locality` `settlement` `home_?town` `hometown` `place_?of_?birth` `birth_?place` `birthplace` `residence_?(town|village)` `town_?of_?residence` `post_?town` `postal_?town`

50% `city` `municipality`

VALUE FORMAT — UK_L2_TOWN – Values

PATTERN	STRENGTH	NOTES
One of 4,514 known values, ignoring case — Barnsley, Darton, Hoyland Nether, Wombwell, Bolton upon Dearne, Cudworth, …	80%	

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

A place of fewer than 20,000 people becomes the main town of its local authority — Windermere and Keswick become Carlisle, Portree becomes Inverness, a small part of London becomes London. Places of 20,000 or more, and the main towns, keep their names.

`UK_TOWN` DATA CLEANSING

Replaces each value found in `uk-town-generalized.txt` (9,078 pairs) by its counterpart; values not in the file stay as they are.

EXAMPLES

VALUE		MASKED
Windermere	→	**Carlisle**
Portree	→	**Inverness**
Manchester	→	**Manchester**

8.8 Nationality

UK_L2_NATIONALITY

COLUMN NAME | VALUE FORMAT | SECURE LOOKUP

WHAT IT IS

Nationality, citizenship and country of birth.

HOW IT IS FOUND

COLUMN NAME — UK_L2_NATIONALITY – Path

80% `nationality` `citizenship` `country_?of_?(birth|origin|citizenship|nationality)` `birth_?country`

VALUE FORMAT UK_L2_NATIONALITY — Values

PATTERN	STRENGTH	NOTES
One of 56 known values, ignoring case — British, Irish, Indian, Polish, Pakistani, Romanian, …	70%	

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

Replaced by keyed hash with one of the twenty most common nationalities of UK residents.

`UK_NATIONALITY` SECURE LOOKUP

Replaces the value with one of the 21 values in `uk-nationalities.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Latvian	→	**Portuguese**

8.9 Marital and civil partnership status

UK_L2_MARITAL_STATUS

COLUMN NAME VALUE FORMAT SECURE LOOKUP

WHAT IT IS

Legal partnership status as the 2021 census asks for it — with civil partnerships next to marriages — and cohabitation.

HOW IT IS FOUND

COLUMN NAME UK_L2_MARITAL_STATUS — Path

85% `marital_?status` `civil_?status` `legal_?partnership_?status` `partnership_?status` `marriage_?status` `relationship_?status` `living_?arrangements?`

VALUE FORMAT UK_L2_MARITAL_STATUS — Values

PATTERN	STRENGTH	NOTES
One of 21 known values, ignoring case — Never married or civil partnered, Married, In a civil partnership, Separated, Divorced, Formerly in a civil partnership, …	70%	

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

Replaced from the census categories by keyed hash.

`UK_MARITAL_STATUS` SECURE LOOKUP

Replaces the value with one of the 9 values in `uk-marital-status.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
In a civil partnership	→	**Never married or civil partnered**

8.10 Occupation

UK_L2_OCCUPATION

COLUMN NAME SECURE LOOKUP

WHAT IT IS

Occupation, job title and SOC code.

HOW IT IS FOUND

COLUMN NAME UK_L2_OCCUPATION — Path

80% `occupation` `profession` `job_?title` `designation` `post_?title` `position_?title` `soc_?(code|2020)`

50% `job` `position` `post` `trade` *(except when followed by `name`, `mark`, `licen[cs]e`)*

HOW IT IS MASKED

Replaced from a list of forty common occupations. A rare job title in a small organisation identifies on its own.

`UK_OCCUPATION` SECURE LOOKUP

Replaces the value with one of the 40 values in `uk-occupations.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
`Regional Operations Manager, North West`	→	**`HGV driver`**

8.11 Employer

UK_L2_EMPLOYER

COLUMN NAME SECURE LOOKUP

WHAT IT IS

Name of the employer or workplace.

HOW IT IS FOUND

COLUMN NAME UK_L2_EMPLOYER — Path

80% `employer(_?name)?` `name_?of_?employer` `employer_?business_?name` `company_?of_?employment` `workplace` `work_?place` `place_?of_?(work|employment)` `employed_?(at|by|with)` `current_?employer` `paye_?employer`

HOW IT IS MASKED

Replaced from a list of fictitious employers.

`UK_EMPLOYER` SECURE LOOKUP

Replaces the value with one of the 25 values in `uk-employers.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
`Acme Widgets plc`	→	**`North Sea Fisheries Ltd`**

8.12 Highest qualification

UK_L2_EDUCATION

COLUMN NAME VALUE FORMAT SECURE LOOKUP

WHAT IT IS

Highest qualification or level, in the terms of the Regulated Qualifications Framework and the Scottish framework: GCSEs and Standard Grades, A levels and Highers, NVQs, BTECs, apprenticeships, degrees.

HOW IT IS FOUND

COLUMN NAME UK_L2_EDUCATION — Path

80% `education(_?(level|attainment|status|completed))?` `educational_?(level|attainment)` `highest_?(level_?(of_?)?)?(education|qualification|grade)` `level_?of_?education` `qualifications?` `schooling` `rqf_?level` `scqf_?level` `nvq_?level`

VALUE FORMAT UK_L2_EDUCATION — Values

PATTERN	STRENGTH	NOTES
One of 40 known values, ignoring case — No qualifications, GCSE or equivalent, A level or equivalent, Scottish Highers, Apprenticeship, NVQ level 3, …	60%	

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

Replaced from a list of levels by keyed hash.

`UK_EDUCATION` SECURE LOOKUP

Replaces the value with one of the 11 values in `uk-education.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
`A level or equivalent`	→	**`Master's degree`**

8.13 School

UK_L2_SCHOOL

COLUMN NAME SECURE LOOKUP

WHAT IT IS

School, college or university attended.

HOW IT IS FOUND

COLUMN NAME UK_L2_SCHOOL — Path

75% `school(_?name)?` `name_?of_?school` `high_?school` `primary_?school` `secondary_?school` `academy(_?name)?` `sixth_?form` `college(_?name)?` `university(_?name)?` `institution_?attended` `alma_?mater` `urn_?school` `dfe_?(no|number)`

HOW IT IS MASKED

Replaced from a list of fictitious schools and colleges.

`UK_SCHOOL` SECURE LOOKUP

Replaces the value with one of the 15 values in `uk-schools.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
`Eton College`	→	**`Greenfield University`**

8.14 Number of children or dependants

UK_L2_DEPENDANTS

COLUMN NAME | DATA TYPE | REGEX DECOMPOSE

WHAT IT IS

Number of children, dependants or household members.

HOW IT IS FOUND

> **COLUMN NAME** UK_L2_DEPENDANTS — Path
>
> 80% `dependants?` `dependents?` `no_?of_?(children|dependants|dependents|kids)` `num(ber)?_?(of_?)?(children|dependants|dependents|kids)` `children` `household_?(size|members)` `persons_?in_?household`

> **DATA TYPE** UK_L2_DEPENDANTS — Type
>
> Accepts number (length up to 4) or text (length up to 4); any other type rules the data class out.

HOW IT IS MASKED

Kept up to 5; anything above becomes 5. Large households are rare, and a rare count identifies.

`UK_CAPPED_COUNT` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

EXAMPLES

VALUE		MASKED
9	→	5

PART 9

Special category data

The categories of Article 9 that are not health — which has a group of its own: racial or ethnic origin, political opinions, religious or philosophical beliefs, trade union membership, genetic data, biometric data used for identification, sex life and sexual orientation — and gender identity, which the Gender Recognition Act 2004 protects. Categorical columns keep their shape: a yes/no flag stays a flag, a numeric code gets other digits, and any other value is replaced from the categories of the 2021 census, or suppressed where there is no shared vocabulary.

9.1 Genetic data

UK_L3_GENETIC

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

WHAT IT IS

Personal data relating to the inherited or acquired genetic characteristics of a person, resulting from the analysis of a biological sample (Article 4(13)): DNA profiles, sequences, genotypes, BRCA and other genetic tests, newborn blood spot results, carrier status. A sequence is also found by its letters.

HOW IT IS FOUND

COLUMN NAME — UK_L3_GENETIC — Path

90%

`dna` `genetic[a-z_]*` `genome` `genomic[a-z_]*` `genotyp[a-z_]*` `gene_?(test|result|marker|panel)` `genetic_?(test|marker|profile|result)` `mutation` `brca[12]?` `karyotype` `paternity_?test` `dna_?(profile|sample|index)` `snp` `haplotype` `newborn_?(blood_?spot|screening)` `heel_?prick` `sickle_?cell_?(genotype|trait)` `carrier_?status`

VALUE FORMAT — UK_L3_GENETIC — Regex

PATTERN	STRENGTH	NOTES
`[ACGTUN]{24,}`	90%	

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Every character becomes `X`.

`UK_REDACT` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

EXAMPLES

VALUE		MASKED
`BRCA1 positive`	→	`XXXXXXXXXXXXXX`

9.2 Biometric data

UK_L3_BIOMETRIC

COLUMN NAME | DATA TYPE | REGEX DECOMPOSE

WHAT IT IS

Data resulting from processing the physical, physiological or behavioural characteristics of a person that allows or confirms their unique identification, such as facial images and fingerprints (Article 4(14)): fingerprints, face templates, photographs and custody images, iris and vein scans, voice prints, signatures, keystroke and gait patterns.

HOW IT IS FOUND

COLUMN NAME UK_L3_BIOMETRIC – Path

90% `fingerprints?` `finger_?prints?` `thumb_?prints?` `biometric[a-z_]*` `face_?(template|encoding|id|print|hash|image)` `facial_?(template|recognition|id|image|image_?hash)` `photo(graph)?(_?(image|file|path|blob|data|url))?` `passport_?photo` `selfie` `headshot` `mugshot` `custody_?image` `iris(_?(scan|template|code))?` `retina(_?scan)?` `vein_?(pattern|scan)` `palm_?(print|vein)` `voice_?(print|template|sample|recording)` `signature_?(image|biometric|capture|pad|specimen|scan)` `specimen_?signature` `handwritten_?signature` `keystroke(_?(dynamics|pattern))?` `typing_?pattern` `gait(_?(pattern|analysis))?` `minutiae`

DATA TYPE UK_L3_BIOMETRIC – Type

Accepts text; any other type rules the data class out.

HOW IT IS MASKED

Every character becomes `X` — a file path to a photograph included.

`UK_REDACT` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

EXAMPLES

VALUE		MASKED
`/photos/12345.jpg`	→	`XXXXXXXXXXXXXXXXX`

9.3 Racial or ethnic origin

UK_L3_ETHNICITY

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

Ethnic group, in the categories of the 2021 census of England and Wales (Scotland and Northern Ireland ask nearly the same question): White British, Irish, Gypsy or Irish Traveller, Roma, the Mixed groups, Indian, Pakistani, Bangladeshi, Chinese, African, Caribbean, Arab.

HOW IT IS FOUND

COLUMN NAME UK_L3_ETHNICITY – Path

90% `ethnic(ity|_?group|_?origin|_?background|_?identity|_?category|_?code)?` `race` `racial_?(origin|group|background)` `traveller(_?status)?`

VALUE FORMAT UK_L3_ETHNICITY – Values

PATTERN	STRENGTH	NOTES
One of 47 known values, ignoring case — White British, White Irish, Gypsy or Irish Traveller, Roma, Other White, White and Black Caribbean, …	90%	
One of 6 known values, ignoring case — y, n, yes, no, 0, 1	30%	

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

Replaced from the census categories by keyed hash; a yes/no flag stays a flag and a numeric code gets other digits.

`UK_ETHNICITY` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_ETHNICITY_LIST`.

> `UK_FLAG` REGEX DECOMPOSE
>
> Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `UK_FLAG_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-yes-no.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-true-false.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-y-n.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-0-1.txt`, chosen by a keyed hash.
>
> `UK_CM_ALNUM` CHARACTER MAPPING
>
> Maps each character to another of its group (5 groups), keeping length and format.
>
> `UK_ETHNICITY_LIST` SECURE LOOKUP
>
> Replaces the value with one of the 20 values in `uk-ethnic-groups.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Kashmiri	→	**Other White**
Y	→	**N**

9.4 Religious beliefs

UK_L3_RELIGION

COLUMN NAME · VALUE FORMAT · REGEX DECOMPOSE

WHAT IT IS

Religion and denomination: the answers of the 2021 census (No religion, Christian, Muslim, Hindu, Sikh, Jewish, Buddhist) and the main churches — Church of England, Church of Scotland, Roman Catholic, Church of Ireland, Presbyterian, Methodist.

HOW IT IS FOUND

> COLUMN NAME UK_L3_RELIGION – Path
>
> 95% `religion[a-z_]*` `religious_?(affiliation|denomination|belief|beliefs|preference|group)` `denomination` `faith` `church(_?(name|affiliation|member|membership|attended))?` `creed` `congregation`

> VALUE FORMAT UK_L3_RELIGION – Values
>
> | PATTERN | STRENGTH | NOTES |
> |---|---|---|
> | One of 45 known values, ignoring case — No religion, Christian, Church of England, Church of Scotland, Roman Catholic, Church of Ireland, … | 90% | |
>
> A value that matches none counts 40% against the data class.

HOW IT IS MASKED

Replaced from the list by keyed hash; flags and codes keep their shape.

`UK_RELIGION` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_RELIGION_LIST`.

> `UK_FLAG` REGEX DECOMPOSE
>
> Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.

UK_FLAG_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `uk-flag-yes-no.txt`, chosen by a keyed hash.

UK_FLAG_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in `uk-flag-true-false.txt`, chosen by a keyed hash.

UK_FLAG_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in `uk-flag-y-n.txt`, chosen by a keyed hash.

UK_FLAG_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in `uk-flag-0-1.txt`, chosen by a keyed hash.

UK_CM_ALNUM CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

UK_RELIGION_LIST SECURE LOOKUP

Replaces the value with one of the 18 values in `uk-religions.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Quaker	→	**Roman Catholic**

9.5 Philosophical beliefs

UK_L3_BELIEF

COLUMN NAME REGEX DECOMPOSE

WHAT IT IS

Philosophical beliefs and convictions, conscientious objection. Employment tribunals have recognized ethical veganism, among others, as a philosophical belief protected by the Equality Act 2010.

HOW IT IS FOUND

COLUMN NAME UK_L3_BELIEF – Path

90% `beliefs?` `philosophical_?(beliefs?|convictions?)` `conscientious_?objection` `conscientious_?objector` `moral_?(beliefs?|convictions?)` `personal_?convictions?` `worldview` `ethical_?veganism`

HOW IT IS MASKED

Suppressed to `Not stated`: the values have no shared vocabulary to replace from.

UK_SUPPRESSED_CATEGORY REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: UK_SUPPRESS.

UK_FLAG REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.

UK_FLAG_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `uk-flag-yes-no.txt`, chosen by a keyed hash.

UK_FLAG_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in `uk-flag-true-false.txt`, chosen by a keyed hash.

UK_FLAG_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in `uk-flag-y-n.txt`, chosen by a keyed hash.

UK_FLAG_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in `uk-flag-0-1.txt`, chosen by a keyed hash.

UK_CM_ALNUM CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

UK_SUPPRESS SECURE LOOKUP

Replaces the value with one of the 1 values in `uk-not-stated.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Ethical veganism	→	**Not stated**

9.6 Political opinions

UK_L3_POLITICAL_OPINION

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

Political leaning, voting intention, referendum vote.

HOW IT IS FOUND

COLUMN NAME UK_L3_POLITICAL_OPINION — Path

90% `political_?(opinion|opinions|view|views|orientation|leaning|leanings|preference|ideology)` `ideology` `voting_?intention` `vote_?intention` `intended_?vote` `swing_?voter` `brexit_?vote` `referendum_?vote` `leave_?or_?remain`

VALUE FORMAT UK_L3_POLITICAL_OPINION — Values

PATTERN	STRENGTH	NOTES
One of 22 known values, ignoring case — Left, Centre-left, Centre, Centre-right, Right, No preference, …	70%	

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

Replaced from a scale of leanings by keyed hash.

`UK_POLITICAL_OPINION` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_POLITICAL_OPINION_LIST`.

> `UK_FLAG` REGEX DECOMPOSE
>
> Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `UK_FLAG_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-yes-no.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-true-false.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-y-n.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-0-1.txt`, chosen by a keyed hash.
>
> `UK_CM_ALNUM` CHARACTER MAPPING
>
> Maps each character to another of its group (5 groups), keeping length and format.
>
> `UK_POLITICAL_OPINION_LIST` SECURE LOOKUP
>
> Replaces the value with one of the 7 values in `uk-political-opinions.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Centre-right	→	**Right**

9.7 Political party

UK_L3_POLITICAL_AFFILIATION

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

WHAT IT IS

The party a person belongs to or supports: Labour, Conservative, Liberal Democrats, Reform UK, the Greens, the SNP, Plaid Cymru, Sinn Féin, the DUP, Alliance, the SDLP, the UUP.

HOW IT IS FOUND

COLUMN NAME — UK_L3_POLITICAL_AFFILIATION – Path

95%	`political_?(party|body|membership|affiliation|group)` `party_?(membership|member|affiliation|card|group|supporter|preference)` `preferred_?party` `member_?of_?(party|political_?(party|body))` `party_?(of_?)?choice` `vote_?(2019|2024)` `general_?election_?vote`
50%	`party` *(except when followed by list E5)*

VALUE FORMAT — UK_L3_POLITICAL_AFFILIATION – Values

PATTERN	STRENGTH	NOTES
One of 35 known values, ignoring case — Labour, Conservative, Liberal Democrats, Reform UK, Green Party, Scottish National Party, …	80%	

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

Replaced from the list of parties by keyed hash; flags and codes keep their shape.

`UK_POLITICAL_PARTY` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_POLITICAL_PARTY_LIST`.

> `UK_FLAG` REGEX DECOMPOSE
>
> Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `UK_FLAG_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-yes-no.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-true-false.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-y-n.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-0-1.txt`, chosen by a keyed hash.
>
> `UK_CM_ALNUM` CHARACTER MAPPING
>
> Maps each character to another of its group (5 groups), keeping length and format.
>
> `UK_POLITICAL_PARTY_LIST` SECURE LOOKUP
>
> Replaces the value with one of the 14 values in `uk-political-parties.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Scottish National Party	→	**Sinn Féin**

9.8 Trade union membership

UK_L3_TRADE_UNION

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

WHAT IT IS

Union membership, name and subscription (check-off): Unison, Unite, GMB, the RCN, the NEU, NASUWT, PCS, Usdaw, the CWU, the UCU, the BMA, the RMT and others.

HOW IT IS FOUND

COLUMN NAME — UK_L3_TRADE_UNION — Path

95% `trade_?union(_?(member|membership|name|subs|subscription|dues))?` `union_?(member|membership|affiliation|name|subs|subscription|dues|fees|code|branch|rep|representative)` `unioni[sz]ed` `labou?r_?union` `bargaining_?unit` `collective_?(agreement|bargaining)` `check_?off` `staff_?association` `shop_?steward`

VALUE FORMAT — UK_L3_TRADE_UNION — Values

PATTERN	STRENGTH	NOTES
One of 32 known values, ignoring case — Unison, Unite, GMB, RCN, Royal College of Nursing, NEU, …	60%	

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Suppressed to `Not stated`; a yes/no flag stays a flag.

`UK_SUPPRESSED_CATEGORY` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_SUPPRESS`.

> `UK_FLAG` REGEX DECOMPOSE
>
> Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `UK_FLAG_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-yes-no.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-true-false.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-y-n.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-0-1.txt`, chosen by a keyed hash.
>
> `UK_CM_ALNUM` CHARACTER MAPPING
>
> Maps each character to another of its group (5 groups), keeping length and format.
>
> `UK_SUPPRESS` SECURE LOOKUP
>
> Replaces the value with one of the 1 values in `uk-not-stated.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Unison	→	**Not stated**
Y	→	**N**

9.9 Sexual orientation

UK_L3_SEXUAL_ORIENTATION

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

WHAT IT IS

Sexual orientation, in the categories of the 2021 census.

HOW IT IS FOUND

COLUMN NAME UK_L3_SEXUAL_ORIENTATION — Path

95% `sexual_?(orientation|preference|identity)` `orientation_?sexual` `orient_?sexual` `msm(_?(status|flag))?` `lgbt(q|qi|qia)?(_?(status|flag))?`

VALUE FORMAT UK_L3_SEXUAL_ORIENTATION — Values

PATTERN	STRENGTH	NOTES
One of 19 known values, ignoring case — Straight or heterosexual, Gay or lesbian, Bisexual, Pansexual, Asexual, Queer, …	80%	

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

Replaced from the census categories by keyed hash.

`UK_SEXUAL_ORIENTATION` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_SEXUAL_ORIENTATION_LIST`.

> `UK_FLAG` REGEX DECOMPOSE
>
> Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `UK_FLAG_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-yes-no.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-true-false.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-y-n.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-0-1.txt`, chosen by a keyed hash.
>
> `UK_CM_ALNUM` CHARACTER MAPPING
>
> Maps each character to another of its group (5 groups), keeping length and format.
>
> `UK_SEXUAL_ORIENTATION_LIST` SECURE LOOKUP
>
> Replaces the value with one of the 8 values in `uk-sexual-orientations.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Bisexual	→	**Other sexual orientation**

9.10 Sex life

UK_L3_SEXUAL_LIFE

COLUMN NAME | REGEX DECOMPOSE

WHAT IT IS

Sexual activity, partners, practices and history.

HOW IT IS FOUND

COLUMN NAME UK_L3_SEXUAL_LIFE — Path

90% `sex(ual)?_?life` `sexual_?(activity|behaviou?r|history|partners?|practices?)` `sexually_?active` `number_?of_?sexual_?partners` `age_?(at_?)?first_?(sex|intercourse)` `sex_?work(er)?`

HOW IT IS MASKED

Suppressed to `Not stated`.

`UK_SUPPRESSED_CATEGORY` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_SUPPRESS`.

`UK_FLAG` REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.

`UK_FLAG_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `uk-flag-yes-no.txt`, chosen by a keyed hash.

`UK_FLAG_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `uk-flag-true-false.txt`, chosen by a keyed hash.

`UK_FLAG_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `uk-flag-y-n.txt`, chosen by a keyed hash.

`UK_FLAG_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `uk-flag-0-1.txt`, chosen by a keyed hash.

`UK_CM_ALNUM` CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

`UK_SUPPRESS` SECURE LOOKUP

Replaces the value with one of the 1 values in `uk-not-stated.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Active	→	**Not stated**

9.11 Gender identity

UK_L3_GENDER_IDENTITY

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

Gender identity as the 2021 census asks for it (whether it is the sex registered at birth), trans status, pronouns, and gender recognition certificates. Section 22 of the Gender Recognition Act 2004 makes it an offence to disclose, from official information, that a person applied for or holds a certificate.

HOW IT IS FOUND

COLUMN NAME UK_L3_GENDER_IDENTITY — Path

95% `gender_?(identity|expression|recognition(_?(certificate|cert|status))?|reassignment|same_?as_?birth)` `grc(_?(holder|status|issued|date|no|number))?` `preferred_?pronouns?` `pronouns?` `transgender` `trans_?(status|history)` `gender_?same_?as_?sex`

VALUE FORMAT `UK_L3_GENDER_IDENTITY – Values`

PATTERN	STRENGTH	NOTES
One of 15 known values, ignoring case — Same as sex registered at birth, Trans woman, Trans man, Non-binary, Other gender identity, Prefer not to say, …	70%	
One of 6 known values, ignoring case — y, n, yes, no, 0, 1	30%	

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

Replaced from the census categories by keyed hash; a flag (holds a certificate) stays a flag.

`UK_GENDER_IDENTITY` **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_GENDER_IDENTITY_LIST`.

> `UK_FLAG` **REGEX DECOMPOSE**
>
> Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `UK_FLAG_YES_NO` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `uk-flag-yes-no.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_TRUE_FALSE` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `uk-flag-true-false.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_Y_N` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `uk-flag-y-n.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_0_1` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `uk-flag-0-1.txt`, chosen by a keyed hash.
>
> `UK_CM_ALNUM` **CHARACTER MAPPING**
>
> Maps each character to another of its group (5 groups), keeping length and format.
>
> `UK_GENDER_IDENTITY_LIST` **SECURE LOOKUP**
>
> Replaces the value with one of the 6 values in `uk-gender-identities.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
`Trans man`	→	**`Prefer not to say`**

PART 10

Data concerning health

Special category data under Article 9, defined as data about a person's physical or mental health, including the provision of health care, that reveals information about their health status (Article 4(15)). Health records carry, besides, the common law duty of confidentiality of the NHS. A diagnosis, an HIV result, an antidepressant or a pregnancy exposes a person as much as any other category; the NHS number itself, though, is in the direct identifiers, since it follows a person well beyond the clinic.

10.1 Health record identifiers

UK_HEALTH_ID

COLUMN NAME | CHARACTER MAPPING

WHAT IT IS

Hospital and medical record numbers, case note and PAS numbers, GHIC and EHIC numbers, prescription and fit note numbers, and GP practice codes.

HOW IT IS FOUND

COLUMN NAME — UK_HEALTH_ID — Path

85% `medical_?record(_?(no|num|number))?` `mrn` `hospital_?(no|num|number|id)` `patient_?(id|no|num|number)` `case_?note_?(no|number)` `pas_?(id|no|number)` `chart_?(no|num|number)` `health_?(record|card|id)(_?(no|num|number))?` `ehic(_?(no|number))?` `ghic(_?(no|number))?` `admission_?(no|num|number)` `prescription_?(no|num|number)` `fit_?note_?(no|number|ref)` `gp_?(practice_?)?(code|id)` `ods_?code` `practice_?code`

55% `chart` `admission` `episode`

HOW IT IS MASKED

Character by character, keeping the format.

`UK_CM_ALNUM` CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
RX123456	→	**MZ999874**

10.2 Diagnosis

UK_HEALTH_DIAGNOSIS

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

WHAT IT IS

Diagnoses, conditions, allergies and causes of death or absence, as ICD-10 codes (`F32.9`), SNOMED CT concepts (`44054006`) or text.

HOW IT IS FOUND

COLUMN NAME UK_HEALTH_DIAGNOSIS – Path

85% `diagnos(is|es|tic)[a-z_]*` `dx` `icd_?(10|11)?(_?code)?` `snomed(_?(ct|code|concept))?` `read_?code` `ctv3` `medical_?(condition|history|problem|diagnosis)` `health_?(condition|problem)` `long_?term_?condition` `chronic_?(condition|disease|illness)` `illness(es)?` `disease[a-z_]*` `comorbidit(y|ies)` `allerg(y|ies)` `cause_?of_?(death|admission|illness|injury|absence)` `reason_?for_?(visit|admission|referral|absence|sickness)` `presenting_?complaint` `chief_?complaint` `pre_?existing_?conditions?`

VALUE FORMAT UK_HEALTH_DIAGNOSIS – Regex

PATTERN	STRENGTH	NOTES
`[A-TV-Za-tv-z]\d{2}(\.\d{1,2}|\d)?`	50%	

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

An ICD-10 code becomes another common ICD-10 code of the same shape, a SNOMED CT concept gets other digits, and text becomes a common diagnosis — a rare one identifies.

`UK_DIAGNOSIS` **REGEX DECOMPOSE**

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `UK_DIAGNOSIS_TEXT`.

> `UK_ICD10_DECIMAL` **SECURE LOOKUP**
>
> Replaces the value with one of the 33 values in `uk-icd10-decimal.txt`, chosen by a keyed hash.
>
> `UK_ICD10` **SECURE LOOKUP**
>
> Replaces the value with one of the 37 values in `uk-icd10.txt`, chosen by a keyed hash.
>
> `UK_CM_ALNUM` **CHARACTER MAPPING**
>
> Maps each character to another of its group (5 groups), keeping length and format.
>
> `UK_FLAG` **REGEX DECOMPOSE**
>
> Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `UK_FLAG_YES_NO` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `uk-flag-yes-no.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_TRUE_FALSE` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `uk-flag-true-false.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_Y_N` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `uk-flag-y-n.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_0_1` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `uk-flag-0-1.txt`, chosen by a keyed hash.
>
> `UK_DIAGNOSIS_TEXT` **SECURE LOOKUP**
>
> Replaces the value with one of the 30 values in `uk-diagnoses.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
F32.9	→	**J02.9**
44054006	→	**36556707**

10.3 Medication

UK_HEALTH_MEDICATION

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

WHAT IT IS

Prescribed medicines and repeat prescriptions. Some reveal the condition on their own: an antiretroviral, PrEP, an antipsychotic, methadone.

HOW IT IS FOUND

> **COLUMN NAME** UK_HEALTH_MEDICATION — Path
>
> 85% `medications?` `medicines?` `drugs?_?(name|prescribed)` `prescribed_?(drugs?|medications?|items?)` `prescription_?(drug|medication|item)s?` `repeat_?prescriptions?` `active_?ingredient` `dosage` `treatment_?(drug|medication)`

> **VALUE FORMAT** UK_HEALTH_MEDICATION — Values
>
PATTERN	STRENGTH	NOTES
One of 51 known values, ignoring case — Paracetamol, Ibuprofen, Naproxen, Co-codamol, Metformin, Gliclazide, …	70%	
>
> A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Replaced by a common medicine of primary care.

`UK_MEDICATION` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_MEDICATION_LIST`.

> `UK_FLAG` REGEX DECOMPOSE
>
> Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `UK_FLAG_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-yes-no.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-true-false.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-y-n.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-0-1.txt`, chosen by a keyed hash.
>
> `UK_CM_ALNUM` CHARACTER MAPPING
>
> Maps each character to another of its group (5 groups), keeping length and format.
>
> `UK_MEDICATION_LIST` SECURE LOOKUP
>
> Replaces the value with one of the 26 values in `uk-medications.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
`Sertraline 50 mg`	→	**`Ferrous sulfate`**

10.4 Clinical notes

UK_HEALTH_CLINICAL_TEXT

COLUMN NAME | DATA TYPE | STRING ALGORITHM CHAIN

WHAT IT IS

Consultation and progress notes, discharge summaries, clinic and referral letters, care plans.

HOW IT IS FOUND

COLUMN NAME UK_HEALTH_CLINICAL_TEXT – Path

85% `clinical_?notes?` `progress_?notes?` `nurs(e|ing)_?notes?` `doctors?_?notes?` `gp_?notes?` `consultation_?notes?` `medical_?notes?` `discharge_?(summary|letter|notes?)` `clinic_?letter` `referral_?letter` `clinical_?summary` `history_?of_?present(ing)?_?illness` `hpi` `physical_?exam(ination)?` `assessment_?(and_?)?plan` `treatment_?plan` `care_?plan` `triage_?notes?` `soap_?notes?`

DATA TYPE UK_HEALTH_CLINICAL_TEXT – Type

Accepts text (length 30 or more); any other type rules the data class out.

HOW IT IS MASKED

The same two steps as free text: identifiers, spaced or not, become `[DATA]` and names `[NAME]`. The clinical content stays and still deserves care.

`UK_FREE_TEXT_ALL` STRING ALGORITHM CHAIN

Applies `UK_TEXT_SPACED` → `UK_FREE_TEXT`, in this order.

`UK_TEXT_SPACED` REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.

`UK_FREE_TEXT` FREE TEXT REDACTION

Replaces the text matching 10 patterns with `[DATA]`, and the 1,266 terms of `uk-text-names.txt` with `[NAME]`.

EXAMPLES

VALUE		MASKED
`Patient Smith, NHS 943 476 5919, seen for headache.`	→	**`Patient [NAME], NHS [DATA], seen for headache.`**

10.5 Test results

UK_HEALTH_TEST_RESULT

COLUMN NAME CHARACTER MAPPING

WHAT IT IS

Laboratory and pathology results, HIV status, viral load, HbA1c, eGFR, cervical screening, COVID-19 tests.

HOW IT IS FOUND

COLUMN NAME UK_HEALTH_TEST_RESULT – Path

80% `lab_?(result|test|value)s?` `test_?results?` `pathology_?(result|report)s?` `laboratory_?(result|test)s?` `hiv(_?(status|test|result))?` `viral_?load` `cd4(_?count)?` `blood_?(sugar|glucose|pressure)` `glucose` `hba1c` `cholesterol` `egfr` `bmi` `pregnancy_?test` `drug_?(test|screen)` `covid_?(test|result)` `pcr_?result` `lateral_?flow(_?result)?` `smear_?(test|result)` `cervical_?screening(_?result)?` `mammogram_?result` `x_?ray_?result` `psa(_?result)?`

50% `result` `test`

HOW IT IS MASKED

Character by character, keeping the format.

`UK_CM_ALNUM` CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
Positive	→	**Haheceki**

10.6 Blood group

UK_HEALTH_BLOOD_GROUP

COLUMN NAME · VALUE FORMAT · SECURE LOOKUP

WHAT IT IS

ABO group and Rh factor.

HOW IT IS FOUND

COLUMN NAME UK_HEALTH_BLOOD_GROUP — Path

90% `blood_?(type|group)` `abo(_?rh)?` `rh_?factor` `rh` `rhesus`

VALUE FORMAT UK_HEALTH_BLOOD_GROUP — Values

PATTERN	STRENGTH	NOTES
One of 26 known values, ignoring case — A, A-, B, B-, AB, AB-, …	90%	

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

BloodType SL, which comes with Delphix: one of the eight groups, by keyed hash.

`dlpx-core:BloodType SL` SECURE LOOKUP

Comes with Delphix and is used as it ships. Replaces the blood group with one of the eight groups, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
0+	→	**0−**

10.7 Disability

UK_HEALTH_DISABILITY

COLUMN NAME · VALUE FORMAT · REGEX DECOMPOSE

WHAT IT IS

Disability as the 2021 census asks for it (the Equality Act 2010 definition: a condition that limits day-to-day activities a lot or a little), impairment types, special educational needs and EHC plans, Blue Badge, reasonable adjustments.

HOW IT IS FOUND

COLUMN NAME UK_HEALTH_DISABILITY — Path

90% `disabilit(y|ies)[a-z_]*` `disabled` `impairments?` `handicap(ped)?` `special_?(educational_?)?needs` `sen(d)?(_?(status|support|provision))?` `ehcp` `education_?health_?(and_?)?care_?plan` `blue_?badge` `reasonable_?adjustments?` `limited_?(a_?lot|a_?little)` `day_?to_?day_?activities_?limited` `access_?to_?work`

VALUE FORMAT UK_HEALTH_DISABILITY — Values

PATTERN	STRENGTH	NOTES
One of 29 known values, ignoring case — Disabled limited a lot, Disabled limited a little, Not disabled, Visual impairment, Hearing impairment, Mobility impairment, …	60%	

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Replaced from the list by keyed hash; flags and codes keep their shape.

`UK_DISABILITY` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_DISABILITY_LIST`.

> `UK_FLAG` REGEX DECOMPOSE
>
> Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `UK_FLAG_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-yes-no.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-true-false.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-y-n.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-0-1.txt`, chosen by a keyed hash.
>
> `UK_CM_ALNUM` CHARACTER MAPPING
>
> Maps each character to another of its group (5 groups), keeping length and format.
>
> `UK_DISABILITY_LIST` SECURE LOOKUP
>
> Replaces the value with one of the 10 values in `uk-disabilities.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
`Disabled: limited a lot`	→	**`Learning disability`**

10.8 Reproductive and sexual health

UK_HEALTH_REPRODUCTIVE

COLUMN NAME REGEX DECOMPOSE

WHAT IT IS

Pregnancy, due date, terminations, miscarriages, contraception, fertility treatment, sexually transmitted infections.

HOW IT IS FOUND

COLUMN NAME UK_HEALTH_REPRODUCTIVE — Path

90% `pregnan(t|cy)[a-z_]*` `gestation(al)?(_?age)?` `weeks_?(of_?)?(pregnancy|gestation)` `lmp` `last_?menstrual_?period` `edd` `estimated_?due_?date` `abortions?` `termination_?of_?pregnancy` `miscarriages?` `contracepti(on|ve)[a-z_]*` `family_?planning(_?method)?` `fertility` `infertility` `ivf` `gravida` `parity` `antenatal` `maternity_?(status|care)` `postnatal` `postpartum` `sti` `sexually_?transmitted(_?infection)?` `sexual_?health`

HOW IT IS MASKED

Suppressed to `Not stated`; flags and codes keep their shape.

`UK_SUPPRESSED_CATEGORY` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_SUPPRESS`.

> `UK_FLAG` REGEX DECOMPOSE
>
> Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `UK_FLAG_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-yes-no.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-true-false.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-y-n.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-0-1.txt`, chosen by a keyed hash.
>
> `UK_CM_ALNUM` CHARACTER MAPPING
>
> Maps each character to another of its group (5 groups), keeping length and format.
>
> `UK_SUPPRESS` SECURE LOOKUP
>
> Replaces the value with one of the 1 values in `uk-not-stated.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
`12 weeks pregnant`	→	**`Not stated`**

10.9 Mental health

UK_HEALTH_MENTAL

COLUMN NAME REGEX DECOMPOSE

WHAT IT IS

Mental health conditions, detention under the Mental Health Act, self-harm, substance misuse, talking therapies and CAMHS.

HOW IT IS FOUND

> COLUMN NAME UK_HEALTH_MENTAL — Path
>
> **90%** `mental_?(health|illness|disorder|condition)` `psychiatr[a-z_]*` `psycholog[a-z_]*` `depression` `anxiety` `suicid[a-z_]*` `self_?harm` `sectioned` `mental_?health_?act(_?(section|status))?` `substance_?(use|abuse|misuse)` `alcohol(ism|_?use|_?abuse|_?dependence)` `drug_?(use|abuse|misuse|addiction|dependence)` `addiction[a-z_]*` `iapt` `talking_?therapies` `camhs`

HOW IT IS MASKED

Suppressed to `Not stated`.

`UK_SUPPRESSED_CATEGORY` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_SUPPRESS`.

> `UK_FLAG` REGEX DECOMPOSE
>
> Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `UK_FLAG_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-yes-no.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-true-false.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-y-n.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_0_1` SECURE LOOKUP

> Replaces the value with one of the 2 values in `uk-flag-0-1.txt`, chosen by a keyed hash.

`UK_CM_ALNUM` **CHARACTER MAPPING**

Maps each character to another of its group (5 groups), keeping length and format.

`UK_SUPPRESS` **SECURE LOOKUP**

Replaces the value with one of the 1 values in `uk-not-stated.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
`Bipolar disorder`	→	**`Not stated`**

PART 11

Financial data

The UK GDPR does not make financial data special category. It is personal data all the same, and other rules close it further: banks keep their customers' affairs under the duty of confidence recognized in Tournier v National Provincial and Union Bank of England (1924), and credit information circulates among the credit reference agencies under the Consumer Credit Act 1974. Several benefits reveal health: Personal Independence Payment, Disability Living Allowance, Employment and Support Allowance and Attendance Allowance are paid because of a condition. Categories are replaced within their own notation; amounts keep their number of digits, with no link to the real value.

11.1 Credit file and insolvency

UK_FIN_CREDIT_REFERENCE

COLUMN NAME | REGEX DECOMPOSE

WHAT IT IS

Credit scores, missed payments, defaults, County Court Judgments, Individual Voluntary Arrangements, Debt Relief Orders and bankruptcy.

HOW IT IS FOUND

COLUMN NAME — UK_FIN_CREDIT_REFERENCE — Path

90% `credit_?(score|rating|history|report|status|file|reference|risk|grade|standing)` `experian(_?(score|report|status))?` `equifax(_?(score|report|status))?` `transunion(_?(score|report|status))?` `ccj(s)?` `county_?court_?judg(e)?ments?` `iva` `individual_?voluntary_?arrangement` `debt_?relief_?order` `dro` `bankrupt(cy)?(_?(status|flag|date))?` `insolven(cy|t)(_?(status|flag))?` `delinquen(cy|t)(_?(status|days))?` `days_?(past_?due|in_?arrears|overdue)` `dpd` `arrears(_?(status|days))?` `missed_?payments?` `default_?(status|flag|indicator)` `defaulted` `written_?off` `write_?off(_?(status|flag))?` `collections?_?status` `bad_?debt` `loan_?(status|classification|grade)`

HOW IT IS MASKED

A score gets other digits of the same length; a status is replaced from a list; flags stay flags.

`UK_CREDIT` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_CREDIT_STATUS`.

> `UK_CM_ALNUM` CHARACTER MAPPING
>
> Maps each character to another of its group (5 groups), keeping length and format.
>
> `UK_FLAG` REGEX DECOMPOSE
>
> Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `UK_FLAG_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-yes-no.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-true-false.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-y-n.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-0-1.txt`, chosen by a keyed hash.
>
> `UK_CREDIT_STATUS` SECURE LOOKUP
>
> Replaces the value with one of the 13 values in `uk-credit-statuses.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
County Court Judgment	→	**3 payments missed**
745	→	**471**

11.2 Debt

UK_FIN_DEBT

COLUMN NAME | DATA TYPE | NUMERIC MAPPING

WHAT IT IS

Loan, mortgage and overdraft balances, amounts owed and in arrears, credit limits, instalments.

HOW IT IS FOUND

COLUMN NAME UK_FIN_DEBT – Path

85% `loan_?(amount|balance|principal|outstanding)` `outstanding_?(balance|amount|debt|loan|principal)` `debt(_?(amount|total|balance))?` `amount_?(owed|due|outstanding|in_?arrears|overdue)` `balance_?(owed|outstanding|due)` `arrears_?amount` `credit_?limit` `overdraft(_?(limit|balance))?` `mortgage_?(balance|amount)` `instal?ment(_?amount)?` `monthly_?(payment|repayment)`

DATA TYPE UK_FIN_DEBT – Type

Accepts number; any other type rules the data class out.

HOW IT IS MASKED

CM Numeric, which comes with Delphix: each digit is mapped, so the amount keeps its number of digits and its order of magnitude.

`dlpx-core:CM Numeric` NUMERIC MAPPING

Comes with Delphix and is used as it ships. Maps each digit to another, keeping the number of digits, the sign, the decimal places and the zeros on the left — an amount keeps its order of magnitude.

EXAMPLES

VALUE		MASKED
850000	→	**847556**

11.3 Income

UK_FIN_INCOME

COLUMN NAME | DATA TYPE | NUMERIC MAPPING

WHAT IT IS

Salary, wages, earnings, bonus and household income.

HOW IT IS FOUND

COLUMN NAME UK_FIN_INCOME – Path

85% `salary` `salaries` `wages?` `income[a-z_]*` `gross_?(pay|salary|income|earnings)` `net_?(pay|salary|income|earnings)` `(monthly|annual|weekly|fortnightly)_?(income|salary|earnings|pay|wage)` `earnings` `remuneration` `basic_?(pay|salary)` `take_?home_?pay` `household_?income` `taxable_?pay` `p60_?pay` `bonus(_?amount)?`

DATA TYPE UK_FIN_INCOME – Type

Accepts number; any other type rules the data class out.

HOW IT IS MASKED

CM Numeric, which comes with Delphix: the order of magnitude stays, the value does not.

`dlpx-core:CM Numeric` NUMERIC MAPPING

Comes with Delphix and is used as it ships. Maps each digit to another, keeping the number of digits, the sign, the decimal places and the zeros on the left — an amount keeps its order of magnitude.

EXAMPLES

VALUE		MASKED
42000	→	**36672**

11.4 Assets

UK_FIN_ASSETS

COLUMN NAME DATA TYPE NUMERIC MAPPING

WHAT IT IS

Property and house values, savings and ISA balances, investments, net worth.

HOW IT IS FOUND

COLUMN NAME UK_FIN_ASSETS – Path

85% `net_?worth` `assets?_?(value|total)` `total_?assets` `property_?value` `house_?value` `market_?value` `valuation(_?amount)?` `savings(_?balance)?` `account_?balance` `current_?balance` `deposit_?balance` `isa_?(balance|value)` `investment_?(balance|value|amount)` `share_?(holding|value)` `capital_?(savings|value)`

DATA TYPE UK_FIN_ASSETS – Type

Accepts number; any other type rules the data class out.

HOW IT IS MASKED

CM Numeric, which comes with Delphix.

`dlpx-core:CM Numeric` NUMERIC MAPPING

Comes with Delphix and is used as it ships. Maps each digit to another, keeping the number of digits, the sign, the decimal places and the zeros on the left — an amount keeps its order of magnitude.

EXAMPLES

VALUE		MASKED
385000	→	**797256**

11.5 Pension

UK_FIN_PENSION

COLUMN NAME DATA TYPE NUMERIC MAPPING

WHAT IT IS

State Pension and private pension amounts, pension pots, annuities, lump sums and transfer values.

HOW IT IS FOUND

COLUMN NAME UK_FIN_PENSION – Path

90% `pension(_?(amount|payment|monthly|weekly|annual|benefit_?amount|balance|pot|value))?` `monthly_?pension` `state_?pension(_?amount)?` `retirement_?(pension|benefit|amount|balance)` `pension_?pot` `annuity(_?amount)?` `lump_?sum` `transfer_?value` `superannuation(_?(balance|amount))?`

DATA TYPE UK_FIN_PENSION – Type

Accepts number; any other type rules the data class out.

HOW IT IS MASKED

CM Numeric, which comes with Delphix.

`dlpx-core:CM Numeric` NUMERIC MAPPING

Comes with Delphix and is used as it ships. Maps each digit to another, keeping the number of digits, the sign, the decimal places and the zeros on the left — an amount keeps its order of magnitude.

EXAMPLES

VALUE		MASKED
11502	→	36359

11.6 Benefits

UK_FIN_BENEFIT

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

Benefits of the DWP, HMRC and the councils: Universal Credit, Personal Independence Payment, Disability Living Allowance, Employment and Support Allowance, Jobseeker's Allowance, Pension Credit, State Pension, Child Benefit, Carer's Allowance, Attendance Allowance, Housing Benefit, Council Tax Reduction, and free school meals. Several reveal a disability or a health condition.

HOW IT IS FOUND

COLUMN NAME UK_FIN_BENEFIT – Path

90% `benefits?_?(type|claim|category|name|code|status|received|entitlement)` `claim_?type` `universal_?credit(_?(status|claimant|flag))?` `uc_?(claimant|status|flag)` `pip(_?(award|rate|status))?` `dla(_?(award|rate|status))?` `esa(_?(status|group))?` `jsa(_?(status))?` `pension_?credit` `child_?benefit` `carers?_?allowance` `attendance_?allowance` `housing_?benefit` `council_?tax_?(reduction|support|benefit)` `dwp_?(benefit|claim)` `in_?receipt_?of_?benefits?` `welfare_?(benefit|status)` `free_?school_?meals` `fsm(_?(eligible|status|flag))?`

VALUE FORMAT UK_FIN_BENEFIT – Values

PATTERN	STRENGTH	NOTES
One of 30 known values, ignoring case — Universal Credit, Personal Independence Payment, Disability Living Allowance, Employment and Support Allowance, Jobseeker s Allowance, Pension Credit, ...	80%	
One of 6 known values, ignoring case — y, n, yes, no, 0, 1	30%	

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

Replaced from the list by keyed hash; a flag stays a flag.

`UK_BENEFIT` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_BENEFIT_LIST`.

> `UK_FLAG` REGEX DECOMPOSE
>
> Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `UK_FLAG_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-yes-no.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-true-false.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-y-n.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-0-1.txt`, chosen by a keyed hash.
>
> `UK_CM_ALNUM` CHARACTER MAPPING
>
> Maps each character to another of its group (5 groups), keeping length and format.
>
> `UK_BENEFIT_LIST` SECURE LOOKUP
>
> Replaces the value with one of the 13 values in `uk-benefits.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
`Personal Independence Payment`	→	**`Housing Benefit`**
`Y`	→	**`N`**

11.7 Deprivation and socio-economic class

UK_FIN_DEPRIVATION

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

Indices of Multiple Deprivation of the four nations (decile, quintile, rank), IDACI, NS-SEC, geodemographic segments (ACORN, Mosaic) and income bands.

HOW IT IS FOUND

> COLUMN NAME UK_FIN_DEPRIVATION — Path
>
> **90%** `imd(_?(decile|quintile|rank|score))?` `simd(_?(decile|quintile|rank))?` `wimd(_?(decile|quintile|rank))?` `nimdm(_?(decile|rank))?` `index_?of_?multiple_?deprivation` `deprivation(_?(decile|quintile|rank|score|index|level))?` `idaci(_?(decile|score))?` `socio_?economic_?(status|class|classification|group|index)` `ns_?sec` `acorn(_?(category|group))?` `mosaic(_?(group|type))?` `income_?(quintile|decile|bracket|band)` `quintile` `decile`

> VALUE FORMAT UK_FIN_DEPRIVATION — Values
>
PATTERN	STRENGTH	NOTES
One of 9 known values, ignoring case — most deprived, deprived, least deprived, less deprived, average, affluent, …	40%	
>
> A value that matches none counts 30% against the data class.

HOW IT IS MASKED

A decile or quintile is replaced within its range, a rank gets other digits, and a label is replaced from a list.

`UK_DEPRIVATION` REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: `UK_DEPRIVATION_STATUS`.

> `UK_QUINTILE` SECURE LOOKUP
>
> Replaces the value with one of the 5 values in `uk-quintile.txt`, chosen by a keyed hash.
>
> `UK_DECILE_HIGH` SECURE LOOKUP
>
> Replaces the value with one of the 5 values in `uk-decile-high.txt`, chosen by a keyed hash.
>
> `UK_DECILE` SECURE LOOKUP
>
> Replaces the value with one of the 10 values in `uk-decile.txt`, chosen by a keyed hash.
>
> `UK_CM_ALNUM` CHARACTER MAPPING
>
> Maps each character to another of its group (5 groups), keeping length and format.
>
> `UK_DEPRIVATION_STATUS` SECURE LOOKUP
>
> Replaces the value with one of the 5 values in `uk-deprivation-statuses.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
`Decile 2`	→	**`Decile 4`**
`Most deprived`	→	**`Most deprived`**

11.8 Housing tenure

UK_FIN_HOUSING

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

Tenure as the 2021 census asks for it — owned outright or with a mortgage, shared ownership, social or private rented, rent free — and homelessness and temporary accommodation.

HOW IT IS FOUND

> COLUMN NAME UK_FIN_HOUSING – Path
>
> 95% `housing_?(tenure|status|type|situation|need)` `dwelling_?(tenure|ownership|type)` `home_?ownership` `own_?or_?rent` `rent_?or_?own` `owns_?(home|house)` `homeless(ness)?(_?(status|duty|flag))?` `temporary_?accommodation` `council_?tenant` `social_?housing`
>
> 55% `tenure`

> VALUE FORMAT UK_FIN_HOUSING – Values
>
PATTERN	STRENGTH	NOTES
One of 24 known values, ignoring case — Owned outright, Owned with a mortgage, Shared ownership, Social rented from council, Social rented from housing association, Private rented, …	60%	
>
> A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Replaced from the census categories by keyed hash.

`UK_HOUSING` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_HOUSING_LIST`.

> `UK_FLAG` REGEX DECOMPOSE
>
> Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `UK_FLAG_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-yes-no.txt`, chosen by a keyed hash.

> `UK_FLAG_TRUE_FALSE` **SECURE LOOKUP**
>
> Replaces the value with one of the 2 values in `uk-flag-true-false.txt`, chosen by a keyed hash.
>
> `UK_FLAG_Y_N` **SECURE LOOKUP**
>
> Replaces the value with one of the 2 values in `uk-flag-y-n.txt`, chosen by a keyed hash.
>
> `UK_FLAG_0_1` **SECURE LOOKUP**
>
> Replaces the value with one of the 2 values in `uk-flag-0-1.txt`, chosen by a keyed hash.

`UK_CM_ALNUM` **CHARACTER MAPPING**

Maps each character to another of its group (5 groups), keeping length and format.

`UK_HOUSING_LIST` **SECURE LOOKUP**

Replaces the value with one of the 9 values in `uk-housing-tenure.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
`Private rented`	→	**`Other`**

PART 12

Criminal offence data

Article 10 of the UK GDPR, which section 11(2) of the Data Protection Act 2018 extends to the alleged commission of offences and to proceedings and their outcome — an arrest that ended in no further action is as protected as a conviction. Under the Rehabilitation of Offenders Act 1974 most convictions and cautions become spent after a time, and a spent conviction is treated as never having happened; a test copy that keeps it undoes that. DBS checks say whether a record exists at all.

12.1 Criminal record

UK_PENAL_CRIMINAL_RECORD

COLUMN NAME | REGEX DECOMPOSE

WHAT IT IS

Convictions, spent and unspent, cautions, penalty notices, charges, arrests, bail, remand, sentences, probation and licence conditions, acquittals and no further action.

HOW IT IS FOUND

COLUMN NAME — UK_PENAL_CRIMINAL_RECORD — Path

95%

```
criminal_?(record|history|conviction|convictions|status|background|charges?|proceedings|case_?(status|outcome)) police_?(record|caution|report_?status) convictions?(_?(status|record|details))? unspent_?convictions? spent_?convictions? convicted previous_?convictions cautions?(_?(status|details|record))? offen[cs]es?(_?(type|committed|code|description|details))? charged_?with arrests?(_?(record|history))? sentence(_?(imposed|type|length|details))? custodial_?sentence incarcerat(ed|ion) imprisonment prison(_?(status|release))? remand(ed)?(_?status)? bail(_?status)? probation(_?status)? licen[cs]e_?conditions parole verdict acquitt(al|ed) nfa no_?further_?action
```

55%

```
court_?(outcome|disposal|decision|status) case_?(outcome|disposal) disposal
```

HOW IT IS MASKED

Replaced by a generic status from a list; flags stay flags.

`UK_CRIMINAL_RECORD` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_CRIMINAL_RECORD_LIST`.

> `UK_FLAG` REGEX DECOMPOSE
>
> Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `UK_FLAG_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-yes-no.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-true-false.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-y-n.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `uk-flag-0-1.txt`, chosen by a keyed hash.
>
> `UK_CM_ALNUM` CHARACTER MAPPING
>
> Maps each character to another of its group (5 groups), keeping length and format.
>
> `UK_CRIMINAL_RECORD_LIST` SECURE LOOKUP

> Replaces the value with one of the 12 values in `uk-criminal-record.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Spent conviction	→	**No further action**

12.2 Police and court references

UK_PENAL_CASE_NUMBER

COLUMN NAME | VALUE FORMAT | CHARACTER MAPPING

WHAT IT IS

Police National Computer identifiers (`2019/0123456X`), CRO numbers, crime and incident references, custody numbers, court case numbers and prisoner numbers.

HOW IT IS FOUND

> COLUMN NAME UK_PENAL_CASE_NUMBER — Path
>
> | | |
> |---|---|
> | 90% | `court_?case_?(no\|num\|number\|ref)` `criminal_?case_?(no\|num\|number)` `indictment_?(no\|num\|number)` `pnc_?(id\|no\|number\|ref)` `police_?national_?computer_?(id\|no\|number)` `cro_?(no\|number)` `crime_?(ref\|reference\|no\|number)` `crime_?report_?(no\|number)` `police_?(ref\|reference\|report_?(no\|number)\|incident_?(no\|number))` `custody_?(no\|number\|record_?(no\|number))` `urn_?(police\|case)` `case_?urn` `prisoner_?(no\|number)` `prison_?(no\|number)` `noms_?(no\|number\|id)` `nomis_?(id\|no\|number)` |
> | 55% | `case_?(no\|num\|number\|ref\|reference)` |

> VALUE FORMAT UK_PENAL_CASE_NUMBER — Regex
>
> | PATTERN | STRENGTH | NOTES |
> |---|---|---|
> | `0*(?:\d{2,4}/\d{1,7}[A-HJ-NP-RT-Z])` | 70% | |
> | `[TSA]\d{8}` | 50% | |
>
> A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Character by character, keeping the format.

`UK_CM_ALNUM` CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
2019/0123456X	→	**2998/2920867R**

12.3 DBS checks

UK_PENAL_DBS

COLUMN NAME | REGEX DECOMPOSE

WHAT IT IS

Disclosure and Barring Service checks (Disclosure Scotland and AccessNI in the other nations): the certificate number, the level — basic, standard, enhanced, with the barred lists — and the outcome.

HOW IT IS FOUND

COLUMN NAME UK_PENAL_DBS — Path

95% `dbs(_?(cert|certificate|no|num|number|ref|reference|check|status|level|result|outcome|date|update_?service))?` `crb(_?(cert|check|no|number|status))?` `disclosure_?(cert|certificate|no|number|level|result|outcome)` `disclosure_?scotland` `pvg(_?(no|number|scheme|membership|status))?` `access_?ni(_?(no|number|check|status))?` `barred_?list(_?(check|status))?` `barring_?status`

HOW IT IS MASKED

A certificate number gets other digits; a level or outcome is replaced from a list; flags stay flags.

`UK_DBS` **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_DBS_OUTCOME`.

> `UK_CM_ALNUM` **CHARACTER MAPPING**
>
> Maps each character to another of its group (5 groups), keeping length and format.
>
> `UK_FLAG` **REGEX DECOMPOSE**
>
> Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `UK_FLAG_YES_NO` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `uk-flag-yes-no.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_TRUE_FALSE` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `uk-flag-true-false.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_Y_N` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `uk-flag-y-n.txt`, chosen by a keyed hash.
> >
> > `UK_FLAG_0_1` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `uk-flag-0-1.txt`, chosen by a keyed hash.
>
> `UK_DBS_OUTCOME` **SECURE LOOKUP**
>
> Replaces the value with one of the 7 values in `uk-dbs-outcomes.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
`001234567890`	→	**`001713678115`**
`Enhanced with barred list`	→	**`Standard`**

13

Where people live: postcodes and towns

Two columns place a person on the map, and both are generalized rather than replaced, so the data keeps its geography at a level where it no longer points at a household.

- The **postcode** keeps its outward code: `SW1A 2AA` stays in `SW1A`, the district, and its inward code is masked. A postcode district holds thousands of addresses; the full postcode, about fifteen.
- A **town or village** of fewer than 20,000 people becomes the **main town of its local authority** — its council, unitary authority or county: Windermere and Keswick become Carlisle, Whitby becomes Harrogate, Portree becomes Inverness, Tenby becomes Haverfordwest. 3,963 of the 4,643 places are generalized.
- Parts of London count as places of their own: Wimbledon, Brixton and Peckham keep their names, and the smaller ones — Highgate, Soho — become **London**.
- Places of 20,000 people or more keep their names, and so do the main towns, even the small ones (Lerwick, Kirkwall, Stornoway): a main town in a masked column says the local authority, no more.
- A name shared by several places (there are Newports in Wales, on the Isle of Wight and in Shropshire) is kept when any of them is large, and otherwise generalized as the largest of them is. Names written with `St` or `Saint`, or with spaces for hyphens, are recognized too.

The populations come from the GeoNames list of places of more than 1,000 people, checked by hand where it gave a district's population to a single village. Names that are also given names or surnames are left out of the detection list, so that a surname column is not taken for towns.

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Limits

- **Binary columns** — scans of passports, driving licences and residence permits, photographs, custody images, clinical PDFs in BLOBs — cannot be masked by these algorithms. Inventory them and report them as residual risk.
- **Places smaller than 1,000 people** are not in the table: a column of hamlets is found by its name, but values the table does not know pass unchanged. Where such a column sits next to a birth date and sex, generalize it to the local authority.
- **The NHS number repair** moves about one masked number in eleven up by one step, so a few different NHS numbers may mask to the same one. Where the NHS number is a key that must stay unique, check the masked copy for duplicates.
- **A CHI number keeps a real date**, but a different one: its date part no longer matches the date of birth column, which is shifted independently.
- **The National Insurance suffix and prefix stay.** They say nothing about the person, but a temporary number (`TN`) stays recognizable as one.
- **Passport, permit and licence numbers** are masked character by character; no check digit of theirs is recomputed.
- **Data Cleansing leaves unknown values as they are**: a town spelled in a way the list does not have passes unchanged.
- **Free text** is redacted pattern by pattern: up to four spaced identifiers in a value, then word by word. A number split across lines or written in an unusual way is not found, and a given name that is also a common word (`Grace` , `Rose` , `Hope`) is redacted wherever it appears.

- **Colour and trade words are surnames.** Brown, Green, Gray, White, Baker, Cook and Wood are common British surnames: with only the essential pack loaded, a column of such words may be taken for names.
- **Children** — a birth date moves at most 60 days, so few people cross their 13th, 16th or 18th birthday, but some do. Split the rule by age group with Multi-Column Condition where that matters.
- **The engine key** links masked values back to production for whoever holds both; keep it away from the users of the masked copy — re-identifying without the controller's consent is an offence under section 171.

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Verification

Before publication the policy was checked outside Delphix:

- Every classifier configuration was reviewed against what Delphix accepts.
- Discovery was simulated on 168 columns with the whole policy at its assignment threshold — well named, badly named, ten with the short names of packaged systems and 36 columns that are not personal data — again on 157 of them padded with spaces or zeros, and again with the classifiers of the essential pack alone.
- The table that generalizes towns was audited for all 4,643 places.
- The algorithms, run with the Delphix algorithm plugin, masked the sample values and placeholders such as `N/A`, checking each output against its vocabulary and every masked NHS number, UTR and IBAN against its check digits; the same values again padded with spaces or zeros, which had to mask as their bare value; and 150 National Insurance and NHS numbers each for collisions. Each of the 105 algorithms also masked its own sample input.

Every column was assigned as expected and every value masked as expected. Before using the policy on real data, run a discovery job and a masking job on a representative copy.

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Sources

- UK GDPR and Data Protection Act 2018, as amended by the Data (Use and Access) Act 2025 — legislation.gov.uk. Commencement regulations of the 2025 Act and the abolition of the Information Commissioner's Office — S.I. 2026/82 and S.I. 2026/1015.
- Anonymisation, pseudonymisation and privacy enhancing technologies guidance, and guidance on special category and criminal offence data — Information Commissioner's Office, now the Information Commission.
- Rehabilitation of Offenders Act 1974; Gender Recognition Act 2004; Equality Act 2010; Consumer Credit Act 1974 — legislation.gov.uk.
- National Insurance numbers — HMRC National Insurance Manual. Unique Taxpayer Reference — HMRC. NHS number — NHS England data dictionary; CHI number — Public Health Scotland; H&C number — Business Services Organisation, Northern Ireland.
- Driving licence numbers and vehicle registration marks — DVLA. Postcodes — Royal Mail and the ONS Postcode Directory. IBAN — ISO 13616 and the UK sort code directory.
- Ethnic group, religion, sexual orientation, gender identity, legal partnership status, disability and tenure — 2021 Census, Office for National Statistics; Scotland's Census 2022; Census 2021, NISRA.

- Indices of Multiple Deprivation — Ministry of Housing, Communities and Local Government, Scottish Government, Welsh Government, NISRA. Benefits — Department for Work and Pensions.
- Towns, villages and their populations — GeoNames (cities1000), with local authorities from its administrative divisions. Surname and given-name frequency — ONS baby names and published surname studies.

APPENDIX A

Algorithms

Every algorithm of the policy in alphabetical order, with its configuration. The one a data class names is the top of its tree; the others are building blocks.

UK_ADDRESS SECURE LOOKUP

Replaces the value with one of the 3,000 values in `uk-addresses.txt`, chosen by a keyed hash.

Used by: `UK_L1_ADDRESS`

CONFIGURATION

lookupFile	`uk-addresses.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_LOOKUP_FILE`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

UK_AGE REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `UK_L2_AGE`

PATTERNS, TRIED IN ORDER

`(0*?)(?:([1-9]\d{2,}))`	keep · replace with `90`
`(0*?)(?:([1-9])(\d)(\.\d+))`	keep · keep · `UK_CM_ALNUM` · keep
`(0*?)(?:([1-9])(\d))`	keep · keep · `UK_CM_ALNUM`
`(0*?)(?:(\d))`	keep · `UK_CM_ALNUM`
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

UK_BENEFIT REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_BENEFIT_LIST`.

Used by: `UK_FIN_BENEFIT`

PATTERNS, TRIED IN ORDER

`(?i)(yes|no|true|false|[yn01])`	`UK_FLAG`
`(0*?)(?:(\d{1,6}))`	keep · `UK_CM_ALNUM`
otherwise	`UK_BENEFIT_LIST`

CONFIGURATION

trimInput	yes
requireMask	no

UK_BENEFIT_LIST SECURE LOOKUP

Replaces the value with one of the 13 values in `uk-benefits.txt`, chosen by a keyed hash.

Used by: `UK_BENEFIT`

CONFIGURATION

`lookupFile`	`uk-benefits.txt`
`hashMethod`	`SHA256`
`maskedValueCase`	`PRESERVE_INPUT`
`inputCaseSensitive`	no
`trimWhitespaceFromInput`	yes
`trimWhitespaceInLookupFile`	yes

`UK_BIRTH_DATE` DATE SHIFT

Moves the date by -60 to 60 days.

Used by: `UK_L2_BIRTH_DATE`

CONFIGURATION

`minRange`	`-60`
`maxRange`	`60`
`unit`	`DAYS`
`roll`	no

`UK_CAPPED_COUNT` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `UK_L2_DEPENDANTS`

PATTERNS, TRIED IN ORDER

`(0*?)(?:([0-5]))`	keep · keep
`(0*?)(?:(\d+))`	keep · replace with `5`
`otherwise`	keep

CONFIGURATION

`trimInput`	yes
`requireMask`	no

`UK_CARD` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: `UK_L1_PAYMENT_CARD`

PATTERNS, TRIED IN ORDER

`(0*?)(?:(\d[\d\s\-]{11,22}\d))`	keep · `dlpx-core:Credit Card`
`otherwise`	keep

CONFIGURATION

`trimInput`	yes
`requireMask`	no

`UK_CHI_DAY` SECURE LOOKUP

Replaces the value with one of the 28 values in `uk-chi-day.txt`, chosen by a keyed hash.

Used by: `UK_NHS_BODY`

CONFIGURATION

lookupFile	`uk-chi-day.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

UK_CHI_MONTH SECURE LOOKUP

Replaces the value with one of the 12 values in `uk-chi-month.txt`, chosen by a keyed hash.

Used by: `UK_NHS_BODY`

CONFIGURATION

lookupFile	`uk-chi-month.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

UK_CM_ALNUM CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

Used by: `UK_L1_PASSPORT`, `UK_L1_DRIVING_LICENCE`, `UK_L1_IMMIGRATION_DOCUMENT`, `UK_L1_VITAL_RECORD`, `UK_L1_ADDRESS_DETAIL`, `UK_L1_BANK_ACCOUNT`, `UK_L1_VIN`, `UK_L1_PROPERTY`, `UK_L1_CONTRACT_NUMBER`, `UK_L1_USERNAME`, `UK_HEALTH_ID`, `UK_HEALTH_TEST_RESULT`, `UK_PENAL_CASE_NUMBER`, `UK_SUPPRESSED_CATEGORY`, `UK_NINO`, `UK_NHS_COLUMN`, `UK_UTR`, `UK_GIVEN_NAMES`, `UK_SURNAMES`, `UK_FULL_NAME`, `UK_EMAIL`, `UK_PHONE`, `UK_SORT_CODE`, `UK_IBAN`, `UK_COORDINATE`, `UK_YEAR`, `UK_AGE`, `UK_ETHNICITY`, `UK_RELIGION`, `UK_POLITICAL_OPINION`, `UK_POLITICAL_PARTY`, `UK_SEXUAL_ORIENTATION`, `UK_GENDER_IDENTITY`, `UK_DIAGNOSIS`, `UK_MEDICATION`, `UK_DISABILITY`, `UK_CREDIT`, `UK_BENEFIT`, `UK_DEPRIVATION`, `UK_HOUSING`, `UK_CRIMINAL_RECORD`, `UK_DBS`

CONFIGURATION

characterGroups	`0123456789` `AEIOU` `BCDFGHJKLMNPQRSTVWXYZ` `aeiou` `bcdfghjklmnpqrstvwxyz`
caseSensitive	yes
minMaskedPositions	`0`
preserveLeadingZeros	yes

UK_CM_DIGITS CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

Used by: `UK_NINO`, `UK_NHS_BODY`, `UK_UTR_CD`, `UK_SORT_CODE`

CONFIGURATION

characterGroups	`0123456789`
caseSensitive	yes
minMaskedPositions	`0`

UK_CM_HEX CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

Used by: `UK_L1_DEVICE_ID`, `UK_IP`

CONFIGURATION

characterGroups	`0123456789` `abcdef` `ABCDEF` `ghijklmnopqrstuvwxyz` `GHIJKLMNOPQRSTUVWXYZ`
caseSensitive	yes
minMaskedPositions	`0`
preserveLeadingZeros	yes

UK_CM_INWARD CHARACTER MAPPING

Maps each character to another of its group (3 groups), keeping length and format.

Used by: `UK_POSTCODE`

CONFIGURATION

characterGroups	`0123456789` `ABDEFGHJLNPQRSTUWXYZ` `abdefghjlnpqrstuwxyz`
caseSensitive	yes
minMaskedPositions	`0`
preserveLeadingZeros	no

UK_CM_PLATE CHARACTER MAPPING

Maps each character to another of its group (3 groups), keeping length and format.

Used by: `UK_PLATE`

CONFIGURATION

characterGroups	`0123456789` `ABCDEFGHJKLMNOPRSTUVWXYZ` `abcdefghjklmnoprstuvwxyz`
caseSensitive	yes
minMaskedPositions	`0`
preserveLeadingZeros	no

UK_COORDINATE REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `UK_L1_GEOLOCATION`

PATTERNS, TRIED IN ORDER

`(-?\d{1,3}\.\d)(\d+)\s*,\s*(-?\d{1,3}\.\d)(\d+)`	keep · `UK_CM_ALNUM` · keep · `UK_CM_ALNUM`
`(-?\d{1,3}\.\d)(\d+)`	keep · `UK_CM_ALNUM`
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

UK_CREDIT REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_CREDIT_STATUS`.

Used by: `UK_FIN_CREDIT_REFERENCE`

PATTERNS, TRIED IN ORDER

`(0*?)(?:(\d{2,4}))`	keep · `UK_CM_ALNUM`
`(?i)(yes|no|true|false|[yn01])`	`UK_FLAG`
otherwise	`UK_CREDIT_STATUS`

CONFIGURATION

trimInput	yes
requireMask	no

UK_CREDIT_STATUS SECURE LOOKUP

Replaces the value with one of the 13 values in `uk-credit-statuses.txt`, chosen by a keyed hash.

Used by: `UK_CREDIT`

CONFIGURATION

lookupFile	`uk-credit-statuses.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

UK_CRIMINAL_RECORD REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_CRIMINAL_RECORD_LIST`.

Used by: `UK_PENAL_CRIMINAL_RECORD`

PATTERNS, TRIED IN ORDER

`(?i)(yes|no|true|false|[yn01])`	`UK_FLAG`
`(0*?)(?:(\d{1,6}))`	keep · `UK_CM_ALNUM`
otherwise	`UK_CRIMINAL_RECORD_LIST`

CONFIGURATION

trimInput	yes
requireMask	no

UK_CRIMINAL_RECORD_LIST SECURE LOOKUP

Replaces the value with one of the 12 values in `uk-criminal-record.txt`, chosen by a keyed hash.

Used by: `UK_CRIMINAL_RECORD`

CONFIGURATION

lookupFile	`uk-criminal-record.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

UK_DBS REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_DBS_OUTCOME`.

Used by: `UK_PENAL_DBS`

PATTERNS, TRIED IN ORDER

`(0*?)(?:(\d{6,14}))`	keep · `UK_CM_ALNUM`
`(?i)(yes|no|true|false|[yn01])`	`UK_FLAG`

otherwise	`UK_DBS_OUTCOME`

CONFIGURATION

trimInput	yes
requireMask	no

UK_DBS_OUTCOME SECURE LOOKUP

Replaces the value with one of the 7 values in `uk-dbs-outcomes.txt`, chosen by a keyed hash.

Used by: `UK_DBS`

CONFIGURATION

lookupFile	`uk-dbs-outcomes.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

UK_DECILE SECURE LOOKUP

Replaces the value with one of the 10 values in `uk-decile.txt`, chosen by a keyed hash.

Used by: `UK_DEPRIVATION`

CONFIGURATION

lookupFile	`uk-decile.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

UK_DECILE_HIGH SECURE LOOKUP

Replaces the value with one of the 5 values in `uk-decile-high.txt`, chosen by a keyed hash.

Used by: `UK_DEPRIVATION`

CONFIGURATION

lookupFile	`uk-decile-high.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

UK_DEPRIVATION REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: `UK_DEPRIVATION_STATUS`.

Used by: `UK_FIN_DEPRIVATION`

PATTERNS, TRIED IN ORDER

`(0*?)(?:([1-5]))`	keep · `UK_QUINTILE`

`(0*?)(?:([6-9]|10))`	keep · `UK_DECILE_HIGH`
`(?i)(decile\s*)(\d{1,2})`	keep · `UK_DECILE`
`(?i)(quintile\s*)([1-5])`	keep · `UK_QUINTILE`
`(0*?)(?:(\d{2,5}))`	keep · `UK_CM_ALNUM`
otherwise	`UK_DEPRIVATION_STATUS`

CONFIGURATION

trimInput	yes
requireMask	no

`UK_DEPRIVATION_STATUS` SECURE LOOKUP

Replaces the value with one of the 5 values in `uk-deprivation-statuses.txt`, chosen by a keyed hash.

Used by: `UK_DEPRIVATION`

CONFIGURATION

lookupFile	`uk-deprivation-statuses.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_LOOKUP_FILE`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

`UK_DIAGNOSIS` REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `UK_DIAGNOSIS_TEXT`.

Used by: `UK_HEALTH_DIAGNOSIS`

PATTERNS, TRIED IN ORDER

`([A-TV-Za-tv-z]\d{2}\.\d{1,2})`	`UK_ICD10_DECIMAL`
`([A-TV-Za-tv-z]\d{2,3})`	`UK_ICD10`
`(0*?)(?:(\d{6,18}))`	keep · `UK_CM_ALNUM`
`(?i)(yes|no|true|false|[yn01])`	`UK_FLAG`
otherwise	`UK_DIAGNOSIS_TEXT`

CONFIGURATION

trimInput	yes
requireMask	no

`UK_DIAGNOSIS_TEXT` SECURE LOOKUP

Replaces the value with one of the 30 values in `uk-diagnoses.txt`, chosen by a keyed hash.

Used by: `UK_DIAGNOSIS`

CONFIGURATION

lookupFile	`uk-diagnoses.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

UK_DISABILITY REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_DISABILITY_LIST`.

Used by: `UK_HEALTH_DISABILITY`

PATTERNS, TRIED IN ORDER

`(?i)(yes|no|true|false|[yn01])`	`UK_FLAG`
`(0*?)(?:(\d{1,6}))`	keep · `UK_CM_ALNUM`
otherwise	`UK_DISABILITY_LIST`

CONFIGURATION

trimInput	yes
requireMask	no

UK_DISABILITY_LIST SECURE LOOKUP

Replaces the value with one of the 10 values in `uk-disabilities.txt`, chosen by a keyed hash.

Used by: `UK_DISABILITY`

CONFIGURATION

lookupFile	`uk-disabilities.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

UK_EDUCATION SECURE LOOKUP

Replaces the value with one of the 11 values in `uk-education.txt`, chosen by a keyed hash.

Used by: `UK_L2_EDUCATION`

CONFIGURATION

lookupFile	`uk-education.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

UK_EMAIL REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `UK_CM_ALNUM`.

Used by: `UK_L1_EMAIL`

PATTERNS, TRIED IN ORDER

`([^@\s]+)@([^@\s]+)\.([A-Za-z]{2,24})`	`UK_CM_ALNUM` · `UK_CM_ALNUM` · replace with `test`
otherwise	`UK_CM_ALNUM`

CONFIGURATION

trimInput	yes

requireMask	no

UK_EMPLOYER SECURE LOOKUP

Replaces the value with one of the 25 values in `uk-employers.txt`, chosen by a keyed hash.

Used by: `UK_L2_EMPLOYER`

CONFIGURATION

lookupFile	`uk-employers.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

UK_ETHNICITY REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_ETHNICITY_LIST`.

Used by: `UK_L3_ETHNICITY`

PATTERNS, TRIED IN ORDER

`(?i)(yes|no|true|false|[yn01])`	`UK_FLAG`
`(0*?)(?:(\d{1,6}))`	keep · `UK_CM_ALNUM`
otherwise	`UK_ETHNICITY_LIST`

CONFIGURATION

trimInput	yes
requireMask	no

UK_ETHNICITY_LIST SECURE LOOKUP

Replaces the value with one of the 20 values in `uk-ethnic-groups.txt`, chosen by a keyed hash.

Used by: `UK_ETHNICITY`

CONFIGURATION

lookupFile	`uk-ethnic-groups.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

UK_EVENT_DATE DATE SHIFT

Moves the date by -30 to 30 days.

Used by: `UK_L2_EVENT_DATE`

CONFIGURATION

minRange	`-30`
maxRange	`30`
unit	`DAYS`
roll	no

UK_FLAG REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: UK_SUPPRESSED_CATEGORY, UK_ETHNICITY, UK_RELIGION, UK_POLITICAL_OPINION, UK_POLITICAL_PARTY, UK_SEXUAL_ORIENTATION, UK_GENDER_IDENTITY, UK_DIAGNOSIS, UK_MEDICATION, UK_DISABILITY, UK_CREDIT, UK_BENEFIT, UK_HOUSING, UK_CRIMINAL_RECORD, UK_DBS

PATTERNS, TRIED IN ORDER

(?i)(yes|no)	UK_FLAG_YES_NO
(?i)(true|false)	UK_FLAG_TRUE_FALSE
(?i)([yn])	UK_FLAG_Y_N
([01])	UK_FLAG_0_1
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

UK_FLAG_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in uk-flag-0-1.txt, chosen by a keyed hash.

Used by: UK_FLAG

CONFIGURATION

lookupFile	uk-flag-0-1.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

UK_FLAG_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in uk-flag-true-false.txt, chosen by a keyed hash.

Used by: UK_FLAG

CONFIGURATION

lookupFile	uk-flag-true-false.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

UK_FLAG_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in uk-flag-y-n.txt, chosen by a keyed hash.

Used by: UK_FLAG

CONFIGURATION

lookupFile	uk-flag-y-n.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT

inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

UK_FLAG_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `uk-flag-yes-no.txt`, chosen by a keyed hash.

Used by: `UK_FLAG`

CONFIGURATION

lookupFile	`uk-flag-yes-no.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

UK_FREE_TEXT FREE TEXT REDACTION

Replaces the text matching 10 patterns with `[DATA]`, and the 1,266 terms of `uk-text-names.txt` with `[NAME]`.

Used by: `UK_FREE_TEXT_ALL`

PATTERNS, TRIED IN ORDER

`(?<![A-Za-z])[A-CEGHJ-PR-TW-Z][A-CEGHJ-NPR-TW-Z]\s?\d{2}\s?\d{2}\s?\d{2}\s?[A-D]?(?![A-Za-z\d])`

`(?<![\d\-])\d{3}[\s\-]?\d{3}[\s\-]?\d{4}(?![\d\-])`

`(?<!\d)\d{5}\s?\d{5}(?!\d)`

`[\w.+\-]+@[\w\-]+(\.[\w\-]+)+`

`(?<![\d\-])(\+?44\s?(\(0\)\s?)?|\(?0)[1-35789]\d{1,3}\)?[\s\-.]?\d{3,4}[\s\-.]?\d{3,4}(?![\d\-])`

`(?<!\d)\d{13,19}(?!\d)`

`(?i)(?<![A-Za-z\d])GB\d{2}\s?[A-Z]{4}(\s?\d{4}){3}\s?\d{2}(?![A-Za-z\d])`

`(?<![A-Za-z\d])[A-Z]{1,2}\d[A-Z\d]?\s?\d[ABD-HJLNP-UW-Z]{2}(?![A-Za-z\d])`

`(?<![A-Za-z\d])[A-Z]{2}\d{2}\s?[A-Z]{3}(?![A-Za-z\d])`

`(?<!\d)(\d{1,3}\.){3}\d{1,3}(?!\d)`

CONFIGURATION

regExRedactValue	`[DATA]`
lookupFile	`uk-text-names.txt`
lookupFileRedactValue	`[NAME]`
isDenyList	yes

UK_FREE_TEXT_ALL STRING ALGORITHM CHAIN

Applies `UK_TEXT_SPACED` → `UK_FREE_TEXT`, in this order.

Used by: `UK_L1_FREE_TEXT`, `UK_HEALTH_CLINICAL_TEXT`

CONFIGURATION

algorithmReferences[1]	`UK_TEXT_SPACED`
algorithmReferences[2]	`UK_FREE_TEXT`

UK_FULL_NAME REGEX DECOMPOSE

Tries 9 patterns in order and gives each part of the value its own action; when none matches: `UK_CM_ALNUM`.

Used by: `UK_L1_FULL_NAME`

PATTERNS, TRIED IN ORDER

`([^,]+),\s*(.+)`	`UK_SURNAMES` · `UK_GIVEN_NAMES`
`(?i)((?:Mr|Mrs|Ms|Miss|Mx|Dr|Prof|Sir|Dame|Rev|Revd|Lady|Lord)\.?\s+)(\S+)`	keep · `UK_SURNAMES`
`(?i)((?:Mr|Mrs|Ms|Miss|Mx|Dr|Prof|Sir|Dame|Rev|Revd|Lady|Lord)\.?\s+)(\S+)\s+(\S+)`	keep · `UK_GIVEN_NAME` · `UK_SURNAMES`
`(?i)((?:Mr|Mrs|Ms|Miss|Mx|Dr|Prof|Sir|Dame|Rev|Revd|Lady|Lord)\.?\s+)(\S+)\s+(\S+)\s+(\S+)`	keep · `UK_GIVEN_NAME` · `UK_GIVEN_NAME` · `UK_SURNAMES`
`(\S+)`	`UK_GIVEN_NAME`
`(\S+)\s+(\S+)`	`UK_GIVEN_NAME` · `UK_SURNAMES`
`(\S+)(\s+)([A-Za-z]\.?)(\s+)(\S+)`	`UK_GIVEN_NAME` · keep · keep · keep · `UK_SURNAMES`
`(\S+)\s+(\S+)\s+(\S+)`	`UK_GIVEN_NAME` · `UK_GIVEN_NAME` · `UK_SURNAMES`
`(\S+)\s+(\S+)\s+(\S+)\s+(\S+)`	`UK_GIVEN_NAME` · `UK_GIVEN_NAME` · `UK_GIVEN_NAME` · `UK_SURNAMES`
otherwise	`UK_CM_ALNUM`

CONFIGURATION

trimInput	yes
requireMask	no

`UK_GENDER_IDENTITY` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_GENDER_IDENTITY_LIST`.

Used by: `UK_L3_GENDER_IDENTITY`

PATTERNS, TRIED IN ORDER

`(?i)(yes|no|true|false|[yn01])`	`UK_FLAG`
`(0*?)(?:(\d{1,6}))`	keep · `UK_CM_ALNUM`
otherwise	`UK_GENDER_IDENTITY_LIST`

CONFIGURATION

trimInput	yes
requireMask	no

`UK_GENDER_IDENTITY_LIST` SECURE LOOKUP

Replaces the value with one of the 6 values in `uk-gender-identities.txt`, chosen by a keyed hash.

Used by: `UK_GENDER_IDENTITY`

CONFIGURATION

lookupFile	`uk-gender-identities.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

UK_GIVEN_NAME NAME

Replaces the name with one of the 349 names in `uk-given-names.txt`.

Used by: `UK_GIVEN_NAMES`, `UK_FULL_NAME`

CONFIGURATION

lookupFile	`uk-given-names.txt`
maskedValueCase	`PRESERVE_INPUT`
filterAccent	yes
inputCaseSensitive	no

UK_GIVEN_NAMES REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: `UK_CM_ALNUM`.

Used by: `UK_L1_GIVEN_NAME`, `UK_FULL_NAME`

PATTERNS, TRIED IN ORDER

`(\S+)`	`UK_GIVEN_NAME`
`(\S+)\s+(\S+)`	`UK_GIVEN_NAME` · `UK_GIVEN_NAME`
`(\S+)\s+(\S+)\s+(\S+)`	`UK_GIVEN_NAME` · `UK_GIVEN_NAME` · `UK_GIVEN_NAME`
otherwise	`UK_CM_ALNUM`

CONFIGURATION

trimInput	yes
requireMask	no

UK_HOUSING REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_HOUSING_LIST`.

Used by: `UK_FIN_HOUSING`

PATTERNS, TRIED IN ORDER

`(?i)(yes|no|true|false|[yn01])`	`UK_FLAG`
`(0*?)(?:(\d{1,6}))`	keep · `UK_CM_ALNUM`
otherwise	`UK_HOUSING_LIST`

CONFIGURATION

trimInput	yes
requireMask	no

UK_HOUSING_LIST SECURE LOOKUP

Replaces the value with one of the 9 values in `uk-housing-tenure.txt`, chosen by a keyed hash.

Used by: `UK_HOUSING`

CONFIGURATION

lookupFile	`uk-housing-tenure.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

UK_IBAN REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_CM_ALNUM`.

Used by: `UK_L1_IBAN`

PATTERNS, TRIED IN ORDER

`(?i)(GB\d{2}[A-Z]{4}\d{14})`	`UK_IBAN_COMPACT`
`(?i)(GB\d{2} [A-Z]{4} \d{4} \d{4} \d{4} \d{2})`	`UK_IBAN_PRINTED`
otherwise	`UK_CM_ALNUM`

CONFIGURATION

trimInput	yes
requireMask	no

UK_IBAN_COMPACT IBAN

Masks the last 14 characters of the IBAN and recomputes its two ISO 13616 check digits; the country code and the characters before stay.

Used by: `UK_IBAN`

CONFIGURATION

validateInput	no
numCharsToMask	`14`

UK_IBAN_PRINTED IBAN

Masks the last 17 characters of the IBAN and recomputes its two ISO 13616 check digits; the country code and the characters before stay.

Used by: `UK_IBAN`

CONFIGURATION

validateInput	no
numCharsToMask	`17`

UK_ICD10 SECURE LOOKUP

Replaces the value with one of the 37 values in `uk-icd10.txt`, chosen by a keyed hash.

Used by: `UK_DIAGNOSIS`

CONFIGURATION

lookupFile	`uk-icd10.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_LOOKUP_FILE`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

UK_ICD10_DECIMAL SECURE LOOKUP

Replaces the value with one of the 33 values in `uk-icd10-decimal.txt`, chosen by a keyed hash.

Used by: `UK_DIAGNOSIS`

CONFIGURATION

lookupFile	`uk-icd10-decimal.txt`

hashMethod	SHA256
maskedValueCase	PRESERVE_LOOKUP_FILE
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

UK_IDENTITY REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: UK_NHS_AUX, UK_NHS_RECOMPUTE, UK_UTR_AUX

PATTERNS, TRIED IN ORDER

(.*)	keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

UK_IP REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: UK_CM_HEX.

Used by: UK_L1_IP_ADDRESS

PATTERNS, TRIED IN ORDER

(0*?)(?:(\d{1,3})\.(\d{1,3})\.(\d{1,3})\.(\d{1,3}))	keep · UK_OCTET · UK_OCTET · UK_OCTET · UK_OCTET
otherwise	UK_CM_HEX

CONFIGURATION

trimInput	yes
requireMask	no

UK_MARITAL_STATUS SECURE LOOKUP

Replaces the value with one of the 9 values in uk-marital-status.txt, chosen by a keyed hash.

Used by: UK_L2_MARITAL_STATUS

CONFIGURATION

lookupFile	uk-marital-status.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

UK_MEDICATION REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: UK_MEDICATION_LIST.

Used by: UK_HEALTH_MEDICATION

PATTERNS, TRIED IN ORDER

(?i)(yes|no|true|false|[yn01])	UK_FLAG

`(0*?)(?:(\d{1,6}))`	keep · `UK_CM_ALNUM`
otherwise	`UK_MEDICATION_LIST`

CONFIGURATION

trimInput	yes
requireMask	no

UK_MEDICATION_LIST SECURE LOOKUP

Replaces the value with one of the 26 values in `uk-medications.txt`, chosen by a keyed hash.

Used by: `UK_MEDICATION`

CONFIGURATION

lookupFile	`uk-medications.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

UK_NATIONALITY SECURE LOOKUP

Replaces the value with one of the 21 values in `uk-nationalities.txt`, chosen by a keyed hash.

Used by: `UK_L2_NATIONALITY`

CONFIGURATION

lookupFile	`uk-nationalities.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

UK_NHS_AUX CHECK DIGIT

Masks the digits with `UK_IDENTITY` and recomputes a modulus 11 check digit with weights `0 4 6 8 10 1 3 5 7 9` from the right.

Used by: `UK_NHS_NUMBER`

CONFIGURATION

weightList	`0` `4` `6` `8` `10` `1` `3` `5` `7` `9`
modulusNumber	`11`
checkDigitIndex	`10`
numDigitsForCheckdigitCalculation	`10`
calculateChecksumRightToLeft	yes
numericAlgorithm	`UK_IDENTITY`
alphaNumericAlgorithm	`UK_IDENTITY`
preserveRegex	`[.\-\s]`
inputHandlingConfig.characterHandling	`STANDARD`
inputHandlingConfig.invalidInputHandling	`ERROR`
inputHandlingConfig.shortInputHandling	`FALLBACK`

inputHandlingConfig.padCharacter	0
inputHandlingConfig.trimWhitespace	yes

UK_NHS_BODY REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: UK_CM_DIGITS.

Used by: UK_NHS_CD

PATTERNS, TRIED IN ORDER

(0[1–9]|[12]\d|3[01])(0[1–9]|1[0–2])(\d{5})	UK_CHI_DAY · UK_CHI_MONTH · UK_CM_DIGITS
(\d)(\d{8})	keep · UK_CM_DIGITS
otherwise	UK_CM_DIGITS

CONFIGURATION

trimInput	yes
requireMask	no

UK_NHS_CD CHECK DIGIT

Masks the digits with UK_NHS_BODY and recomputes a modulus 11 check digit with weights 2 3 4 5 6 7 8 9 10 from the right.

Used by: UK_NHS_NUMBER

CONFIGURATION

weightList	2 3 4 5 6 7 8 9 10
modulusNumber	11
checkDigitIndex	9
numDigitsForCheckdigitCalculation	9
calculateChecksumRightToLeft	yes
numericAlgorithm	UK_NHS_BODY
alphaNumericAlgorithm	UK_NHS_BODY
preserveRegex	[.\–\s]
inputHandlingConfig.characterHandling	STANDARD
inputHandlingConfig.invalidInputHandling	ERROR
inputHandlingConfig.shortInputHandling	FALLBACK
inputHandlingConfig.padCharacter	0
inputHandlingConfig.trimWhitespace	yes

UK_NHS_COLUMN REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: UK_CM_ALNUM.

Used by: UK_L1_NHS_NUMBER

PATTERNS, TRIED IN ORDER

(0*?)(?:(\d{3}[\s\–]?\d{3}[\s\–]?\d{4}))	keep · UK_NHS_NUMBER
otherwise	UK_CM_ALNUM

CONFIGURATION

trimInput	yes
requireMask	no

UK_NHS_NUMBER STRING ALGORITHM CHAIN

Applies `UK_NHS_CD` → `UK_NHS_WIDEN` → `UK_NHS_AUX` → `UK_NHS_RESOLVE` → `UK_NHS_REPAIR` → `UK_NHS_RECOMPUTE`, in this order.

Used by: `UK_NHS_COLUMN`

CONFIGURATION

algorithmReferences[1]	`UK_NHS_CD`
algorithmReferences[2]	`UK_NHS_WIDEN`
algorithmReferences[3]	`UK_NHS_AUX`
algorithmReferences[4]	`UK_NHS_RESOLVE`
algorithmReferences[5]	`UK_NHS_REPAIR`
algorithmReferences[6]	`UK_NHS_RECOMPUTE`

UK_NHS_RECOMPUTE CHECK DIGIT

Masks the digits with `UK_IDENTITY` and recomputes a modulus 11 check digit with weights `2 3 4 5 6 7 8 9 10` from the right.

Used by: `UK_NHS_NUMBER`

CONFIGURATION

weightList	`2 3 4 5 6 7 8 9 10`
modulusNumber	`11`
checkDigitIndex	`9`
numDigitsForCheckdigitCalculation	`9`
calculateChecksumRightToLeft	yes
numericAlgorithm	`UK_IDENTITY`
alphaNumericAlgorithm	`UK_IDENTITY`
preserveRegex	`[.\-\s]`
inputHandlingConfig.characterHandling	`STANDARD`
inputHandlingConfig.invalidInputHandling	`ERROR`
inputHandlingConfig.shortInputHandling	`FALLBACK`
inputHandlingConfig.padCharacter	`0`
inputHandlingConfig.trimWhitespace	yes

UK_NHS_REPAIR REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: `UK_NHS_NUMBER`

PATTERNS, TRIED IN ORDER

`(\d{3}[\s\-]?\d{3}[\s\-]?\d{3})(Z)`	`UK_NHS_STEP` · replace with `0`
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

UK_NHS_RESOLVE REGEX DECOMPOSE

Tries 11 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `UK_NHS_NUMBER`

PATTERNS, TRIED IN ORDER

`(\d{3}[\s\-]?\d{3}[\s\-]?\d{3})(00)`	keep · replace with `0`
`(\d{3}[\s\-]?\d{3}[\s\-]?\d{3})(0\d)`	keep · replace with `Z`
`(\d{3}[\s\-]?\d{3}[\s\-]?\d{3})(1\d)`	keep · replace with `1`
`(\d{3}[\s\-]?\d{3}[\s\-]?\d{3})(2\d)`	keep · replace with `2`
`(\d{3}[\s\-]?\d{3}[\s\-]?\d{3})(3\d)`	keep · replace with `3`
`(\d{3}[\s\-]?\d{3}[\s\-]?\d{3})(4\d)`	keep · replace with `4`
`(\d{3}[\s\-]?\d{3}[\s\-]?\d{3})(5\d)`	keep · replace with `5`
`(\d{3}[\s\-]?\d{3}[\s\-]?\d{3})(6\d)`	keep · replace with `6`
`(\d{3}[\s\-]?\d{3}[\s\-]?\d{3})(7\d)`	keep · replace with `7`
`(\d{3}[\s\-]?\d{3}[\s\-]?\d{3})(8\d)`	keep · replace with `8`
`(\d{3}[\s\-]?\d{3}[\s\-]?\d{3})(9\d)`	keep · replace with `9`
`otherwise`	keep

CONFIGURATION

`trimInput`	yes
`requireMask`	no

UK_NHS_STEP REGEX DECOMPOSE

Tries 10 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `UK_NHS_REPAIR`

PATTERNS, TRIED IN ORDER

`(\d{3}[\s\-]?\d{3}[\s\-]?\d{2})(0)`	keep · replace with `1`
`(\d{3}[\s\-]?\d{3}[\s\-]?\d{2})(1)`	keep · replace with `2`
`(\d{3}[\s\-]?\d{3}[\s\-]?\d{2})(2)`	keep · replace with `3`
`(\d{3}[\s\-]?\d{3}[\s\-]?\d{2})(3)`	keep · replace with `4`
`(\d{3}[\s\-]?\d{3}[\s\-]?\d{2})(4)`	keep · replace with `5`
`(\d{3}[\s\-]?\d{3}[\s\-]?\d{2})(5)`	keep · replace with `6`
`(\d{3}[\s\-]?\d{3}[\s\-]?\d{2})(6)`	keep · replace with `7`
`(\d{3}[\s\-]?\d{3}[\s\-]?\d{2})(7)`	keep · replace with `8`
`(\d{3}[\s\-]?\d{3}[\s\-]?\d{2})(8)`	keep · replace with `9`
`(\d{3}[\s\-]?\d{3}[\s\-]?\d{2})(9)`	keep · replace with `0`
`otherwise`	keep

CONFIGURATION

`trimInput`	yes
`requireMask`	no

UK_NHS_WIDEN REGEX DECOMPOSE

Tries 10 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `UK_NHS_NUMBER`

PATTERNS, TRIED IN ORDER

`(\d{3}[\s\-]?\d{3}[\s\-]?\d{3})(0)`	keep · replace with `00`
`(\d{3}[\s\-]?\d{3}[\s\-]?\d{3})(1)`	keep · replace with `10`
`(\d{3}[\s\-]?\d{3}[\s\-]?\d{3})(2)`	keep · replace with `20`
`(\d{3}[\s\-]?\d{3}[\s\-]?\d{3})(3)`	keep · replace with `30`

`(\d{3}[\s\-]?\d{3}[\s\-]?\d{3})(4)`	keep · replace with `40`
`(\d{3}[\s\-]?\d{3}[\s\-]?\d{3})(5)`	keep · replace with `50`
`(\d{3}[\s\-]?\d{3}[\s\-]?\d{3})(6)`	keep · replace with `60`
`(\d{3}[\s\-]?\d{3}[\s\-]?\d{3})(7)`	keep · replace with `70`
`(\d{3}[\s\-]?\d{3}[\s\-]?\d{3})(8)`	keep · replace with `80`
`(\d{3}[\s\-]?\d{3}[\s\-]?\d{3})(9)`	keep · replace with `90`
`otherwise`	keep

CONFIGURATION

`trimInput`	yes
`requireMask`	no

UK_NINO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `UK_CM_ALNUM`.

Used by: `UK_L1_NINO`

PATTERNS, TRIED IN ORDER

`(?i)([a-z]{2})(\s?\d{2}\s?\d{2}\s?\d{2}\s?)([a-d]?)`	keep · `UK_CM_DIGITS` · keep
`otherwise`	`UK_CM_ALNUM`

CONFIGURATION

`trimInput`	yes
`requireMask`	no

UK_OCCUPATION SECURE LOOKUP

Replaces the value with one of the 40 values in `uk-occupations.txt`, chosen by a keyed hash.

Used by: `UK_L2_OCCUPATION`

CONFIGURATION

`lookupFile`	`uk-occupations.txt`
`hashMethod`	`SHA256`
`maskedValueCase`	`PRESERVE_INPUT`
`inputCaseSensitive`	no
`trimWhitespaceFromInput`	yes
`trimWhitespaceInLookupFile`	yes

UK_OCTET NUMERIC MAPPING

Maps the number to another between 1 and 254.

Used by: `UK_IP`

CONFIGURATION

`minValue`	`1`
`maxValue`	`254`

UK_PHONE REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_CM_ALNUM`.

Used by: `UK_L1_PHONE`

PATTERNS, TRIED IN ORDER

`(0*?)(?:(\+?44\s?(?:\(0\)\s?)?|\(?0)(\d{2})((?:[\s\-.)]*\d){7,8}))`	keep · keep · keep · `UK_CM_ALNUM`
`(0*?)(?:(\d{3})([\s\-.]?)(\d{4}))`	keep · `UK_CM_ALNUM` · keep · `UK_CM_ALNUM`
otherwise	`UK_CM_ALNUM`

CONFIGURATION

trimInput	yes
requireMask	no

UK_PLATE REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: `UK_CM_PLATE`.

Used by: `UK_L1_LICENCE_PLATE`

PATTERNS, TRIED IN ORDER

`(?i)([a-z]{2}\d{2})(\s?)([a-z]{3})`	keep · keep · `UK_CM_PLATE`
`(?i)([a-z])(\d{1,3})(\s?)([a-z]{3})`	keep · `UK_CM_PLATE` · keep · `UK_CM_PLATE`
`(?i)([a-z]{3})(\s?)(\d{1,3})([a-z])`	`UK_CM_PLATE` · keep · `UK_CM_PLATE` · keep
otherwise	`UK_CM_PLATE`

CONFIGURATION

trimInput	yes
requireMask	no

UK_POLITICAL_OPINION REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_POLITICAL_OPINION_LIST`.

Used by: `UK_L3_POLITICAL_OPINION`

PATTERNS, TRIED IN ORDER

`(?i)(yes|no|true|false|[yn01])`	`UK_FLAG`
`(0*?)(?:(\d{1,6}))`	keep · `UK_CM_ALNUM`
otherwise	`UK_POLITICAL_OPINION_LIST`

CONFIGURATION

trimInput	yes
requireMask	no

UK_POLITICAL_OPINION_LIST SECURE LOOKUP

Replaces the value with one of the 7 values in `uk-political-opinions.txt`, chosen by a keyed hash.

Used by: `UK_POLITICAL_OPINION`

CONFIGURATION

lookupFile	`uk-political-opinions.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

UK_POLITICAL_PARTY REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_POLITICAL_PARTY_LIST`.

Used by: `UK_L3_POLITICAL_AFFILIATION`

PATTERNS, TRIED IN ORDER

`(?i)(yes|no|true|false|[yn01])`	`UK_FLAG`
`(0*?)(?:(\d{1,6}))`	keep · `UK_CM_ALNUM`
otherwise	`UK_POLITICAL_PARTY_LIST`

CONFIGURATION

trimInput	yes
requireMask	no

UK_POLITICAL_PARTY_LIST SECURE LOOKUP

Replaces the value with one of the 14 values in `uk-political-parties.txt`, chosen by a keyed hash.

Used by: `UK_POLITICAL_PARTY`

CONFIGURATION

lookupFile	`uk-political-parties.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

UK_POSTCODE REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: `UK_L2_POSTCODE`

PATTERNS, TRIED IN ORDER

`(?i)([a-z]{1,2}\d[a-z\d]?)(\s*)(\d[a-z]{2})`	keep · keep · `UK_CM_INWARD`
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

UK_QUINTILE SECURE LOOKUP

Replaces the value with one of the 5 values in `uk-quintile.txt`, chosen by a keyed hash.

Used by: `UK_DEPRIVATION`

CONFIGURATION

lookupFile	`uk-quintile.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes

trimWhitespaceInLookupFile	yes

UK_REDACT REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: UK_L1_CREDENTIAL , UK_L3_GENETIC , UK_L3_BIOMETRIC

PATTERNS, TRIED IN ORDER

(.+)	each character → X
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

UK_RELIGION REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: UK_RELIGION_LIST .

Used by: UK_L3_RELIGION

PATTERNS, TRIED IN ORDER

(?i)(yes|no|true|false|[yn01])	UK_FLAG
(0*?)(?:(\d{1,6}))	keep · UK_CM_ALNUM
otherwise	UK_RELIGION_LIST

CONFIGURATION

trimInput	yes
requireMask	no

UK_RELIGION_LIST SECURE LOOKUP

Replaces the value with one of the 18 values in uk-religions.txt , chosen by a keyed hash.

Used by: UK_RELIGION

CONFIGURATION

lookupFile	uk-religions.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

UK_SCHOOL SECURE LOOKUP

Replaces the value with one of the 15 values in uk-schools.txt , chosen by a keyed hash.

Used by: UK_L2_SCHOOL

CONFIGURATION

lookupFile	uk-schools.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes

trimWhitespaceInLookupFile	yes

UK_SEX REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: UK_L2_SEX

PATTERNS, TRIED IN ORDER

(?i)(female|male)	UK_SEX_FEMALE_MALE
(?i)(woman|man)	UK_SEX_WOMAN_MAN
(?i)([fm])	UK_SEX_F_M
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

UK_SEX_F_M SECURE LOOKUP

Replaces the value with one of the 2 values in uk-sex-f-m.txt, chosen by a keyed hash.

Used by: UK_SEX

CONFIGURATION

lookupFile	uk-sex-f-m.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

UK_SEX_FEMALE_MALE SECURE LOOKUP

Replaces the value with one of the 2 values in uk-sex-female-male.txt, chosen by a keyed hash.

Used by: UK_SEX

CONFIGURATION

lookupFile	uk-sex-female-male.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

UK_SEX_WOMAN_MAN SECURE LOOKUP

Replaces the value with one of the 2 values in uk-sex-woman-man.txt, chosen by a keyed hash.

Used by: UK_SEX

CONFIGURATION

lookupFile	uk-sex-woman-man.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no

trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

UK_SEXUAL_ORIENTATION REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_SEXUAL_ORIENTATION_LIST`.

Used by: `UK_L3_SEXUAL_ORIENTATION`

PATTERNS, TRIED IN ORDER

`(?i)(yes|no|true|false|[yn01])`	`UK_FLAG`
`(0*?)(?:(\d{1,6}))`	keep · `UK_CM_ALNUM`
otherwise	`UK_SEXUAL_ORIENTATION_LIST`

CONFIGURATION

trimInput	yes
requireMask	no

UK_SEXUAL_ORIENTATION_LIST SECURE LOOKUP

Replaces the value with one of the 8 values in `uk-sexual-orientations.txt`, chosen by a keyed hash.

Used by: `UK_SEXUAL_ORIENTATION`

CONFIGURATION

lookupFile	`uk-sexual-orientations.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_INPUT`
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

UK_SORT_CODE REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `UK_CM_ALNUM`.

Used by: `UK_L1_SORT_CODE`

PATTERNS, TRIED IN ORDER

`(0*?)(?:(\d{2})([\s\-]?\d{2}[\s\-]?\d{2}))`	keep · keep · `UK_CM_DIGITS`
otherwise	`UK_CM_ALNUM`

CONFIGURATION

trimInput	yes
requireMask	no

UK_SUPPRESS SECURE LOOKUP

Replaces the value with one of the 1 values in `uk-not-stated.txt`, chosen by a keyed hash.

Used by: `UK_SUPPRESSED_CATEGORY`

CONFIGURATION

lookupFile	`uk-not-stated.txt`
hashMethod	`SHA256`
maskedValueCase	`PRESERVE_LOOKUP_FILE`

inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

UK_SUPPRESSED_CATEGORY REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `UK_SUPPRESS`.

Used by: `UK_L3_BELIEF`, `UK_L3_TRADE_UNION`, `UK_L3_SEXUAL_LIFE`, `UK_HEALTH_REPRODUCTIVE`, `UK_HEALTH_MENTAL`

PATTERNS, TRIED IN ORDER

`(?i)(yes|no|true|false|[yn01])`	`UK_FLAG`
`(0*?)(?:(0*?)(?:(0*?)(?:(0*?)(?:(0*?)(?:(\d{1,6})))))))`	keep · keep · keep · keep · keep · `UK_CM_ALNUM`
otherwise	`UK_SUPPRESS`

CONFIGURATION

trimInput	yes
requireMask	no

UK_SURNAME NAME

Replaces the name with one of the 294 names in `uk-surnames.txt`.

Used by: `UK_SURNAMES`

CONFIGURATION

lookupFile	`uk-surnames.txt`
maskedValueCase	`PRESERVE_INPUT`
filterAccent	yes
inputCaseSensitive	no

UK_SURNAMES REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `UK_CM_ALNUM`.

Used by: `UK_L1_SURNAME`, `UK_FULL_NAME`

PATTERNS, TRIED IN ORDER

`([^\s\-]+)(\-)([^\s\-]+)`	`UK_SURNAME` · keep · `UK_SURNAME`
`(\S+)`	`UK_SURNAME`
`(\S+)\s+(\S+)`	`UK_SURNAME` · `UK_SURNAME`
`(\S+)\s+(\S+)\s+(\S+)`	`UK_SURNAME` · `UK_SURNAME` · `UK_SURNAME`
otherwise	`UK_CM_ALNUM`

CONFIGURATION

trimInput	yes
requireMask	no

UK_TEXT_SPACED REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `UK_FREE_TEXT_ALL`

PATTERNS, TRIED IN ORDER

`([\s\S]*?)((?<![A-Za-z\d])(?:[A-CEGHJ-PR-TW-Z][A-CEGHJ-NPR-TW-Z] ?\d{2} \d{2} ?\d{2} ?[A-D]?|\d{3} \d{3} \d{4}|\d{5} \d{5}|(?:\+44 ?(?:\(0\) ?)?|\(?0)\d{2,4}\)? \d{3,4} ?\d{3,4}|GB\d{2} [A-Z]{4}(?: \d{4}){3} \d{2}|[A-Z]{1,2}\d[A-Z\d]? \d[ABD-HJLNP-UW-Z]{2}|[A-Z]{2}\d{2} [A-Z]{3})(?![A-Za-z\d]))([\s\S]*?)((?<![A-Za-z\d])(?:[A-CEGHJ-PR-TW-Z][A-CEGHJ-NPR-TW-Z] ?\d{2} \d{2} ?\d{2} ?[A-D]?|\d{3} \d{3} \d{4}|\d{5} \d{5}|(?:\+44 ?(?:\(0\) ?)?|\(?0)\d{2,4}\)? \d{3,4} ?\d{3,4}|GB\d{2} [A-Z]{4}(?: \d{4}){3} \d{2}|[A-Z]{1,2}\d[A-Z\d]? \d[ABD-HJLNP-UW-Z]{2}|[A-Z]{2}\d{2} [A-Z]{3})(?![A-Za-z\d]))([\s\S]*?)((?<![A-Za-z\d])(?:[A-CEGHJ-PR-TW-Z][A-CEGHJ-NPR-TW-Z] ?\d{2} \d{2} ?\d{2} ?[A-D]?|\d{3} \d{3} \d{4}|\d{5} \d{5}|(?:\+44 ?(?:\(0\) ?)?|\(?0)\d{2,4}\)? \d{3,4} ?\d{3,4}|GB\d{2} [A-Z]{4}(?: \d{4}){3} \d{2}|[A-Z]{1,2}\d[A-Z\d]? \d[ABD-HJLNP-UW-Z]{2}|[A-Z]{2}\d{2} [A-Z]{3})(?![A-Za-z\d]))([\s\S]*)`	keep · replace with `[DATA]` · keep · replace with `[DATA]` · keep · replace with `[DATA]` · keep · replace with `[DATA]` · keep
`([\s\S]*?)((?<![A-Za-z\d])(?:[A-CEGHJ-PR-TW-Z][A-CEGHJ-NPR-TW-Z] ?\d{2} \d{2} ?\d{2} ?[A-D]?|\d{3} \d{3} \d{4}|\d{5} \d{5}|(?:\+44 ?(?:\(0\) ?)?|\(?0)\d{2,4}\)? \d{3,4} ?\d{3,4}|GB\d{2} [A-Z]{4}(?: \d{4}){3} \d{2}|[A-Z]{1,2}\d[A-Z\d]? \d[ABD-HJLNP-UW-Z]{2}|[A-Z]{2}\d{2} [A-Z]{3})(?![A-Za-z\d]))([\s\S]*?)((?<![A-Za-z\d])(?:[A-CEGHJ-PR-TW-Z][A-CEGHJ-NPR-TW-Z] ?\d{2} \d{2} ?\d{2} ?[A-D]?|\d{3} \d{3} \d{4}|\d{5} \d{5}|(?:\+44 ?(?:\(0\) ?)?|\(?0)\d{2,4}\)? \d{3,4} ?\d{3,4}|GB\d{2} [A-Z]{4}(?: \d{4}){3} \d{2}|[A-Z]{1,2}\d[A-Z\d]? \d[ABD-HJLNP-UW-Z]{2}|[A-Z]{2}\d{2} [A-Z]{3})(?![A-Za-z\d]))([\s\S]*?)((?<![A-Za-z\d])(?:[A-CEGHJ-PR-TW-Z][A-CEGHJ-NPR-TW-Z] ?\d{2} \d{2} ?\d{2} ?[A-D]?|\d{3} \d{3} \d{4}|\d{5} \d{5}|(?:\+44 ?(?:\(0\) ?)?|\(?0)\d{2,4}\)? \d{3,4} ?\d{3,4}|GB\d{2} [A-Z]{4}(?: \d{4}){3} \d{2}|[A-Z]{1,2}\d[A-Z\d]? \d[ABD-HJLNP-UW-Z]{2}|[A-Z]{2}\d{2} [A-Z]{3})(?![A-Za-z\d]))([\s\S]*)`	keep · replace with `[DATA]` · keep · replace with `[DATA]` · keep · replace with `[DATA]` · keep

`([\s\S]*?)((?<![A-Za-z\d])(?:[A-CEGHJ-PR-TW-Z][A-CEGHJ-NPR-TW-Z] ?\d{2} \d{2} ?\d{2} ?[A-D]?|\d{3} \d{3} \d{4}|\d{5} \d{5}|(?:\+44 ?(?:\(0\) ?)?|\(?0)\d{2,4}\)? \d{3,4} ?\d{3,4}|GB\d{2} [A-Z]{4}(?: \d{4}){3} \d{2}|[A-Z]{1,2}\d[A-Z\d]? \d[ABD-HJLNP-UW-Z]{2}|[A-Z]{2}\d{2} [A-Z]{3})(?![A-Za-z\d]))([\s\S]*)(?<![A-Za-z\d])(?:[A-CEGHJ-PR-TW-Z][A-CEGHJ-NPR-TW-Z] ?\d{2} \d{2} ?\d{2} ?[A-D]?|\d{3} \d{3} \d{4}|\d{5} \d{5}|(?:\+44 ?(?:\(0\) ?)?|\(?0)\d{2,4}\)? \d{3,4} ?\d{3,4}|GB\d{2} [A-Z]{4}(?: \d{4}){3} \d{2}|[A-Z]{1,2}\d[A-Z\d]? \d[ABD-HJLNP-UW-Z]{2}|[A-Z]{2}\d{2} [A-Z]{3})(?![A-Za-z\d]))([\s\S]*)`	keep · replace with `[DATA]` · keep · replace with `[DATA]` · keep
`([\s\S]*?)((?<![A-Za-z\d])(?:[A-CEGHJ-PR-TW-Z][A-CEGHJ-NPR-TW-Z] ?\d{2} \d{2} ?\d{2} ?[A-D]?|\d{3} \d{3} \d{4}|\d{5} \d{5}|(?:\+44 ?(?:\(0\) ?)?|\(?0)\d{2,4}\)? \d{3,4} ?\d{3,4}|GB\d{2} [A-Z]{4}(?: \d{4}){3} \d{2}|[A-Z]{1,2}\d[A-Z\d]? \d[ABD-HJLNP-UW-Z]{2}|[A-Z]{2}\d{2} [A-Z]{3})(?![A-Za-z\d]))([\s\S]*)`	keep · replace with `[DATA]` · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

UK_TOWN DATA CLEANSING

Replaces each value found in `uk-town-generalized.txt` (9,078 pairs) by its counterpart; values not in the file stay as they are.

Used by: `UK_L2_TOWN`

CONFIGURATION

lookupFile	`uk-town-generalized.txt`
delimiter	`|`
caseSensitive	yes
trimWhitespace	yes

UK_UTR REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `UK_CM_ALNUM`.

Used by: `UK_L1_UTR`

PATTERNS, TRIED IN ORDER

`(0*?)(?:(\d[\s\-]?\d{4}[\s\-]?\d{5})(\s?[Kk]?))`	keep · `UK_UTR_DIGITS` · keep
otherwise	`UK_CM_ALNUM`

CONFIGURATION

trimInput	yes
requireMask	no

UK_UTR_AUX CHECK DIGIT

Masks the digits with `UK_IDENTITY` and recomputes a modulus 11 check digit with weights `4 6 8 10 9 7 5 3 1 0` from the right.

Used by: `UK_UTR_DIGITS`

CONFIGURATION

weightList	`4 6 8 10 9 7 5 3 1 0`
modulusNumber	`11`
checkDigitIndex	`1`
numDigitsForCheckdigitCalculation	`10`
calculateChecksumRightToLeft	yes
numericAlgorithm	`UK_IDENTITY`
alphaNumericAlgorithm	`UK_IDENTITY`
preserveRegex	`[.\-\s]`
inputHandlingConfig.characterHandling	`STANDARD`
inputHandlingConfig.invalidInputHandling	`ERROR`
inputHandlingConfig.shortInputHandling	`FALLBACK`
inputHandlingConfig.padCharacter	`0`
inputHandlingConfig.trimWhitespace	yes

UK_UTR_CD CHECK DIGIT

Masks the digits with `UK_CM_DIGITS` and recomputes a modulus 11 check digit with weights `2 3 4 5 10 9 8 7 6` from the right.

Used by: `UK_UTR_DIGITS`

CONFIGURATION

weightList	`2 3 4 5 10 9 8 7 6`
modulusNumber	`11`
checkDigitIndex	`0`
numDigitsForCheckdigitCalculation	`9`
calculateChecksumRightToLeft	yes
numericAlgorithm	`UK_CM_DIGITS`
alphaNumericAlgorithm	`UK_CM_DIGITS`
preserveRegex	`[.\-\s]`
inputHandlingConfig.characterHandling	`STANDARD`
inputHandlingConfig.invalidInputHandling	`ERROR`
inputHandlingConfig.shortInputHandling	`FALLBACK`
inputHandlingConfig.padCharacter	`0`
inputHandlingConfig.trimWhitespace	yes

UK_UTR_DIGITS STRING ALGORITHM CHAIN

Applies `UK_UTR_CD` → `UK_UTR_WIDEN` → `UK_UTR_AUX` → `UK_UTR_RESOLVE`, in this order.

Used by: `UK_UTR`

CONFIGURATION

algorithmReferences[1]	`UK_UTR_CD`
algorithmReferences[2]	`UK_UTR_WIDEN`

algorithmReferences[3]	`UK_UTR_AUX`
algorithmReferences[4]	`UK_UTR_RESOLVE`

UK_UTR_RESOLVE REGEX DECOMPOSE

Tries 11 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `UK_UTR_DIGITS`

PATTERNS, TRIED IN ORDER

`(00)([\s\-]?\d{4}[\s\-]?\d{5})`	replace with `2` · keep
`(0\d)([\s\-]?\d{4}[\s\-]?\d{5})`	replace with `1` · keep
`(1\d)([\s\-]?\d{4}[\s\-]?\d{5})`	replace with `1` · keep
`(2\d)([\s\-]?\d{4}[\s\-]?\d{5})`	replace with `2` · keep
`(3\d)([\s\-]?\d{4}[\s\-]?\d{5})`	replace with `3` · keep
`(4\d)([\s\-]?\d{4}[\s\-]?\d{5})`	replace with `4` · keep
`(5\d)([\s\-]?\d{4}[\s\-]?\d{5})`	replace with `5` · keep
`(6\d)([\s\-]?\d{4}[\s\-]?\d{5})`	replace with `6` · keep
`(7\d)([\s\-]?\d{4}[\s\-]?\d{5})`	replace with `7` · keep
`(8\d)([\s\-]?\d{4}[\s\-]?\d{5})`	replace with `8` · keep
`(9\d)([\s\-]?\d{4}[\s\-]?\d{5})`	replace with `9` · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

UK_UTR_WIDEN REGEX DECOMPOSE

Tries 10 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `UK_UTR_DIGITS`

PATTERNS, TRIED IN ORDER

`(0)([\s\-]?\d{4}[\s\-]?\d{5})`	replace with `00` · keep
`(1)([\s\-]?\d{4}[\s\-]?\d{5})`	replace with `10` · keep
`(2)([\s\-]?\d{4}[\s\-]?\d{5})`	replace with `20` · keep
`(3)([\s\-]?\d{4}[\s\-]?\d{5})`	replace with `30` · keep
`(4)([\s\-]?\d{4}[\s\-]?\d{5})`	replace with `40` · keep
`(5)([\s\-]?\d{4}[\s\-]?\d{5})`	replace with `50` · keep
`(6)([\s\-]?\d{4}[\s\-]?\d{5})`	replace with `60` · keep
`(7)([\s\-]?\d{4}[\s\-]?\d{5})`	replace with `70` · keep
`(8)([\s\-]?\d{4}[\s\-]?\d{5})`	replace with `80` · keep
`(9)([\s\-]?\d{4}[\s\-]?\d{5})`	replace with `90` · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

UK_YEAR REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: `UK_L2_BIRTH_YEAR`

PATTERNS, TRIED IN ORDER

`(0*?)(?:(\d{3})(\d))`	keep · keep · `UK_CM_ALNUM`
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

APPENDIX B

Files

The lookup and detection files the policy reads. In Delphix each one is uploaded with the object that reads it — the lookup file of a Secure Lookup or Data Cleansing algorithm, the value list of a List classifier — and referenced by that object's configuration.

FILE	LINES	USED BY	FIRST LINES
`uk-addresses.txt`	3,000	`UK_ADDRESS`	8 Meadow View · Meadow View, Station Road · 122 Shore Road · 80 North Street, Derry
`uk-benefits.txt`	13	`UK_BENEFIT_LIST`	Universal Credit · Personal Independence Payment · Disability Living Allowance · Employment and Support Allowance
`uk-chi-day.txt`	28	`UK_CHI_DAY`	01 · 02 · 03 · 04
`uk-chi-month.txt`	12	`UK_CHI_MONTH`	01 · 02 · 03 · 04
`uk-credit-statuses.txt`	13	`UK_CREDIT_STATUS`	Up to date · 1 payment missed · 2 payments missed · 3 payments missed
`uk-criminal-record.txt`	12	`UK_CRIMINAL_RECORD_LIST`	No convictions · Unspent conviction · Spent conviction · Caution
`uk-dbs-outcomes.txt`	7	`UK_DBS_OUTCOME`	Basic · Standard · Enhanced · Enhanced with barred list
`uk-decile-high.txt`	5	`UK_DECILE_HIGH`	6 · 7 · 8 · 9
`uk-decile.txt`	10	`UK_DECILE`	1 · 2 · 3 · 4
`uk-deprivation-statuses.txt`	5	`UK_DEPRIVATION_STATUS`	Most deprived · Deprived · Average · Less deprived
`uk-detect-given-names.txt`	356	`UK_L1_GIVEN_NAME – List` `UK_L1_FULL_NAME – List`	Aarav · Abena · Abigail · Adaeze
`uk-detect-sex-letter.txt`	2	`UK_L2_SEX – List`	f · m
`uk-detect-sex.txt`	5	`UK_L2_SEX – List`	female · male · woman · man
`uk-detect-surnames.txt`	295	`UK_L1_SURNAME – List` `UK_L1_FULL_NAME – List`	Abdi · Adams · Adeyemi · Ahmed
`uk-diagnoses.txt`	30	`UK_DIAGNOSIS_TEXT`	Essential hypertension · Type 2 diabetes mellitus · Asthma · Low back pain
`uk-disabilities.txt`	10	`UK_DISABILITY_LIST`	Disabled: limited a lot · Disabled: limited a little · Not disabled · Visual impairment
`uk-education.txt`	11	`UK_EDUCATION`	No qualifications · GCSE or equivalent · A level or equivalent · Scottish Highers
`uk-employers.txt`	25	`UK_EMPLOYER`	Northfield Trading Ltd · Riverside Services Ltd · Albion Construction Ltd · Pennine Transport Ltd
`uk-ethnic-groups.txt`	20	`UK_ETHNICITY_LIST`	White British · White Irish · Gypsy or Irish Traveller · Roma

FILE	LINES	USED BY	FIRST LINES
uk-flag-0-1.txt	2	UK_FLAG_0_1	0 · 1
uk-flag-true-false.txt	2	UK_FLAG_TRUE_FALSE	true · false
uk-flag-y-n.txt	2	UK_FLAG_Y_N	Y · N
uk-flag-yes-no.txt	2	UK_FLAG_YES_NO	Yes · No
uk-gender-identities.txt	6	UK_GENDER_IDENTITY_LIST	Same as sex registered at birth · Trans woman · Trans man · Non-binary
uk-given-names.txt	349	UK_GIVEN_NAME	Aarav · Abena · Abigail · Adaeze
uk-housing-tenure.txt	9	UK_HOUSING_LIST	Owned outright · Owned with a mortgage · Shared ownership · Social rented from council
uk-icd10-decimal.txt	33	UK_ICD10_DECIMAL	E11.9 · E78.5 · J45.9 · J06.9
uk-icd10.txt	37	UK_ICD10	I10 · E11 · E78 · J45
uk-marital-status.txt	9	UK_MARITAL_STATUS	Never married or civil partnered · Married · In a civil partnership · Separated
uk-medications.txt	26	UK_MEDICATION_LIST	Paracetamol · Ibuprofen · Naproxen · Co-codamol
uk-nationalities.txt	21	UK_NATIONALITY	British · Irish · Indian · Polish
uk-not-stated.txt	1	UK_SUPPRESS	Not stated
uk-occupations.txt	40	UK_OCCUPATION	Administrator · Sales assistant · Retail manager · Care worker
uk-political-opinions.txt	7	UK_POLITICAL_OPINION_LIST	Left · Centre-left · Centre · Centre-right
uk-political-parties.txt	14	UK_POLITICAL_PARTY_LIST	Labour · Conservative · Liberal Democrats · Reform UK
uk-quintile.txt	5	UK_QUINTILE	1 · 2 · 3 · 4
uk-religions.txt	18	UK_RELIGION_LIST	No religion · Christian · Church of England · Church of Scotland
uk-schools.txt	15	UK_SCHOOL	Oakfield Primary School · St Mary's Church of England Primary School · Holy Trinity Catholic Primary School · Riverside Infant School
uk-sex-f-m.txt	2	UK_SEX_F_M	F · M
uk-sex-female-male.txt	2	UK_SEX_FEMALE_MALE	Female · Male
uk-sex-woman-man.txt	2	UK_SEX_WOMAN_MAN	Woman · Man

FILE	LINES	USED BY	FIRST LINES
`uk-sexual-orientations.txt`	8	`UK_SEXUAL_ORIENTATION_LIST`	Straight or heterosexual · Gay or lesbian · Bisexual · Pansexual
`uk-surnames.txt`	294	`UK_SURNAME`	Abdi · Adams · Adeyemi · Ahmed
`uk-text-names.txt`	1,266	`UK_FREE_TEXT`	Aarav · AARAV · Abena · ABENA
`uk-town-generalized.txt`	9,078	`UK_TOWN`	Barnsley|Barnsley · BARNSLEY|BARNSLEY · Darton|Darton · DARTON|DARTON

APPENDIX C

Exclusion lists

Words that keep a column-name rule from matching when they follow it — so that `NOMBRE_PRODUCTO` is not taken for a person's name.

E1 `plate` `vehicle` `business` `premises` `liquor` `tv` `television` `software`

E2 `product` `prod` `item` `article` `company` `business` `trade` `brand` `file` `street` `road` `avenue` `village` `town` `city` `county` `district` `borough` `council` `country` `bank` `branch` `plan` `campaign` `project` `service` `table` `column` `field` `user` `host` `server` `domain` `model` `category` `type` `school` `college` `institution` `course` `programme` `program` `subject` `document` `doc` `centre` `center` `unit` `clinic` `hospital` `surgery` `practice` `trust` `area` `department` `dept` `position` `system` `app` `application` `group` `list` `report` `process` `event` `task` `profile` `role` `class` `object` `account` `drug` `medication` `diagnosis` `study` `benefit` `party` `union` `religion` `ethnicity` `language` `place` `store` `shop` `warehouse` `supplier` `vendor` `contract` `policy` `insurance` `org` `organi[sz]ation` `entity` `parameter` `variable` `method` `function` `index` `job` `script` `queue` `resource` `network` `device` `template` `image` `photo` `page` `site` `module` `menu` `option` `action` `permission` `tax` `currency` `issuer` `invoice` `payment` `form` `zone` `route` `line` `sector` `building` `display` `key` `tag` `label` `title` `code` `config` `setting` `firm` `employer` `estate` `parcel` `scheme` `community` `ward` `constituency` `customer` `cust` `client` `patient` `employee` `emp` `staff` `worker` `member` `student` `pupil` `insured` `beneficiary` `holder` `account_?holder` `policy_?holder` `mother` `father` `parent` `guardian` `spouse` `partner` `next_?of_?kin` `doctor` `gp` `physician` `nurse` `carer` `contact` `emergency_?contact` `guarantor` `debtor` `borrower` `applicant` `signatory` `witness` `representative` `owner` `tenant` `landlord` `victim` `complainant` `accused` `defendant` `donor` `heir` `person` `driver` `agent` `citizen` `elector` `voter` `taxpayer` `child` `dependant` `dependent` `relative` `officer` `teacher` `contributor` `pensioner` `claimant` `resident` `occupier` `full` `complete` `last` `first`

E3 `ip` `mac` `email` `e_?mail` `web` `url` `book` `type` `id`

E4 `id` `type` `code` `key` `no` `num` `number` `date` `status` `queue` `count` `log`

E5 `id` `key` `type` `role` `site` `ref` `code` `number` `no` `size`

APPENDIX D

Discovery policy and classifiers

The discovery policy `UK – UK GDPR and DPA 2018 – v1` runs these 130 classifiers, with an assignment threshold of 60%.

UK GDPR and DPA 2018 - personal data

CLASSIFIER	KIND	DATA CLASS
UK_L1_NINO – Path	COLUMN NAME	UK_L1_NINO
UK_L1_NINO – Regex	VALUE FORMAT	UK_L1_NINO
UK_L1_NHS_NUMBER – Path	COLUMN NAME	UK_L1_NHS_NUMBER
UK_L1_NHS_NUMBER – Regex	VALUE FORMAT	UK_L1_NHS_NUMBER
UK_L1_UTR – Path	COLUMN NAME	UK_L1_UTR
UK_L1_UTR – Regex	VALUE FORMAT	UK_L1_UTR
UK_L1_PASSPORT – Path	COLUMN NAME	UK_L1_PASSPORT
UK_L1_PASSPORT – Regex	VALUE FORMAT	UK_L1_PASSPORT
UK_L1_DRIVING_LICENCE – Path	COLUMN NAME	UK_L1_DRIVING_LICENCE
UK_L1_DRIVING_LICENCE – Regex	VALUE FORMAT	UK_L1_DRIVING_LICENCE
UK_L1_IMMIGRATION_DOCUMENT – Path	COLUMN NAME	UK_L1_IMMIGRATION_DOCUMENT
UK_L1_VITAL_RECORD – Path	COLUMN NAME	UK_L1_VITAL_RECORD
UK_L1_GIVEN_NAME – Path	COLUMN NAME	UK_L1_GIVEN_NAME
UK_L1_GIVEN_NAME – List	KNOWN VALUES	UK_L1_GIVEN_NAME
UK_L1_SURNAME – Path	COLUMN NAME	UK_L1_SURNAME
UK_L1_SURNAME – List	KNOWN VALUES	UK_L1_SURNAME
UK_L1_FULL_NAME – Path	COLUMN NAME	UK_L1_FULL_NAME
UK_L1_FULL_NAME – List	KNOWN VALUES	UK_L1_FULL_NAME
UK_L1_EMAIL – Path	COLUMN NAME	UK_L1_EMAIL
UK_L1_EMAIL – Type	DATA TYPE	UK_L1_EMAIL
UK_L1_EMAIL – Regex	VALUE FORMAT	UK_L1_EMAIL
UK_L1_PHONE – Path	COLUMN NAME	UK_L1_PHONE
UK_L1_PHONE – Regex	VALUE FORMAT	UK_L1_PHONE
UK_L1_ADDRESS – Path	COLUMN NAME	UK_L1_ADDRESS
UK_L1_ADDRESS – Type	DATA TYPE	UK_L1_ADDRESS
UK_L1_ADDRESS – Regex	VALUE FORMAT	UK_L1_ADDRESS
UK_L1_ADDRESS_DETAIL – Path	COLUMN NAME	UK_L1_ADDRESS_DETAIL
UK_L1_BANK_ACCOUNT – Path	COLUMN NAME	UK_L1_BANK_ACCOUNT
UK_L1_SORT_CODE – Path	COLUMN NAME	UK_L1_SORT_CODE
UK_L1_SORT_CODE – Regex	VALUE FORMAT	UK_L1_SORT_CODE
UK_L1_IBAN – Path	COLUMN NAME	UK_L1_IBAN

CLASSIFIER	KIND	DATA CLASS
UK_L1_IBAN – Regex	VALUE FORMAT	UK_L1_IBAN
UK_L1_PAYMENT_CARD – Path	COLUMN NAME	UK_L1_PAYMENT_CARD
UK_L1_PAYMENT_CARD – Regex	VALUE FORMAT	UK_L1_PAYMENT_CARD
UK_L1_IP_ADDRESS – Path	COLUMN NAME	UK_L1_IP_ADDRESS
UK_L1_IP_ADDRESS – Regex	VALUE FORMAT	UK_L1_IP_ADDRESS
UK_L1_DEVICE_ID – Path	COLUMN NAME	UK_L1_DEVICE_ID
UK_L1_DEVICE_ID – Regex	VALUE FORMAT	UK_L1_DEVICE_ID
UK_L1_LICENCE_PLATE – Path	COLUMN NAME	UK_L1_LICENCE_PLATE
UK_L1_LICENCE_PLATE – Regex	VALUE FORMAT	UK_L1_LICENCE_PLATE
UK_L1_VIN – Path	COLUMN NAME	UK_L1_VIN
UK_L1_VIN – Regex	VALUE FORMAT	UK_L1_VIN
UK_L1_PROPERTY – Path	COLUMN NAME	UK_L1_PROPERTY
UK_L1_CONTRACT_NUMBER – Path	COLUMN NAME	UK_L1_CONTRACT_NUMBER
UK_L1_USERNAME – Path	COLUMN NAME	UK_L1_USERNAME
UK_L1_CREDENTIAL – Path	COLUMN NAME	UK_L1_CREDENTIAL
UK_L1_CREDENTIAL – Regex	VALUE FORMAT	UK_L1_CREDENTIAL
UK_L1_GEOLOCATION – Path	COLUMN NAME	UK_L1_GEOLOCATION
UK_L1_GEOLOCATION – Regex	VALUE FORMAT	UK_L1_GEOLOCATION
UK_L1_FREE_TEXT – Path	COLUMN NAME	UK_L1_FREE_TEXT
UK_L1_FREE_TEXT – Type	DATA TYPE	UK_L1_FREE_TEXT
UK_L1_FREE_TEXT – Regex	VALUE FORMAT	UK_L1_FREE_TEXT
UK_L2_BIRTH_DATE – Path	COLUMN NAME	UK_L2_BIRTH_DATE
UK_L2_BIRTH_DATE – Type	DATA TYPE	UK_L2_BIRTH_DATE
UK_L2_BIRTH_YEAR – Path	COLUMN NAME	UK_L2_BIRTH_YEAR
UK_L2_BIRTH_YEAR – Type	DATA TYPE	UK_L2_BIRTH_YEAR
UK_L2_AGE – Path	COLUMN NAME	UK_L2_AGE
UK_L2_AGE – Type	DATA TYPE	UK_L2_AGE
UK_L2_EVENT_DATE – Path	COLUMN NAME	UK_L2_EVENT_DATE
UK_L2_EVENT_DATE – Type	DATA TYPE	UK_L2_EVENT_DATE
UK_L2_SEX – Path	COLUMN NAME	UK_L2_SEX
UK_L2_SEX – List	KNOWN VALUES	UK_L2_SEX
UK_L2_POSTCODE – Path	COLUMN NAME	UK_L2_POSTCODE
UK_L2_POSTCODE – Regex	VALUE FORMAT	UK_L2_POSTCODE
UK_L2_TOWN – Path	COLUMN NAME	UK_L2_TOWN
UK_L2_TOWN – Values	VALUE FORMAT	UK_L2_TOWN
UK_L2_NATIONALITY – Path	COLUMN NAME	UK_L2_NATIONALITY
UK_L2_NATIONALITY – Values	VALUE FORMAT	UK_L2_NATIONALITY

CLASSIFIER	KIND	DATA CLASS
UK_L2_MARITAL_STATUS – Path	COLUMN NAME	UK_L2_MARITAL_STATUS
UK_L2_MARITAL_STATUS – Values	VALUE FORMAT	UK_L2_MARITAL_STATUS
UK_L2_OCCUPATION – Path	COLUMN NAME	UK_L2_OCCUPATION
UK_L2_EMPLOYER – Path	COLUMN NAME	UK_L2_EMPLOYER
UK_L2_EDUCATION – Path	COLUMN NAME	UK_L2_EDUCATION
UK_L2_EDUCATION – Values	VALUE FORMAT	UK_L2_EDUCATION
UK_L2_SCHOOL – Path	COLUMN NAME	UK_L2_SCHOOL
UK_L2_DEPENDANTS – Path	COLUMN NAME	UK_L2_DEPENDANTS
UK_L2_DEPENDANTS – Type	DATA TYPE	UK_L2_DEPENDANTS
UK_L3_GENETIC – Path	COLUMN NAME	UK_L3_GENETIC
UK_L3_GENETIC – Regex	VALUE FORMAT	UK_L3_GENETIC
UK_L3_BIOMETRIC – Path	COLUMN NAME	UK_L3_BIOMETRIC
UK_L3_BIOMETRIC – Type	DATA TYPE	UK_L3_BIOMETRIC
UK_L3_ETHNICITY – Path	COLUMN NAME	UK_L3_ETHNICITY
UK_L3_ETHNICITY – Values	VALUE FORMAT	UK_L3_ETHNICITY
UK_L3_RELIGION – Path	COLUMN NAME	UK_L3_RELIGION
UK_L3_RELIGION – Values	VALUE FORMAT	UK_L3_RELIGION
UK_L3_BELIEF – Path	COLUMN NAME	UK_L3_BELIEF
UK_L3_POLITICAL_OPINION – Path	COLUMN NAME	UK_L3_POLITICAL_OPINION
UK_L3_POLITICAL_OPINION – Values	VALUE FORMAT	UK_L3_POLITICAL_OPINION
UK_L3_POLITICAL_AFFILIATION – Path	COLUMN NAME	UK_L3_POLITICAL_AFFILIATION
UK_L3_POLITICAL_AFFILIATION – Values	VALUE FORMAT	UK_L3_POLITICAL_AFFILIATION
UK_L3_TRADE_UNION – Path	COLUMN NAME	UK_L3_TRADE_UNION
UK_L3_TRADE_UNION – Values	VALUE FORMAT	UK_L3_TRADE_UNION
UK_L3_SEXUAL_ORIENTATION – Path	COLUMN NAME	UK_L3_SEXUAL_ORIENTATION
UK_L3_SEXUAL_ORIENTATION – Values	VALUE FORMAT	UK_L3_SEXUAL_ORIENTATION
UK_L3_SEXUAL_LIFE – Path	COLUMN NAME	UK_L3_SEXUAL_LIFE
UK_L3_GENDER_IDENTITY – Path	COLUMN NAME	UK_L3_GENDER_IDENTITY
UK_L3_GENDER_IDENTITY – Values	VALUE FORMAT	UK_L3_GENDER_IDENTITY
UK_HEALTH_ID – Path	COLUMN NAME	UK_HEALTH_ID
UK_HEALTH_DIAGNOSIS – Path	COLUMN NAME	UK_HEALTH_DIAGNOSIS
UK_HEALTH_DIAGNOSIS – Regex	VALUE FORMAT	UK_HEALTH_DIAGNOSIS
UK_HEALTH_MEDICATION – Path	COLUMN NAME	UK_HEALTH_MEDICATION
UK_HEALTH_MEDICATION – Values	VALUE FORMAT	UK_HEALTH_MEDICATION
UK_HEALTH_CLINICAL_TEXT – Path	COLUMN NAME	UK_HEALTH_CLINICAL_TEXT
UK_HEALTH_CLINICAL_TEXT – Type	DATA TYPE	UK_HEALTH_CLINICAL_TEXT
UK_HEALTH_TEST_RESULT – Path	COLUMN NAME	UK_HEALTH_TEST_RESULT

CLASSIFIER	KIND	DATA CLASS
UK_HEALTH_BLOOD_GROUP – Path	COLUMN NAME	UK_HEALTH_BLOOD_GROUP
UK_HEALTH_BLOOD_GROUP – Values	VALUE FORMAT	UK_HEALTH_BLOOD_GROUP
UK_HEALTH_DISABILITY – Path	COLUMN NAME	UK_HEALTH_DISABILITY
UK_HEALTH_DISABILITY – Values	VALUE FORMAT	UK_HEALTH_DISABILITY
UK_HEALTH_REPRODUCTIVE – Path	COLUMN NAME	UK_HEALTH_REPRODUCTIVE
UK_HEALTH_MENTAL – Path	COLUMN NAME	UK_HEALTH_MENTAL
UK_FIN_CREDIT_REFERENCE – Path	COLUMN NAME	UK_FIN_CREDIT_REFERENCE
UK_FIN_DEBT – Path	COLUMN NAME	UK_FIN_DEBT
UK_FIN_DEBT – Type	DATA TYPE	UK_FIN_DEBT
UK_FIN_INCOME – Path	COLUMN NAME	UK_FIN_INCOME
UK_FIN_INCOME – Type	DATA TYPE	UK_FIN_INCOME
UK_FIN_ASSETS – Path	COLUMN NAME	UK_FIN_ASSETS
UK_FIN_ASSETS – Type	DATA TYPE	UK_FIN_ASSETS
UK_FIN_PENSION – Path	COLUMN NAME	UK_FIN_PENSION
UK_FIN_PENSION – Type	DATA TYPE	UK_FIN_PENSION
UK_FIN_BENEFIT – Path	COLUMN NAME	UK_FIN_BENEFIT
UK_FIN_BENEFIT – Values	VALUE FORMAT	UK_FIN_BENEFIT
UK_FIN_DEPRIVATION – Path	COLUMN NAME	UK_FIN_DEPRIVATION
UK_FIN_DEPRIVATION – Values	VALUE FORMAT	UK_FIN_DEPRIVATION
UK_FIN_HOUSING – Path	COLUMN NAME	UK_FIN_HOUSING
UK_FIN_HOUSING – Values	VALUE FORMAT	UK_FIN_HOUSING
UK_PENAL_CRIMINAL_RECORD – Path	COLUMN NAME	UK_PENAL_CRIMINAL_RECORD
UK_PENAL_CASE_NUMBER – Path	COLUMN NAME	UK_PENAL_CASE_NUMBER
UK_PENAL_CASE_NUMBER – Regex	VALUE FORMAT	UK_PENAL_CASE_NUMBER
UK_PENAL_DBS – Path	COLUMN NAME	UK_PENAL_DBS